MTH1002: Mathematical Methods (Part 2)

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Summary of the module

We may be doing mathematics for its own sake, for applications, or for fun, but inevitably we need to be able to calculate. That is what gives the subject its great power. Of course, we could just develop an appropriate method of calculation as and when we need it, but to do that all the time would be rather inefficient. Mathematical Methods covers developed bodies of useful techniques as a sort of toolkit of common knowledge. It emphasises techniques rather than applications.

Course content can be found on the ELE page. The syllabus plan and reading list can be found here.

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0 Advice and Notation

These notes are distinguished from similar texts by the inclusion of reminders, examples, and side comments intended to aid your understanding of the material.

0.1 Some Comments/Advice:

- These lecture notes are, by no means, a replacement to attending the lectures. All the theory and examples in these notes are understood, and better understood, through the explanations given in the lectures.
- Some definitions, theorems and other statements might look like a collection of symbols, but we use such ways to write complex maths ideas in a condensed form.
- When a theorem or proposition is presented, we usually follow it up with its proof. In some cases within these notes, the proof is omitted as it is outside the scope of this module, and will be visited in MTH1001 Mathematical Structures or MTH2008 Real Analysis.
- If you are struggling to understand something, try explaining it to a fellow student in your class. By doing so, you will understand it yourself.
- The examination in January and May will consist of a mixture of statements of definitions, theorems, proofs and exercises/calculations.
- Please email me (l.hakim@exeter.ac.uk) if you need some one to one help.

0.2 Plan for Mathematical Methods (Part 2)

Weeks 1-5: Chapters 1-2

Weeks 6-10: Chapters 3-4

Week 11: Revision

1 Matrices

Matrices (singular: *matrix*) are arrays of numbers, for example,

$$M = \begin{bmatrix} 4 & -\sqrt{3} & \pi \\ -5.2 & 0 & 13 \end{bmatrix},$$

and *vectors* are matrices that have only one column. Usage of vectors and matrices makes the notation and handling of data and variables in large computations more clear and compact, and their study has also led to new concepts and theories. Vectors and matrices are fundamental in mathematics and for applications of mathematical analysis.

In the example below, we will work with a matrix that reflects contact networks (e.g. disease spread), and this matrix can therefore be thought of as an important link that makes disease spread accessible to mathematical analysis. Other networks that are frequently analysed using matrices include: social networks (e.g., to study the spread of news), the world wide web (e.g., Google PageRank), network of cities (minimise traffic).

Example 1.1

1.1 Matrix Arithmetic

Definition 1.2

An $m \times n$ ***matrix*** is an array

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix},$$

where m is the number of rows and n the number of columns. We also refer to such an A as a matrix of *size* $m \times n$.

An equation of the form $A = B$, where A and B are matrices of the same size $m \times n$, means that $a_{ij} = b_{ij}$ for all $i \in \{1, 2, \ldots, m\}, j \in \{1, 2, \ldots, n\}$. Matrices of different sizes can never be equal.

A *square matrix* is a matrix with $m = n$. A particularly important square matrix is the *identity matrix*, denoted I,

$$I = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix},$$

that has entries 1 on the diagonal and 0 everywhere else.

The *transpose* of a matrix leaves the entries $a_{11}, a_{22}, a_{33}, \ldots$ alone and swaps all other entries across them. This operation is denoted with a $^\top$ and changes the size of the matrix unless it is square, cf. the examples below. (The formulas for this are not included in the definitions above since this could cause confusion over the indexing of matrix entries.)

Matrix Addition and Subtraction

Addition and subtraction can only be carried out for matrices of the same size, and then the operation is carried out element-wise:

$$A \pm B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \pm \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} \pm b_{11} & a_{12} \pm b_{12} & \cdots & a_{1n} \pm b_{1n} \\ a_{21} \pm b_{21} & a_{22} \pm b_{22} & \cdots & a_{2n} \pm b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} \pm b_{m1} & a_{m2} \pm b_{m2} & \cdots & a_{mn} \pm b_{mn} \end{bmatrix}.$$

Matrix Multiplication

Multiplying a number with a matrix is called *scalar multiplication*:

$$\lambda A = \lambda \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = \begin{bmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \cdots & \lambda a_{mn} \end{bmatrix}.$$

Multiplying two matrices A and B is called *matrix multiplication* – it can be carried out only if the number of columns of A agrees with the number of rows of B:

$$A\,B = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1k} \\ b_{21} & b_{22} & \cdots & b_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nk} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1k} \\ c_{21} & c_{22} & \cdots & c_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mk} \end{bmatrix},$$

where $c_{ij} = \sum_{s=1}^{n} a_{is} b_{sj}$. Note that the resulting matrix is of size $m \times k$.

Regarding the condition on when two matrices can be multiplied, the following might be useful to remember: "$(m \times n) \times (n \times k)$ works and gives a $m \times k$ matrix." Therefore, the result

of a $1 \times n$ with a $n \times 1$ is a 1×1, which is just a number, also called a *scalar*.

The rules for matrix multiplication might seem quite difficult – a simple special case that is the building block to understanding general matrix multiplication is the above mentioned multiplication of a $1 \times n$ matrix with a $n \times 1$ matrix:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \\ \vdots \\ b_{n1} \end{bmatrix} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} + \cdots + a_{1n}b_{n1}$$

$$= \sum_{s=1}^{n} a_{1s}b_{s1} = c_{11}.$$

In this case, both matrices have the same number of elements, and those entries are just multiplied with each other and then added up. For other matrices, one just repeats this procedure $m \cdot k$ times as follows:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1k} \\ b_{21} & b_{22} & \cdots & b_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nk} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1k} \\ c_{21} & c_{22} & \cdots & c_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mk} \end{bmatrix}.$$

Example 1.3 (i) The 2×2 identity matrix is $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

(ii) The following matrix A is an example of a 3×2 matrix,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix},$$

and its transpose is the 2×3 matrix

$$A^\top = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}.$$

(iii) For the matrix A from (ii), we have

$$A+A=\begin{bmatrix}1&2\\3&4\\5&6\end{bmatrix}+\begin{bmatrix}1&2\\3&4\\5&6\end{bmatrix}=\begin{bmatrix}1+1&2+2\\3+3&4+4\\5+5&6+6\end{bmatrix}=\begin{bmatrix}2&4\\6&8\\10&12\end{bmatrix},$$

which agrees with

$$2A=2\begin{bmatrix}1&2\\3&4\\5&6\end{bmatrix}=\begin{bmatrix}2\cdot1&2\cdot2\\2\cdot3&2\cdot4\\2\cdot5&2\cdot6\end{bmatrix}=\begin{bmatrix}2&4\\6&8\\10&12\end{bmatrix}.$$

(iv) For A and its transpose $A^\top$, we obtain

$$\begin{aligned}AA^\top&=\begin{bmatrix}1&2\\3&4\\5&6\end{bmatrix}\begin{bmatrix}1&3&5\\2&4&6\end{bmatrix}\\&=\begin{bmatrix}1\cdot1+2\cdot2&1\cdot3+2\cdot4&1\cdot5+2\cdot6\\3\cdot1+4\cdot2&3\cdot3+4\cdot4&3\cdot5+4\cdot6\\5\cdot1+6\cdot2&5\cdot3+6\cdot4&5\cdot5+6\cdot6\end{bmatrix}=\begin{bmatrix}5&11&17\\11&25&39\\17&39&61\end{bmatrix},\end{aligned}$$

and

$$\begin{aligned}A^\top A&=\begin{bmatrix}1&3&5\\2&4&6\end{bmatrix}\begin{bmatrix}1&2\\3&4\\5&6\end{bmatrix}\\&=\begin{bmatrix}1\cdot1+3\cdot3+5\cdot5&1\cdot2+3\cdot4+5\cdot6\\2\cdot1+4\cdot3+6\cdot5&2\cdot2+4\cdot4+6\cdot6\end{bmatrix}=\begin{bmatrix}35&44\\44&56\end{bmatrix}.\end{aligned}$$

Properties

(i) Matrix addition is commutative and associative,

$$A+B=B+A,\qquad A+B+C=(A+B)+C=A+(B+C),$$

where all three matrices are of the same size.

(ii) Matrix multiplication is associative as well, but it is not commutative! That is, AB is usually not the same as BA. However, scalars can be swapped with matrices,

$$A(\lambda B)=\lambda AB.$$

(iii) For combinations of the two operations, distributive laws hold,

$$A(B+\widetilde{B}) = AB + A\widetilde{B}, \qquad (A+\widetilde{A})B = AB + \widetilde{A}B,$$

where $A, \widetilde{A}$ are $m \times n$ and $B, \widetilde{B}$ are $n \times k$.

(iv) For the transpose of a matrix multiplication, we have

$$(AB)^\top = B^\top A^\top,$$

where the sizes of A and B are as in the previous property.

Vector-Matrix Multiplication

A n-vector is a $n \times 1$ matrix:

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix}.$$

We also refer to such a $\vec{v}$ as a vector of size n.

Multiplication of a matrix and a vector is carried out according to the rules of matrix multiplication: If the number of columns of A agrees with the size of $\vec{v}$, then

$$A\,v = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix},$$

where $c_i = \sum_{s=1}^{n} a_{is} v_s$. The result is a vector with m entries.

When defining a vector in-line, it is more convenient to write down its transpose, for example, $\vec{v} = \begin{bmatrix} 2 & -1 & 3 \end{bmatrix}^\top$ is a column vector, not a row vector:

$$v = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

Example 1.4 (i)

$$\begin{bmatrix} 11 \\ -6 \\ 6 \end{bmatrix} - \begin{bmatrix} -7 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 18 \\ -5 \\ 3 \end{bmatrix}$$

(ii)

$$\begin{bmatrix} 0 & -2 & 13 \end{bmatrix} \begin{bmatrix} 3 & 2 & -4 \\ 5 & -3 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 13 \end{bmatrix} \begin{bmatrix} 27 \\ -6 \\ -12 \end{bmatrix} = -144$$

Definition 1.5 — Determinants. The ***determinant*** of a square matrix is a scalar and defined as follows:

The determinant of a 1×1 matrix is

$$\det \begin{bmatrix} a_{11} \end{bmatrix} = a_{11}.$$

The determinant of a 2×2 matrix is

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}a_{22} - a_{21}a_{12}.$$

The determinant of a 3×3 matrix is

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

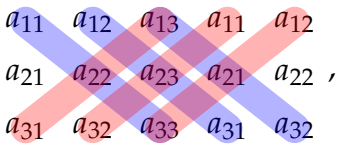
Remark 1.6

(i) As an abbreviation, one can write $|\ldots|$ for $\det \begin{bmatrix} \ldots \end{bmatrix}$.

(ii) Determinants of 1×1 matrices are quite straightforward; for 2×2, subtract the product of the off-

diagonal entries from the product of the diagonal entries; see the following two points for 3×3. Determinants are also defined for larger square matrices, but the formulas become quite bulky then and one would not write them out as in the definition above. See point (v) below for a technique that allows to reduce large determinants to smaller ones.

(iii) Sarrus's rule will help you memorize the formula for the determinant of a 3×3 matrix: For A with entries as in (iv) of the definition above, write down A and then copy the first two columns and append them to the right of the matrix, draw (or imagine) the following lines,



add up the products of the entries in each blue line, and then subtract the three red products.

(iv) Laplace's formula:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Here, we have "developed" the determinant along the first row (a_{11}, a_{12}, a_{13}), but one can develop along a different row as well or even along a col-

umn. The general formulas are:

$$\det A = \sum_{j=1}^{3} (-1)^{i+j} a_{ij} \det \widetilde{A}_{ij} \qquad \text{(dev. along } i\text{-th row)},$$

$$\det A = \sum_{i=1}^{3} (-1)^{i+j} a_{ij} \det \widetilde{A}_{ij} \qquad \text{(dev. along } j\text{-th column)},$$

where $\widetilde{A}_{ij}$ is the 2×2 matrix we obtain by removing the i-th row and the j-th column from A. Choose the row/column wisely! For example, if there is a row or a column with only one non-zero entry, developing along it will shorten the computation.

(v) Laplace's formula generalises to larger square matrices – just replace the maximum index 3 of the sums in (v) with n. However, note that Laplace's formula does not directly compute determinants – it just reduces the problem of finding the determinant of an $n \times n$ matrix to finding n $(n-1) \times (n-1)$ determinants. Sarrus' rule works only for 3×3's; it can not be applied to larger matrices!

Example 1.7 (i)

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \cdot 4 - 2 \cdot 3 = -2$$

(ii)

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1 \cdot 5 \cdot 9 + 2 \cdot 6 \cdot 7 + 3 \cdot 4 \cdot 8 - 7 \cdot 5 \cdot 3 - 8 \cdot 6 \cdot 1 - 9 \cdot 4 \cdot 2 = 0$$

(iii)

$$\begin{vmatrix} 1 & -3 & 2 \\ 0 & 7 & -1 \\ 0 & 0 & -3 \end{vmatrix} = 1 \cdot 7 \cdot (-3) = -21$$

(iv) To compute the determinant of

$$A = \begin{bmatrix} 4 & -1 & 11 & 3 \\ -2 & 3 & -3 & -5 \\ 0 & 1 & -8 & 0 \\ 1 & 0 & -6 & -2 \end{bmatrix},$$

we use Laplace's formula, developing along the row with the most zero entries:

$$\det A = (-1)^{3+1} \cdot 0 \begin{vmatrix} \dots \end{vmatrix} + (-1)^{3+2} \cdot 1 \begin{vmatrix} \dots \end{vmatrix} + (-1)^{3+3} \cdot (-8) \begin{vmatrix} \dots \end{vmatrix}$$

$$+ (-1)^{3+4} \cdot 0 \begin{vmatrix} \dots \end{vmatrix} = - \begin{vmatrix} 4 & 11 & 3 \\ -2 & -3 & -5 \\ 1 & -6 & -2 \end{vmatrix} - 8 \begin{vmatrix} 4 & -1 & 3 \\ -2 & 3 & -5 \\ 1 & 0 & -2 \end{vmatrix}$$

$$= -4 \begin{vmatrix} -3 & -5 \\ -6 & -2 \end{vmatrix} + 11 \begin{vmatrix} -2 & -5 \\ 1 & -2 \end{vmatrix} - 3 \begin{vmatrix} -2 & -3 \\ 1 & -6 \end{vmatrix} - 8 \cdot \begin{vmatrix} \dots \end{vmatrix} = \dots = 342.$$

Properties

Let A be a $n \times n$ matrix and let λ be a scalar.

(i) If A has a row or a column of zeros, then $\det A = 0$.

(ii) If a row of A is a multiple of some other row of A, then $\det A = 0$. Similar for columns.

(iii) If A is an upper triangular matrix,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & & a_{2n} \\ 0 & 0 & a_{33} & & a_{3n} \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & a_{nn} \end{bmatrix},$$

then the determinant is just the product along the diagonal,

$$\det A = a_{11} a_{22} a_{33} \dots a_{nn}.$$

Exercise 1.8 (i) Work through the problems on the first MTH1002 tutorial sheet. You can also just make up some matrices, do the maths, and com-

pare your results to the solutions found by online applets or computer algebra systems; for example: WolframAlpha or bluebit matrix calculator (online), Matlab or Maple (available on work stations in the Harrison building), Octave (freely available to download onto your computer), and many more.

(ii) Pick a 2×3 and a 3×1 matrix and check that property (iv) in 1.1 works. Similarly, verify the statements made in properties 1.1 for a few examples. (Note, however, that verifying examples does not constitute a proof!)

(iii) Let I be the 2×2 identity matrix. Show that for any 2×2 matrix A we have[a]

$$A\,I = I\,A = A.$$

(iv) You have to find the determinant of a 7×7 matrix that has no zero entries. You plan to reduce the problem to finding 3×3's by applying Laplace's formula several times. How many 3×3 determinants will you have to compute[b]?

(v) Find the two scalars λ_1 and λ_2 for which the determinant of

$$A_\lambda = \begin{bmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{bmatrix}$$

is equal to zero[c].

---

[a] Hint: Consider a general matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, multiply with I in both orders, and compare.

[b] Answer: You should obtain a number between 800 and 900.

[c] Hint: Computing this determinant with Sarrus's rule gives a third-order algebraic expression in λ. To find its zeros, one first needs to guess one solution –

property (ii) of 1.1 might help you make a guess. You have solved the problem correctly once you have found two values λ for which $\det A_\lambda = 0$ – there are only two such values.

1.2 Systems of Linear Equations: Gaussian Elimination

Consider the following system of equations:

$$\begin{cases} 2x - 3y &= 7 \\ -2x + y &= 5 \end{cases}.$$

We can solve it by adding the two equations together – this gives $-2y = 12$ which means $y = -6$. Now that we know y, the first equation reads $2x - 3(-6) = 7$, and we obtain $x = -\frac{11}{2}$. Hence the solution of the system is

$$x = -\frac{11}{2},\ y = -6. \tag{1.1}$$

Remark 1.9

(i) The first equation above defines a line in the xy-plane, namely $y = \frac{2}{3}x - \frac{7}{3}$. Similarly, the second equation describes the line $y = 2x + 5$. With the computation above, we have found the intersection of those two lines.

(ii) In order to convince yourself that it is ok to add two equations, think of the following: Suppose we have two libra-style scales, and that each of them is in balance, i.e. for each scale we have equality of the weights on its two arms. Then, taking the two objects from one scale and putting them onto the two arms of the other, it will still be in balance!

(iii) Note that we could have written the solution (1.1) above – which consists of two equations – as one

vector equation:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -\frac{11}{2} \\ -6 \end{bmatrix}.$$

Definition 1.10 — Linear Systems of Equations. A system of equations of the form

$$\begin{array}{ccccccccc} a_{11}\,x_1 & + & a_{12}\,x_2 & + & a_{13}\,x_3 & \dots & + & a_{1n}\,x_n & = b_1 \\ a_{21}\,x_1 & + & a_{22}\,x_2 & + & a_{23}\,x_3 & \dots & + & a_{2n}\,x_n & = b_2 \\ \vdots & & & & \vdots & & & & \vdots \\ a_{m1}\,x_1 & + & a_{m2}\,x_2 & + & a_{m3}\,x_3 & \dots & + & a_{mn}\,x_n & = b_m \end{array}$$

is called a ***linear system of equations***.

Here, $(x_1, x_2, \dots, x_n)$ are the variables, the a_{ij} are the coefficients of the system, and $(b_1, b_2, \dots, b_m)$ is the right-hand side (RHS). If the RHS is zero, $b_1 = b_2 = \cdots = b_m = 0$, then the system is called *homogeneous*. A combination of variables $(x_1^*, x_2^*, \dots, x_n^*)$, that satisfies all m equations simultaneously, is called a *solution* of the system.

Remark 1.11

(i) If $n = 2$ or $n = 3$, one would often denote the variables (x, y) or (x, y, z).

(ii) Each equation demands that some "linear combination" of the variables – that is, the variables multiplied by some coefficients and then added up – be equal to some given value. If powers or square roots or other functions of the variables appear in an equation, then it is not linear and can not be solved with the theory developed in this chapter.

(iii) Such a system can have no solutions, a unique solution, or many solutions.

(iv) We refer to the equations in the system as *rows*, and the following definition lists modifications of a system that do not change the set of solutions. We have already used one such modification in the computation above – adding one row to another.

Definition 1.12 — Elementary Row Operations. The following ***row operations*** on a system of equations do not change the set of its solutions:

(i) Add one row to another.

(ii) Multiply a row by a scalar different from 0.

(iii) Adding the multiple of a row to another row.

(iv) Swapping rows.

Example 1.13 Solve the system

$$\begin{cases} x - y = 3 \\ -3x + 4y + z = -1 \\ 2x + 7z = -3 \end{cases}.$$

We "eliminate" the terms $-3x$ and $2x$ by adding 3 times the first row to the second and by subtracting 2 times the first row from the third. We denote this "$R2 \to R2 + 3 \cdot R1$" and "$R3 \to R3 - 2 \cdot R1$", i.e. the arrow should be read as "is

replaced with":

$$\begin{array}{l} \\ R2 \to R2 + 3 \cdot R1 \\ R3 \to R3 - 2 \cdot R1 \\ \\ \\ R3 \to R3 - 2 \cdot R2 \end{array} \quad \begin{array}{l} \to \\ \\ \\ \\ \to \\ \\ \\ \\ \end{array} \quad \begin{cases} x - y = 3 \\ y + z = 8 \\ 2y + 7z = -9 \end{cases} \quad \begin{cases} x - y = 3 \\ y + z = 8 \\ 5z = -25 \end{cases} \quad . \tag{1.2}$$

The last line gives $z = -5$, and one can then find y and x by back-substituting into the first two equations. We obtain:

$$x = 16,\ y = 13,\ z = -5.$$

Remark 1.14

(i) Note that the system in definition 1.10 could also be written in matrix form – this is more clear and efficient and will be done from the following definition on. Having in mind that an equation of two n-vectors actually amounts to n equations, convince yourself that the following is equivalent to the system in definition 1.10:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

(ii) The transformations in the previous example are called *Gaussian Elimination*. We will now define two standard forms that are the goal of this process, and we then discuss reading off solutions and

the different cases – no solution, one unique solution, many solutions.

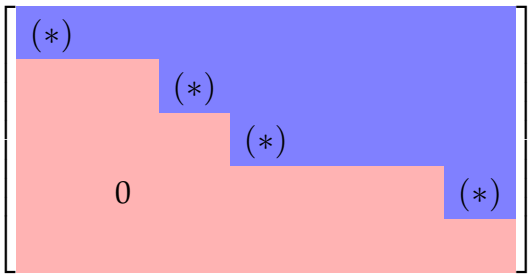
Augmented Matrix, Row Echelon Form, Rank

(i) The *augmented matrix* for the system in definition 1.10 is

$$[A \mid b] = \left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} & b_m \end{array}\right],$$

which can be subjected to elementary row operations the same way as fully written-out systems of linear equations.

(ii) The first non-zero entry in a row of a matrix is called the *leading entry*. A matrix (augmented or not) of the form



where all entries in the red (light) part are 0 and the $(*)$ stand for leading entries, is said to be in *row echelon form* (REF). That is: zero-rows are at the bottom, and for the remaining rows – say there are r of them – we have

$$j_1 < j_2 < j_3 < \cdots < j_r,$$

where j_i is the column index of the leading entry in row i (e.g., in the schematic example above, $j_1 = 1, j_2 = 3, j_3 = 4, j_4 = 7$).

(iii) A matrix in REF is further said to be in *reduced row echelon form* (RREF), if all leading entries are 1 and if each leading entry is the only non-zero entry in its column.

(iv) The *rank* of a matrix is the number of non-zero rows in its REF.

Example 1.15 (i) The matrices

$$\begin{bmatrix} -3 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & -3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

all are in REF and have ranks 2, 2, and 3. If the matrix on the left is the outcome of Gaussian Elimination of some 3×3 matrix without any rows of zeros, then this original matrix has rank 2 as well – the rank does not change under elementary row operations and it can be read off directly from the REF. The matrices

$$\begin{bmatrix} -3 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & 7 \\ 0 & 0 & 8 \end{bmatrix}$$

are not in REF.

(ii) The matrices

$$I, \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

all are in RREF, but the matrices in (i) are not.

(iii) In example 1.13, we have transformed

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 3 \\ -3 & 4 & 1 & -1 \\ 2 & 0 & 7 & -3 \end{array}\right] \overset{(1.2)}{\rightarrow} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 3 \\ 0 & 1 & 1 & 8 \\ 0 & 0 & 5 & -25 \end{array}\right].$$

Hence the augmented matrix has rank 3. The transformed augmented matrix is in REF. Including the vector of variables (that is dropped when writing systems as augmented matrices), the last line reads

$$\begin{bmatrix} 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -25 \end{bmatrix},$$

that is, $5z = -25$. With back-substitution, one can then find x and y, as above. Another option would be to bring the augmented matrix into RREF:

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 3 \\ 0 & 1 & 1 & 8 \\ 0 & 0 & 5 & -25 \end{array}\right] \overset{R3 \to {}^{1}/_{5} \cdot R3}{\to} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 3 \\ 0 & 1 & 1 & 8 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

$$\overset{R2 \to R2 - R3}{\to} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 3 \\ 0 & 1 & 0 & 13 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

$$\overset{R1 \to R1 + R2}{\to} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 16 \\ 0 & 1 & 0 & 13 \\ 0 & 0 & 1 & -5 \end{array}\right].$$

This reads

$$\begin{bmatrix} 16 \\ 13 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

i.e. one can – for this example – directly read off the solution from the RHS of the augmented matrix in RREF.

(iv) Solve the system

$$\begin{cases} -3x & & + & 3z & = & 4 \\ 3x & + & 5y & + & z & = & 0 \\ -x & + & 5y & + & 5z & = & 3 \end{cases},$$

Remark 1.16

As the previous example has shown, if the REF of the augmented matrix has a row of zeros on the A-side, but the corresponding RHS entry is different from zero, then the system has no solutions. A formal way of expressing this is

$$\operatorname{rank}[A \mid b] > \operatorname{rank} A$$

– the rank of coefficient matrix alone is smaller than the rank of the augmented matrix.

Theorem 1.17 — Solutions of Systems of Linear Equations. Suppose a system of m linear equations in n variables is given. That is, the coefficient matrix A of the system is of size $m \times n$. Denote the right-hand side of the system by b. Then the following is true for the set of solutions:

(i) If
$$\operatorname{rank} A < \operatorname{rank}[A \mid b],$$
then the system has no solution.

(ii) If
$$\operatorname{rank} A = \operatorname{rank}[A \mid b] = n,$$
then the system has a unique solution.

(iii) If
$$\operatorname{rank} A = \operatorname{rank}[A \mid b] < n,$$
then the system has many solutions.

(iv) Those cases cover all possibilities, i.e., $\operatorname{rank} A$ is never greater than $\operatorname{rank}[A \mid b]$, and neither of the two can be greater than n.

Example 1.18 (i) Example (iii) of 1.15 corresponds to unique-solution case of theorem 1.17, and example (iv) to the

no-solution case. In the third case, obtaining the answer $\begin{bmatrix} x & y & z \end{bmatrix}^\top$ from the RREF is slightly less straightforward – this will be demonstrated in the following examples.

(ii) We re-do example 1.15 (iv) with the right-hand side $b = \begin{bmatrix} -9 & 8 & -4 \end{bmatrix}^\top$. The same row operations as above lead to

$$\left[\begin{array}{ccc|c} -3 & 0 & 3 & -9 \\ 3 & 5 & 1 & 8 \\ -1 & 5 & 5 & -4 \end{array}\right] \quad \vdots\rightarrow \quad \left[\begin{array}{ccc|c} 1 & -5 & -5 & 4 \\ 0 & 20 & 16 & -4 \\ 0 & -15 & -12 & 3 \end{array}\right].$$

The third row is a multiple of the second, and the next operation, $R3 \to R3 + {}^3/_4 R2$, eliminates the third row altogether. We then continue to bring the system in RREF:

$$\left[\begin{array}{ccc|c} 1 & -5 & -5 & 4 \\ 0 & 20 & 16 & -4 \\ 0 & 0 & 0 & 0 \end{array}\right] \overset{R2 \to {}^1/_{20} R2}{\to} \left[\begin{array}{ccc|c} 1 & -5 & -5 & 4 \\ 0 & 1 & {}^4/_5 & -{}^1/_5 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\overset{R1 \to R1 + 5R2}{\to} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 3 \\ 0 & 1 & {}^4/_5 & -{}^1/_5 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

One obtains a formula for the solution $\begin{bmatrix} x & y & z \end{bmatrix}^\top$ as follows. Variables corresponding to columns that do not have a leading entry can be chosen freely, and this is expressed using a parameter,

$$z = t \qquad (t \in \mathbb{R}).$$

To obtain an expression for y, we use the row whose leading entry is in the column corresponding to y,

$$y + \tfrac{4}{5} z = -\tfrac{1}{5} \to \quad y = -\tfrac{1}{5} - \tfrac{4}{5} t,$$

and similarly for x,

$$x = 3 + t.$$

Therefore, the answer is

$$\begin{cases} x = 3 + t \\ y = -\frac{1}{5} - \frac{4}{5}t \\ z = t \end{cases}, \qquad (t \in \mathbb{R}),$$

which could also be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1/5 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 1 \\ -4/5 \\ 1 \end{bmatrix}, \qquad (t \in \mathbb{R}),$$

and is the equation of a line – namely the line of common points of the three planes $-3x + 3z = -9$, $3x + 5y + z = 8$, $-x + 5y + 5z = -4$!

(iii) For the system

$$\begin{cases} -6x_1 + 6x_2 + 2x_3 - 2x_4 = 2 \\ -9x_1 + 8x_2 + 3x_3 - 2x_4 = 3 \\ -3x_1 + 2x_2 + x_3 = 1 \\ -15x_1 + 14x_2 + 5x_3 - 4x_4 = 5 \end{cases}$$

we obtain a REF (dropping two rows of zeros from our notation)

$$\left[\begin{array}{cccc|c} -3 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & -2 & 0 \end{array}\right]$$

and the RREF

$$\left[\begin{array}{cccc|c} 1 & 0 & -1/3 & -2/3 & -1/3 \\ 0 & 1 & 0 & -1 & 0 \end{array}\right],$$

and therefore the solutions

$$\begin{cases} x_1 = \frac{1}{3}(t + 2s - 1) \\ x_2 = s \\ x_3 = t \\ x_4 = s \end{cases},$$

where $t, s \in \mathbb{R}$.

Exercise 1.19 (i) Fill in the steps in the Gaussian Elimination process in the previous example (careful: the RREF of a matrix is unique, but the REF is not – therefore, your REF might differ from the one given above), and work through the practice problems on the tutorial sheet.

(ii) Find α such that

$$\begin{cases} 5x & -3y & = & 2 \\ -x & +2y & = & 1 \\ -4x & +4y & = & \alpha \end{cases}$$

has a solution[a].

(iii) Find all solutions of the homogeneous system

$$\begin{cases} 4x & -2y & +4z & = & 0 \\ & +y & +2z & = & 0 \\ 3x & -y & +4z & = & 0 \end{cases}.$$

(iv) Convince yourself of the following: Homogeneous systems always have a solution, i.e. at least one solution[b].

(v) As hinted on above, an equation $a_1x + a_2y + a_3z = b$ describes a plane in $\mathbb{R}^3$ (just as $a_1x + a_2y = b$ describes a line in $\mathbb{R}^2$). For example, $y = 0$ describes the xz-plane. Think about this interpretation for a moment, and connect it to solving systems of equations, i.e. finding simultaneous solutions of several equations. For example, try to picture three planes that (i) have no point in common, (ii) exactly one point in common, and (iii) a line of points in common.

[a] Hint: Bring the system in REF using Gaussian elimination as usual. Then choose α such that we are not in the no-solution case of theorem 1.17. Check your answer by trying to solve the system with the α you obtained – there is only one α for which it works.

[b] Hint: You can do this in two ways: (1) Find an fixed x that always solves $Ax = 0$, where the zero on the right stands for the zero vector, not the number zero; (2) Argue the that no-solution case in theorem 1.17 never happens if $b = 0$.

1.3 Eigenvalues and Eigenvectors

Definition 1.20 — Eigenvalues and Eigenvectors. Let A be a $n \times n$ square matrix. A scalar $\lambda \in \mathbb{R}$ is called an ***eigenvalue*** of A, if there exists a n-vector $v \neq 0$ with

$$A\vec{v} = \lambda\vec{v}.$$

Such a vector $\vec{v}$ is called an ***eigenvector*** of A.

Remark 1.21

(i) In this section, all matrices are square, of size $n \times n$, and vectors have the corresponding size n.

(ii) The vector on the left-hand side of the eigenvalue equation above is obtained via matrix multiplication. The vector on the right is found by scalar multiplication, which is much easier to compute. This observation already suggests that the theory of eigenvalues and eigenvectors can be useful for simplifying matrix computations.

(iii) The condition $\vec{v} \neq \vec{0}$ for eigenvectors is crucial. Indeed, writing $\vec{0}$ for the zero vector as usual, we have

$$A\vec{0} = \vec{0} = \lambda\vec{0}$$

for any square matrix A and for any scalar λ. So, eigenvectors are always $\neq 0$. However, eigen*values* can be $= 0$: If there exists $\vec{v} \neq \vec{0}$ such that

$$A\vec{v} = \vec{0},$$

then v is an eigenvector of A with eigenvalue $\lambda = 0$ – note that the equation above can be extended

to

$$A\vec{v} = \vec{0} = 0(\vec{v}) = \lambda\vec{v}$$

(where the "$\vec{0}$" denotes the vector of all zeros).

(iv) If $\vec{v}$ is an eigenvector of A, then any non-zero multiple of $\vec{v}$ is an eigenvector with the same eigenvalue: If we scalar-multiply $\vec{v}$ by $\mu \neq 0$, then

$$A(\mu\vec{v}) = \mu A\vec{v} = \mu(A\vec{v}) = \mu(\lambda\vec{v}) = \lambda(\mu\vec{v}).$$

(v) Next we need a way to actually compute eigenvalues – this will be given by theorem 1.23 below. That result and the results in the preceding lemma provide a good opportunity to present some proofs. In those proofs, we will be relying on results from the previous sections that have not been proven – it is not the goal of MTH1002 to rigorously build up a mathematical theory.

Lemma 1.22 For a $n \times n$ matrix M, we have the following:

(i) There exists $\vec{v} \neq \vec{0}$ with $M\vec{v} = \vec{0} \iff \operatorname{rank} M < n$.

(ii) The determinant of M does not change when M is subjected to elementary row operation (iii) in definition 1.12: adding the multiple of a row to another row.

(iii) Each time two rows are swapped, the determinant changes by a factor of -1.

(iv) $\operatorname{rank} M < n \iff \det M = 0$.

Proof omitted.

Theorem 1.23 — Eigenvalue Equation. If λ is an eigen-

value of A, then

$$\det(A - \lambda I) = 0.$$

The converse is also true: If $\det(A - \lambda I) = 0$, then λ is an eigenvalue of A. The above equation – a condition on the parameter λ – is called the *eigenvalue equation* of A.

Proof.

$$\begin{aligned}
\lambda \text{ eigenvalue of } A &\overset{\text{def.}}{\iff} & Av &= \lambda v && \text{(for some } v \neq 0\text{)} \\
&\iff & Av - \lambda v &= 0 && \text{(for some } v \neq 0\text{)} \\
&\iff & (A - \lambda I)v &= 0 && \text{(for some } v \neq 0\text{)} \\
&\overset{\text{Lemma 1.22 (i)}}{\iff} & \operatorname{rank}(A - \lambda I) &< n \\
&\overset{\text{Lemma 1.22 (iv)}}{\iff} & \det(A - \lambda I) &= 0
\end{aligned}$$

■

Example 1.24 (i) Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 1 & 5 \\ 10 & -4 \end{bmatrix}.$$

Sol.:

(ii) Verify the results from the previous example.

Sol.:

$$Av_1 = \begin{bmatrix} 1 & 5 \\ 10 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1-10 \\ 10+8 \end{bmatrix} = \begin{bmatrix} -9 \\ 18 \end{bmatrix} = -9 \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \lambda_1 v_1 \qquad \checkmark$$

$$Av_2 = \begin{bmatrix} 1 & 5 \\ 10 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+5 \\ 10-4 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \lambda_2 v_2 \qquad \checkmark$$

(iii) Find the eigenvalues and eigenvectors of

$$M = \begin{bmatrix} -5 & 0 & 7 \\ 6 & 2 & -6 \\ -4 & 0 & 6 \end{bmatrix}.$$

Sol.:

$$0 = \det(M - \lambda I) = \begin{vmatrix} -5-\lambda & 0 & 7 \\ 6 & 2-\lambda & -6 \\ -4 & 0 & 6-\lambda \end{vmatrix}$$

$$= (-5-\lambda)(2-\lambda)(6-\lambda) + 0 + 0 - (-4)(2-\lambda)(7) - 0 - 0$$

$$= (2-\lambda)\left[(-\lambda-5)(6-\lambda) + 28\right] = (2-\lambda)\left[(\lambda+5)(\lambda-6) + 28\right]$$

$$= (2-\lambda)\left[\lambda^2 - \lambda - 2\right] = -(\lambda+1)(\lambda-2)(\lambda-2),$$

which gives eigenvalues

$$\begin{cases} \lambda_1 &= -1 \\ \lambda_2 &= 2 \\ \lambda_3 &= 2 \end{cases}.$$

For v_1:

$$\left[\begin{array}{ccc|c} -5-\lambda_1 & 0 & 7 & 0 \\ 6 & 2-\lambda_1 & -6 & 0 \\ -4 & 0 & 6-\lambda_1 & 0 \end{array}\right] = \left[\begin{array}{ccc|c} -4 & 0 & 7 & 0 \\ 6 & 3 & -6 & 0 \\ -4 & 0 & 7 & 0 \end{array}\right]$$

$$\overset{R3\to R3-R1}{\to} \left[\begin{array}{ccc|c} -4 & 0 & 7 & 0 \\ 6 & 3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\overset{R2\to R2+{}^3/_2 R1}{\to} \left[\begin{array}{ccc|c} -4 & 0 & 7 & 0 \\ 0 & 3 & {}^9/_2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\overset{R2\to {}^1/_3 R2,\, R1\to -{}^1/_4 R1}{\to} \left[\begin{array}{ccc|c} 1 & 0 & -{}^7/_4 & 0 \\ 0 & 1 & {}^3/_2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

We now need to find the components x, y, z of v_1. One of them can be chosen freely, say $z = 4$. Then the first equation reads $x - \frac{7}{4}4 = 0$, and we obtain $x = 7$. The second equation reads $y + \frac{3}{2}4 = 0$, and we obtain $y = -6$. This gives

$$v_1 = \begin{bmatrix} 7 \\ -6 \\ 4 \end{bmatrix} \qquad (\text{for } \lambda_1 = -1).$$

For a matrix with three distinct eigenvalues, the computations of v_2 and v_3 would be analogous to the computation of v_1. However, M in this example has a double eigenvalue $\lambda_2 = \lambda_3 = 2$. In this case, one has to solve only one system: $(A - 2I)\vec{v} = \vec{0}$. Due to $\lambda = 2$ being a double eigenvalue, it will have a larger set of solutions, from which one then has to choose two *different* solutions v_2 and v_3. Here, "different" means that v_3 can not be a multiple of v_2. This is called "linear independence", and this notion will be defined formally after this example.

For $v_{2/3}$:

$$\left[\begin{array}{ccc|c} -5-\lambda_{2/3} & 0 & 7 & 0 \\ 6 & 2-\lambda_{2/3} & -6 & 0 \\ -4 & 0 & 6-\lambda_{2/3} & 0 \end{array}\right] = \left[\begin{array}{ccc|c} -7 & 0 & 7 & 0 \\ 6 & 0 & -6 & 0 \\ -4 & 0 & 4 & 0 \end{array}\right]$$

$$\overset{\cdots}{\rightarrow} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

We are left with one equation that imposes a condition on three variables. One simple choice – comparing to what was stated in the previous paragraph – would be:

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \qquad \text{(for } \lambda_2 = \lambda_3 = 2\text{)}.$$

(iv) Find the eigenvalues and eigenvectors of

$$L = \begin{bmatrix} 0 & 0.75 \\ 0.75 & 0.4375 \end{bmatrix}.$$

Sol.: The eigenvalue equation is

$$0 = \det(L - \lambda I) = \begin{vmatrix} 0-\lambda & 0.75 \\ 0.75 & 0.4375-\lambda \end{vmatrix} = \lambda^2 - 0.4375\lambda - 0.5625,$$

which gives eigenvalues (none of the decimals appearing here is rounded)

$$\begin{cases} \lambda_1 &= -0.5625 \\ \lambda_2 &= 1 \end{cases}.$$

We find v_1 by solving $(L - \lambda_1 I)v_1 = 0$:

$$\left[\begin{array}{cc|c} 0-\lambda_1 & 0.75 & 0 \\ 0.75 & 0.4375-\lambda_1 & 0 \end{array}\right] = \left[\begin{array}{cc|c} 0.5625 & 0.75 & 0 \\ 0.75 & 1 & 0 \end{array}\right]$$

$$\overset{R2 \to R2 - \frac{0.75}{0.5625} R1}{\rightarrow} \left[\begin{array}{cc|c} 0.5625 & 0.75 & 0 \\ 0 & 0 & 0 \end{array}\right].$$

During those steps it became apparent that $0.5625 = 0.75^2$. Letting x and y stand for the components of v_1, we now have the equation

$$\left(\frac{3}{4}\right)^2 x + \frac{3}{4} y = 0,$$

which gives $x = -{}^4/_3 y$. As in the previous examples, y can be chosen freely and then x is determined accordingly. We choose

$$v_1 = \begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix} \qquad (\text{for } \lambda_1 = -0.5625),$$

as it has the nice property that its norm is 1:

$$|v_1| = \sqrt{0.8^2 + 0.6^2} = \sqrt{0.64 + 0.36} = 1.$$

Similarly, we obtain the second eigenvector

$$v_2 = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} \qquad (\text{for } \lambda_2 = 1).$$

Definition 1.25 — Linear Independence. Consider a set S of m vectors of size n:

$$S = \{v^{(1)}, v^{(2)}, v^{(3)}, \ldots v^{(m)}\}.$$

(i) A n-vector w is said to be a ***linear combination*** of the vectors in S if it can be written in the form

$$w = c_1\vec{v}^{(1)} + c_2\vec{v}^{(2)} + c_3\vec{v}^{(3)} + \cdots + c_m\vec{v}^{(m)}$$

for some coefficients $c_1, c_2, c_3, \ldots, c_m$.

(ii) The vectors in S are said to be ***linearly independent*** if

$$c_1\vec{v}^{(1)} + c_2\vec{v}^{(2)} + c_3\vec{v}^{(3)} + \cdots + c_m\vec{v}^{(m)} = \vec{0}$$

implies

$$c_1 = 0,\ c_2 = 0,\ c_3 = 0,\ \ldots,\ c_m = 0.$$

That is, the vectors of S are called linearly independent if the zero vector can only be written as the trivial linear combination of vectors in S: the linear combination where all coefficients are zero. Otherwise, the vectors are called ***linearly dependent***.

Remark 1.26

(i) Suppose we have a linearly dependent set of vectors,

$$c_1\vec{v}^{(1)} + c_2\vec{v}^{(2)} + c_3\vec{v}^{(3)} + \cdots + c_m\vec{v}^{(m)} = 0,$$

where not all of the coefficients c_j are equal to zero. Say $c_m \neq 0$; then

$$\vec{v}^{(m)} = -\frac{c_1}{c_m}\vec{v}^{(1)} - \frac{c_2}{c_m}\vec{v}^{(2)} - \frac{c_3}{c_m}\vec{v}^{(3)} - \cdots - \frac{c_{m-1}}{c_m}\vec{v}^{(m-1)}.$$

Hence, in a set of linearly dependent vectors, at least one of the vectors is a linear combination of the others.

(ii) Note that for 3×3 matrices, the eigenvalue equation is a third-order equation, which can not be solved as readily as a second-order equation. One therefore should be careful not to give away any information when writing out the determinant: In the computation in example 1.24 (iii), a factor of $(2 - \lambda)$ was kept rather than multiplied out – multiplying it out would have given an expression of

the form

$$-\lambda^3 + a\lambda^2 + b\lambda + c,$$

of which one then would have to guess a root before being able to continue with polynomial division and the quadratic formula! However, sometimes, solving a third-order equation can not be avoided.

Example 1.27 (i) Show that $\vec{v}_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}^\top$ and $\vec{v}_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}^\top$ are linearly independent.

Sol.:

Exercise 1.28 (i) Practise finding eigenvectors and eigenvalues of 2×2 and 3×3 matrices by working through the additional examples from the lectures and the tutorial sheet. Be reminded that besides talking to your tutor or lecturer, you can also use software to verify the results of your computations (cf. first exercise of this chapter; but be aware that displayed eigenvectors may differ).

(ii) Show that the matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

has no *real* eigenvalues. However, there are complex solutions to the eigenvalue equation, what

are they? (Those solutions will be called "complex eigenvalues" in later modules, but in MTH1002, we only consider the real solutions of the eigenvalue equation.)

(iii) Find all eigenvalues of[a]

$$B = \begin{bmatrix} 3 & 1 & 5 & 5 \\ 0 & -4 & 5 & 0 \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

Can you make a general statement for upper triangular square matrices of any size? Hence argue that two matrices with the same set of eigenvalues need not be the same.

(iv) Consider a set S of n vectors of size n,

$$S = \{v^{(1)}, v^{(2)}, \ldots, v^{(n)}\},$$

and show that

$$S \text{ linearly independent} \iff \det V \neq 0$$

where V is the matrix whose columns are the vectors of S, i.e. the matrix in equation (**??**) of example 1.27 (with $m = n$)[b].

(v) Continuing example 1.24 (iv): Let $p = \begin{bmatrix} 140 & 20 \end{bmatrix}^\top$ and find

$$L^{50}p = L(L(\ldots L(Lp))) \qquad (L \text{ applied 50 times}).$$

As a first step, note that

$$p = 100v_1 + 100v_2,$$

where v_1 and v_2 are the eigenvectors that were found above. Now using (iii) and (ii) of properties

1.1 (in (iii), take $A \leftrightarrow L$, $B \leftrightarrow 100v_1$, $\widetilde{B} \leftrightarrow 100v_2$), we compute:

$$\begin{aligned} L(p) &= L(100v_1 + 100v_2) \\ &\overset{\text{(iii)}}{=} L(100v_1) + L(100v_2) \\ &\overset{\text{(ii)}}{=} 100L(v_1) + 100L(v_2) \\ &= 100\left[L(v_1) + L(v_2)\right] \\ &= 100\left[\lambda_1 v_1 + \lambda_2 v_2\right] \\ &= 100\left[-\left(\tfrac{3}{4}\right)^2 v_1 + v_2\right]. \end{aligned}$$

This allows to find $L^2(p) = L(L(p))$,

$$\begin{aligned} L(L(p)) &= L\left(100\left[-\left(\tfrac{3}{4}\right)^2 v_1 + v_2\right]\right) \\ &= 100\left[-\left(\tfrac{3}{4}\right)^2 L(v_1) + L(v_2)\right] = \ldots. \end{aligned}$$

Repeating this, one eventually finds[a] $L^{50}(p)$. Once you have done that, review "Leslie Matrices" of the interactive MTH1000 worksheet "Applications of Vectors and Matrices". Interpreting L as a Leslie matrix and p as the initial population vector – i.e., initially there are 140 hatchlings and 20 adults – what is the state that the population will converge to?

[a] Hint: One needs the zeros of $\det(B - \lambda I)$ as usual; cf. properties 1.1 for how to compute that determinant.
[b] Hint: Lemma 1.22.
[a] Hint: Next, compute $L^3(p)$ in the same way, compare to $L^2(p)$ and try to find a pattern that allows you to write down $L^k(p)$ for any positive integer k.

1.4 Inverse Matrices

Definition 1.29 If for a square matrix A there exists a matrix B of the same size with

$$AB = BA = I,$$

then we say that A is ***invertible*** and B is its ***inverse***, written $B = A^{-1}$.

Remark 1.30

Not all square matrices are invertible!

Theorem 1.31

$$A \text{ invertible} \iff \det A \neq 0$$

Example 1.32 (i) Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Sol.:

(ii) For

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

we have $\det B = 1 \cdot 4 - 3 \cdot 2 = -2 \neq 0$, and therefore it is invertible. Its inverse is

$$B^{-1} = \begin{bmatrix} -2 & 1 \\ {}^{3}/_{2} & {}^{-1}/_{2} \end{bmatrix}$$

– one can find this inverse either with the naive approach in (i) or with the the more systematic approaches below. Either way, you can verify that the stated B^{-1} is indeed the inverse of B by multiplying the two.

(iii) A 1×1 matrix is just a number, $C = \begin{bmatrix} c_{11} \end{bmatrix}$. The determinant of that trivial matrix is its entry, $\det C = c_{11}$. The above theorem states that this number has an inverse if and only if it is different from zero. You knew that: for $x = 0$ there is no $y \in \mathbb{R}$ with $xy = 1$, but for all other x, there is.

Remark 1.33

We now derive a systematic way to find inverses. If you are not interested in that derivation, you can skip to the the last step, (vi). Let A be an invertible $n \times n$ matrix. That is, we know that A^{-1} exists, and we now need a constructive way to find it. To do that, denote the vectors $\begin{bmatrix} 0 & 0 & \dots & 0 & 1 & 0 & \dots & 0 \end{bmatrix}^\top$ of size n, where the entry 1 is in the j-th place, by e_j.

(i) For any $n \times n$ matrix B, the j-th column of B is Be_j.

(ii) If A maps v to w, then A^{-1} maps w to v:

$$Av = w \quad \Longrightarrow \quad A^{-1}w = A^{-1}Av = Iv = v.$$

(iii) Solving the system / augmented matrix $[A \,|\, w]$ gives the vector v that is mapped to w by A: $Av = w$.

(iv) Now combine those three points to find the columns of A^{-1}:

we want the j-th column of A^{-1}, call it $c_j \xrightarrow{\text{(i)}} c_j = A^{-1}e_j$
$\xrightarrow{\text{(ii)}} Ac_j = e_j$
$\xrightarrow{\text{(iii)}} c_j$ is the solution of $[A|e_j]$

(v) We have to solve $[A \,|\, e_j]$ for all $j \in \{1, 2, 3, \dots, n\}$, and this can actually be done with one Gaussian elimination by augmenting all n vectors at once. Now note the following:

- Putting all the e_j together – in the correct order – gives the identity matrix.
- If the left-hand side of an augmented matrix is I, then the solution can be read off directly: b is the solution of $[I \mid b]$.
- The RREF of A is the identity matrix, since A has a non-zero determinant and therefore full rank (cf. lemma 1.22). That is, A can be transformed into I.

Hence we have found the following inversion algorithm:

(vi) To find the inverse of the invertible matrix A, augment the identity matrix I and bring A in reduced row echelon form using elementary row operations. The transformation of I on the right is then the inverse of A:

$$[A \mid I] \overset{\text{elementary row operations}}{\rightsquigarrow} [I \mid A^{-1}].$$

Example 1.34 Check whether

$$A = \begin{bmatrix} -5 & 0 & 7 \\ 6 & 2 & -6 \\ -4 & 0 & 6 \end{bmatrix}$$

is invertible and find its inverse in case it is.

Sol.:

Remark 1.35

As a first simple example to do on your own, you can try to find B^{-1} from remark 1.32 (ii) with that method. However, for 2×2 matrices there is a direct formula, which you have already seen in MTH1000:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad \Longrightarrow \qquad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

You can verify this formula directly by multiplication. Of course, one can apply it only when the given matrix A is invertible – what happens if you try use it for a matrix that is not invertible (e.g. for the matrix in example 1.32 (i))?

Exercise 1.36 (i) Work through the problems on inverses and invertibility on the tutorial sheet. Note that you actually have an unlimited amount of practice problems: Make up a random 2×2 or 3×3 matrix A (or even 4×4 if you like), find its inverse A^{-1}, and check whether $AA^{-1} = I$ (either by hand or using software, cf. exercise (i) of the first section).

(ii) Find the inverse of

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & 7 \end{bmatrix}.$$

(iii) Find the inverse of

$$C = \begin{bmatrix} -5/18 & 1/18 & 7/18 \\ 1/18 & 7/18 & -5/18 \\ 7/18 & -5/18 & 1/18 \end{bmatrix}.$$

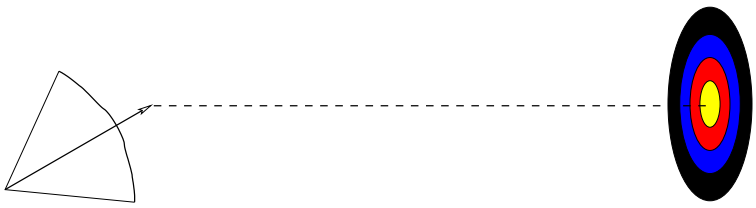
(iv) Convince yourself that theorem 1.31 is true[a].

[a] Hint: Lemma 1.22 should be helpful here. The following is an alternative approach. When $\det A = 0$, we have a certain eigenvalue, $\lambda = \ldots$. Which one? Now try to apply the $v \leftrightarrow$ w argument from remark 1.33 to the corresponding eigenvector. Does that work?

2 Functions of Several Variables

As with matrices, functions are often the link between real-life problems and their mathematical analysis. However, real-life quantities – for example, the expected profit of a business strategy – rarely depend on one parameter (variable) only, and it is therefore important to also study *functions of several variables* as an extension of the one-variable functions you know from school and from last term. The basic ideas are the same as for single-variable calculus, but you will also encounter new concepts, new difficulties, and new potential in this chapter.

Example 2.1 Alice is practising archery, and she is facing a target that is 30 metres away. She can control the velocity of the arrow and the angle at which she shoots. Note that there are many different ways to hit the target: one could aim a very strong shot directly at the bullseye, or one could take it easier and aim a little higher to make up for the reduced velocity of the arrow. Alice's maths skills are excellent – for any given angle α, she can exactly compute the velocity v she needs for the arrow to hit the bullseye. Of all possible choices, Alice would like to choose the most stable one, that is, the configuration (α^*, v^*) that hits the bullseye and is the least sensitive to trembling, misjudging of the angle or the velocity, etc.



For our modelling, we assume that the tip of the arrow, when it is launched, is exactly level with the bullseye, cf. the sketch above. Then the height of the arrow when it hits the target, as a function of the angle α and the velocity v, is

$$h(\alpha, v) = -\frac{4414.5}{\cos^2(\alpha)\, v^2} + 30 \tan \alpha,$$

where $h = 0$ corresponds to hitting the bullseye. There are dif-

ferent possibilities for defining a measurement of "instability", and Alice decides to use

$$f(\alpha, v) = h_\alpha^2 + 3h_v^2,$$

since she finds controlling the velocity more difficult than controlling the angle. The symbols h_α and h_v are the partial derivatives of h, which will be defined in the next section. After familiarising yourself with that concept, convince yourself that f as defined above can be considered a measure of instability of* h! In this chapter, we will learn how to minimise functions like f (instability) under constraints like $h = 0$ (hitting the bullseye). Doing that will give

$$\alpha^* = 44.70^\circ, \qquad v^* = 17.16\tfrac{m}{s}.$$

The fact that those numbers do not agree with how archers usually shoot could be due to our assumptions and our modelling: we neglected wind, we assumed that initially the tip of the arrow and the bullseye are exactly level, we assumed that the archer can do the maths and control the angle and velocity exactly, etc. – and in the absence of these assumptions, a very strong shot almost directly aimed at the bullseye would always be the most straightforward option. However, it is not hard to think of situations in which the above approach is more appropriate than it is for archery (e.g. building mechanical machines; in such applications, one would often have much larger numbers of variables).

2.1 Multivariate Functions and Partial Derivatives

Definition 2.2 — Functions of Several Variables. A ***function of two variables*** is a rule f that assigns to each pair (x, y) in a set $D \subseteq \mathbb{R}^2$ a unique real number $f(x, y)$. The set D is called the *domain* of f (also denoted $D(f)$)

*Hint: What does it mean for a derivative to be small or large at a point?

and its *range* is the set of values it maps to,

$$R(f) = \{z \in \mathbb{R} \mid z = f(x,y) \text{ for some } (x,y) \in D\}.$$

Similarly, a ***function of n variables*** is a rule f that assigns a unique real number $f(x_1, x_2, \ldots, x_n)$ to each $(x_1, x_2, \ldots, x_n) \in D \subseteq \mathbb{R}^n$.

Remark 2.3

(i) Writing $z = f(x,y)$ in the two-variable case, we call x and y the independent variables, and z the dependent variable. Functions of several variables are also called *multivariate functions*.

(ii) Unless specified otherwise, we always choose the domain to be the biggest set of points (x,y) at which the expression that defines f can be evaluated.

(iii) Functions of two variables can be visualised as surfaces/graphs over the xy-plane: For an (x_0, y_0) in the domain, we mark a point of height $f(x_0, y_0)$ over or under (x_0, y_0). That is, a point of the graph with (x,y)-coordinates (x_0, y_0) will have z-coordinate (height above or depth below the xy-plane) $z_0 = f(x_0, y_0)$.

(iv) A very important tool for working with multivariate functions is restriction to lines in the domain, cf. example (ii) below.

Example 2.4 (i) Draw the graph of $f(x,y) = x^2 + y^2$.

Sol.:

(ii) Find the domain and range of $g(x,y) = 1 + \sqrt{y - x^2}$.

Sol.:

(iii) Find the domain and range of $h(x_1, x_2, x_3) = \sin(x_1 x_2 + x_3)$.

Sol.:

Definition 2.5 — Partial Derivatives. For a function f of two variables, the ***partial derivatives*** are the functions

$$\frac{\partial f}{\partial x}(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h},$$
$$\frac{\partial f}{\partial y}(x,y) = \lim_{h\to 0} \frac{f(x,y+h) - f(x,y)}{h}.$$

We also write f_x for the partial derivative $\dfrac{\partial f}{\partial x}$ – that is, the partial derivative of f with respect to x – etc. Similarly, for functions of more than two variables:

$$\frac{\partial f}{\partial x_j}(x_1, x_2, \ldots, x_n) = \lim_{h\to 0} \frac{f(x_1, x_2, \ldots, x_{j-1}, x_j + h, x_{j+1}, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{h}.$$

Example 2.6 Find f_x for the function $f(x,y) = 5x^2y^7$.

Sol.:

Example 2.7 (i) For $f = x^3 + y^5 - 2$, find f_x and f_y.

Sol.:

$$f_x = \frac{\partial}{\partial x}\left(x^3 + y^5 - 2\right) = 3x^2 + 0 + 0 = 3x^2,$$
$$f_y = \frac{\partial}{\partial y}\left(x^3 + y^5 - 2\right) = 0 + 5y^4 + 0 = 5y^4.$$

(ii) For $f = x^3 + x^2y^3 - 2y^2$, find f_x and f_y.

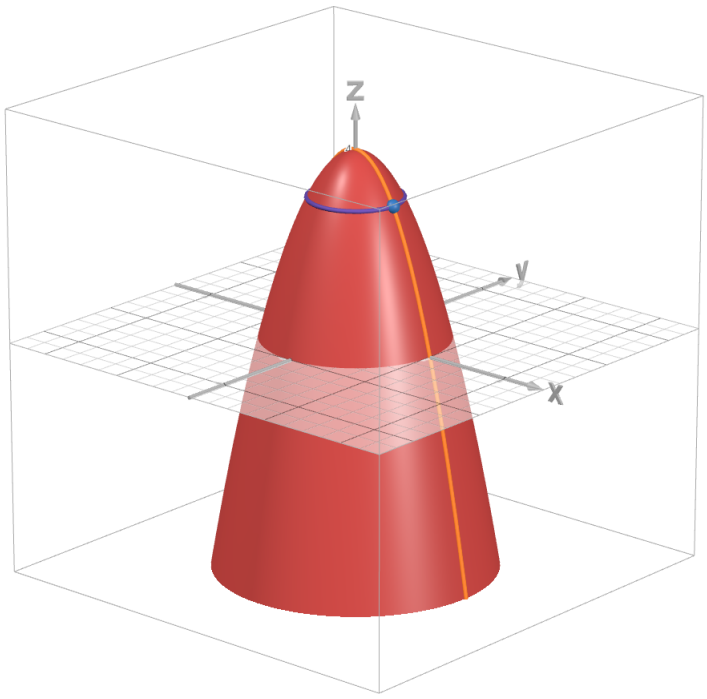
Sol.:

$$f_x = \frac{\partial}{\partial x}\left(x^3\right) + \frac{\partial}{\partial x}\left(x^2\right)y^3 - 0 = 3x^2 + 2xy^3,$$
$$f_y = 0 + x^2\frac{\partial}{\partial y}\left(y^3\right) - 2\frac{\partial}{\partial y}\left(y^2\right) = 3x^2y^2 - 4y.$$

Remark 2.8

(i) For a geometric interpretation, we consider the graph of a function f of two variables, i.e. $z = f(x,y)$. Then $\partial z/\partial x$ is the slope on the surface "when moving in the x direction". That is, x is varied, y is held constant, and we take note of the change of function values. Similarly, $\partial z/\partial y$ is the slope in the y direction.

As an intuitive example, consider a hiker climbing a mountain of the shape $z = 4 - (x^2 + y^2)$.



Suppose she checks her GPS and finds that her coordinates are $(x,y) = (1,0)$. The slopes at this point in both directions are

$$z_x(1,0) = \frac{\partial}{\partial x}\left(4-(x^2+y^2)\right)_{|(x,y)=(1,0)} = (-2x)_{|(x,y)=(1,0)} = -2,$$
$$z_y(1,0) = \frac{\partial}{\partial y}\left(4-(x^2+y^2)\right)_{|(x,y)=(1,0)} = (-2y)_{|(x,y)=(1,0)} = 0.$$

That is, there is no slope in the y direction. The slope in the x direction is negative, because the height decreases as x increases (from where she stands, increasing x would be moving away from the peak).

(ii) In order to be able to work with a larger set of functions – not just sums of powers of variables – we extend the single-variable differentiation rules from Term 1 to the multivariate setting: For example, the power rule becomes

$$(fg)_x = f_x g + f g_x$$

for an x-derivative and similarly for the y-derivative

and the quotient rule.

(iii) In order to find partial derivatives of functions like

$$f(x,y) = \sin(x+y^2),$$

we also need to combine the chain rule for single-variable functions – note that the outer function, $h(t) = \sin(t)$, is a single-variable function – with partial derivatives. Let c be a constant and review the following applications of the single-variable chain rule from Term 1.

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(\sin(t+c)\right) &= \cos(t+c)\frac{\mathrm{d}}{\mathrm{d}t}(t+c) = \cos(t+c),\\ \frac{\mathrm{d}}{\mathrm{d}t}\left(\sin(c+t^2)\right) &= \cos(c+t^2)\frac{\mathrm{d}}{\mathrm{d}t}(c+t^2) = 2t\cos(c+t^2).\end{aligned} \tag{2.1}$$

Now, to the operators $\partial/\partial x$ and $\partial/\partial y$, the terms y^2 and x, respectively, look like constants, and we can therefore apply the single-variable computations (2.1) to the two-variable function f:

$$\begin{aligned}f_x &= \frac{\partial}{\partial x}\left(\sin(x+y^2)\right) = \cos(x+y^2)\frac{\partial}{\partial x}(x+y^2) = \cos(x+y^2),\\ f_y &= \frac{\partial}{\partial y}\left(\sin(x+y^2)\right) = \cos(x+y^2)\frac{\partial}{\partial y}(x+y^2) = 2y\cos(x+y^2).\end{aligned}$$

This can be stated as

$$\frac{\partial}{\partial x}h(g(x,y)) = h'(g(x,y))\cdot g_x(x,y),$$

where h is a single-variable function. Similarly for $\partial/\partial y$. Note that the $'$ is not ambiguous – h has only one variable, and h' is the derivative with respect to it.

Example 2.9 (i) Find the y-derivative of $f(x,y) = \mathrm{e}^{x^2+y^3}$.

Sol.:

Definition 2.10 — Higher-Order Partial Derivatives. The ***second-order partial derivatives*** with respect to x and y of a function f of two variables are written

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right),$$
$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right),$$

and can also be denoted f_{xx} and f_{yy}.

We further have ***mixed*** second-order derivatives:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right),$$
$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

Similarly for functions of more than two variables and for higher-order derivatives; e.g.,

$$f_{xyyz} = \frac{\partial^4 f}{\partial x \partial y \partial y \partial z} = \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial z}\right)\right)\right)$$

is a fourth-order derivative of a three-variable function $f = f(x, y, z)$.

Example 2.11 (i) For

$$f(x, y) = 2x^9 y^5,$$

find all partial derivatives of orders up to 2.

Sol.:

(ii) For

$$f(x,y) = x^2 + 3xy - y^2 + 7x - 4y + 2,$$

find *all* partial derivatives.

Sol.:

$$\begin{aligned} f &= x^2 + 3xy - y^2 + 7x - 4y + 2, \\ f_x &= 2x + 3y + 7, \\ f_y &= 3x - 2y - 4, \\ f_{xx} &= 2, \\ f_{yy} &= -2, \\ f_{xy} &= 3, \\ f_{yx} &= 3. \end{aligned}$$

Since all the second-order derivatives are constant, all derivatives of higher orders are zero,

$$\frac{\partial^k f}{\partial \dots} = 0 \qquad \text{for } k \geq 2.$$

(iii) For

$$f(x,y) = x\,\cos(xy^2),$$

find all partial derivatives of orders up to 2.

Sol.:

$$\begin{aligned} f &= x\cos(xy^2), \\ f_x &= \cos(xy^2) - xy^2\sin(xy^2), \\ f_y &= -2x^2y\sin(xy^2), \\ f_{xx} &= -2y^2\sin(xy^2) - xy^4\cos(xy^2), \\ f_{yy} &= -2x^2\sin(xy^2) - 4x^3y^2\cos(xy^2), \\ f_{xy} &= -4xy\sin(xy^2) - 2x^2y^3\cos(xy^2) = f_{yx}. \end{aligned}$$

Remark 2.12

In all of the previous examples, we had $f_{xy} = f_{yx}$. The following theorem states that this is not a coincidence – it is always true as long as certain technical conditions

are satisfied. You will learn about the conditions of continuity of multivariate functions in MTH2001 (Analysis) next year. For now and for the purpose of MTH1002, you can always work with the conclusion of the following theorem. However, be aware that there are functions that do not meet its requirements – you will encounter examples for that next year – and for which $f_{xy} \neq f_{yx}$.

Theorem 2.13 — Schwarz's Theorem or Clairaut's Theorem. Suppose that $(a,b) \in D$, where D is the domain of $f(x,y)$. If the functions f_{xy} and f_{yx} are both continuous at (a,b), then

$$f_{xy}(a,b) = f_{yx}(a,b).$$

Exercise 2.14 (i) Practise working with multivariate functions – in particular, finding partial derivatives – by working through additional examples from the lectures and by doing the practice problems on the tutorial sheet. Besides tutorials, drop-in surgeries, etc., you can use software to support your studies: For example, you can enter commands like `plot x*cos(x*y^2)` or `partial derivatives of ...` in WolframAlpha. However, be careful not to rely on software, and the latter command you should only use to verify your on-paper computations, not to find partial derivatives.

(ii) Find the domain and the range of[a]

$$f(x,y) = \sqrt{2 \sin x \sin y}.$$

(iii) Let $D \subseteq \mathbb{R}^2$ be the open disk of radius 1, $D =$

$\{x^2 + y^2 < 1\}$. Define a function that is defined for $(x, y) \in D$, and another function that is defined on the *complement* of D,

$$D^c = \mathbb{R}^2 \setminus D = \{x^2 + y^2 \geq 1\}$$

(read this as "all points outside of D")[a].

(iv) Find *all* partial derivatives of[b] $f(x, y) = e^{2x+3y}$.

(v) Consider the function $f(x, y) = x^2(1 - x^2) - y^2$. Find all points for which both the x- and the y-derivative is equal to zero[c], in maths terms: Find (x_0, y_0) with $f_x(x_0, y_0) = f_y(x_0, y_0) = 0$.

Now enter `plot x^2*(1-x^2)-y^2` into WolframAlpha and compare. What roles do the points (x_0, y_0) you have found play in the 3D plot you see? Also, can you make sense of the "contour plot" that is displayed[d]?

[a] Hint: You can check your domain with a WolframAlpha command of the form `plot ... >= 0`.

[a] Hint: E.g., the single-variable functions $\ln(t)$ and $\sqrt{t}$ are suitable as outer functions.

[b] Hint: Use theorem 2.13 to justify that your answer can be given in the form

$$\frac{\partial^n}{\partial y^n}\frac{\partial^m f}{\partial x^m} = \dots,$$

i.e. that one can assume that all x-derivatives are taken first.

[c] Hint: Setting both partial derivatives equal to zero gives you a (non-linear) system of two equations; solving it should lead to three points.

[d] Hint: The 3D surface plot and the contour plot use the same colouring.

2.2 Chain Rule and Implicit Differentiation

Remark 2.15

Curves in the xy-plane can be parametrized by writing x and y as single-variable functions of a third variable, usually t (think of it as "time"):

$$(x, y) = (x(t), y(t)).$$

We will denote such curves γ, i.e. $\gamma(t) = (x(t), y(t)) \in$

$\mathbb{R}^2$. One could then restrict a given function $f = f(x,y)$ to this curve – formally this corresponds to composing f with γ: we define a single-variable function F as

$$F(t) = (f \circ \gamma)(t) = f(\gamma(t)) = f(x(t), y(t)).$$

Finding the derivative of F is now an important and interesting task.

Often, the distinction $f \leftrightarrow F$ is not made, and the single-variable function of t would be called f as well. This simplifies the notation in some of the following formulas, but it also carries potential for confusion, since the symbol "f" would then be used for two different functions.

Example 2.16 (i) Draw the curves

$$\begin{aligned} \gamma_1 &: \quad (x(t), y(t)) = (t, t), \\ \gamma_2 &: \quad (x(t), y(t)) = (3\cos t, 3\sin t), \\ \gamma_3 &: \quad (x(t), y(t)) = (t, 1 - t^2), \end{aligned}$$

and highlight the point corresponding to $t = 0$ in each of them.

Sol.:

(ii) Consider the functions

$$\begin{aligned} f_1 &: \quad f_1(x,y) = x - y, \\ f_2 &: \quad f_2(x,y) = x^2 + y^2, \\ f_3 &: \quad f_3(x,y) = \ln(3 + x^2 - y), \end{aligned}$$

and find the three single-variable functions $F_i = f_i \circ \gamma_i$.

Sol.:

$$F_1(t) = (f_1 \circ \gamma_1)(t) = f_1(t,t) = t - t = 0,$$
$$F_2(t) = (f_2 \circ \gamma_2)(t) = f_2(3\cos t, 3\sin t) = (3\cos t)^2 + (3\sin t)^2 = 9,$$
$$F_3(t) = \ln\left(3 + t^2 - (1 - t^2)\right) = \ln\left(2 + 2t^2\right) = \ln 2 + \ln\left(1 + t^2\right). \tag{2.2}$$

Theorem 2.17 — The Chain Rule I. Let f and F be as in remark 2.15. Then

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}.$$

Example 2.18 (i) The derivatives of the functions F_i in 2.16 are

$$\frac{\mathrm{d}F_1}{\mathrm{d}t} = \frac{\partial f_1}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f_1}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\partial(x-y)}{\partial x}\frac{\mathrm{d}t}{\mathrm{d}t} + \frac{\partial(x-y)}{\partial y}\frac{\mathrm{d}t}{\mathrm{d}t} = 1\cdot 1 + (-1)\cdot 1 = 0,$$
$$\frac{\mathrm{d}F_2}{\mathrm{d}t} = 2x(t)\cdot(-3\sin t) + 2y(t)\cdot 3\cos t = -6\cos t\sin t + 6\sin t\cos t = 0,$$
$$\frac{\mathrm{d}F_3}{\mathrm{d}t} = \frac{2x}{3+x^2-y}\cdot 1 + \frac{-1}{3+x^2-y}\cdot(-2t) = \frac{4t}{2+2t^2} = \frac{2t}{1+t^2}.$$

Alternatively, one could have found these derivatives by directly differentiating the functions F_i in (2.2). However, note that the chain rule allowed us to find them without using the explicit expressions of t on the right-hand sides of (2.2).

(ii) We now present an example in the simplified notation mentioned at the end of remark 2.15, since it is good to get used to this way as well: For $f = x^2y + 3xy^4$, $x = \sin 2t$, $y = \cos t$, find ${}^{\mathrm{d}f}/_{\mathrm{d}t}$ at $t = 0$.

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} = (2xy + 3y^4)(2\cos 2t) + (x^2 + 12xy^3)(-\sin t).$$

The expressions containing the variable t can be evaluated at $t = 0$ directly. The other expressions we evaluate at the x and y values at $t = 0$: $x(0) = 0$, $y(0) = 1$. Therefore, we obtain that the t-derivative of f at $t = 0$

is

$$\frac{\mathrm{d}f}{\mathrm{d}t}(0) = 3 \cdot 2 + 0 \cdot 0 = 6.$$

To give this derivative a geometric interpretation, think of a hiker walking along the path $(x(t), y(t))$ in the "mountain range" that is given by the graph of f. Then our computations show that at time $t = 0$, he will be climbing at a rate of six height units per time unit.

Remark 2.19

Changes of variables in the xy-plane can be realised by writing x and y as functions of *two* other variables,

$$(x, y) = (x(s,t), y(s,t)).$$

Given a function $f = f(x,y)$, one could then define a new function F that has the same values as f, but is written out in term of s and t:

$$F(s,t) = f(x(s,t), y(s,t)).$$

These changes of variables are quite useful – for example, we will use them to solve differential equations in Chapter 4 – and it is important to be able to understand the relation of the partial derivatives of f and F.

As in remark 2.15, the distinction $f \leftrightarrow F$ is sometimes not made, and the function of the new variables would be called f as well. Again, this simplifies the notation but also carries potential for confusion.

Theorem 2.20 — The Chain Rule II. For s, t, x, y, f, F as in remark 2.19, we have

$$\frac{\partial F}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s},$$
$$\frac{\partial F}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t}.$$

Example 2.21 (i) Consider the function $f(x,y) = x - y$ and the change of variables

$$x(s,t) = s + t, \qquad y(s,t) = s - t.$$

For f in the new variables, i.e. $F(s,t) = f(x(s,t), y(s,t))$, find F_s and F_t.

Sol.:

(ii) Consider the paraboloid $f(x,y) = x^2 + y^2$ and the change of variables

$$x(r,\theta) = r\cos\theta, \qquad y(r,\theta) = r\sin\theta. \tag{2.3}$$

For $F(r,\theta) = f(x,y)$, find F_r and F_θ.

Sol.:

(iii) If $g(s,t) = f(s^2 - t^2, t^2 - s^2)$, where f is any differentiable function, show that g satisfies

$$t\frac{\partial g}{\partial s} + s\frac{\partial g}{\partial t} = 0.$$

Sol.: Let $x = s^2 - t^2, y = t^2 - s^2$. Then,

$$\frac{\partial g}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s} = f_x \cdot 2s + f_y \cdot (-2s),$$

$$\frac{\partial g}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t} = f_x \cdot (-2t) + f_y \cdot 2t.$$

Therefore

$$t\frac{\partial g}{\partial s} + s\frac{\partial g}{\partial t} = 0.$$

(iv) As in the examples 2.18 for chain rule I, we now present an example in simplified notation, where the function in the new variables is not given a new name. Consider the expression

$$3\frac{\partial^2 u}{\partial x^2} + \frac{1}{2}\frac{\partial^2 u}{\partial x \partial y} - \frac{1}{2}\frac{\partial^2 u}{\partial y^2}, \tag{2.4}$$

for the function $u = u(x, y)$, and rewrite it in terms of the new variables ξ ("xi") and η ("eta") defined as follows:

$$\xi = x + 3y \quad \text{and} \quad \eta = x - 2y.$$

Sol.: For this, we use chain rule II to rewrite the derivatives of u with respect to x and y as derivatives w.r.t. ξ and η:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi}\cdot 1 + \frac{\partial u}{\partial \eta}\cdot 1 = u_\xi + u_\eta,$$
$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial y} = \frac{\partial u}{\partial \xi}\cdot 3 + \frac{\partial u}{\partial \eta}\cdot(-2) = 3u_\xi - 2u_\eta.$$

These formulas for the first-order derivatives of u are only intermediate steps in our computation – we need the second-order derivatives appearing in (2.4):

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(u_\xi + u_\eta\right) = \frac{\partial u_\xi}{\partial x} + \frac{\partial u_\eta}{\partial x} \\ &= \frac{\partial u_\xi}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u_\xi}{\partial \eta}\frac{\partial \eta}{\partial x} + \frac{\partial u_\eta}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u_\eta}{\partial \eta}\frac{\partial \eta}{\partial x} \\ &= \frac{\partial u_\xi}{\partial \xi}\cdot 1 + \frac{\partial u_\xi}{\partial \eta}\cdot 1 + \frac{\partial u_\eta}{\partial \xi}\cdot 1 + \frac{\partial u_\eta}{\partial \eta}\cdot 1 = u_{\xi\xi} + u_{\eta\xi} + u_{\xi\eta} + u_{\eta\eta}.\end{aligned}$$

Recalling theorem 2.13, we obtain

$$\frac{\partial^2 u}{\partial x^2} = u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}.$$

The second-order derivative w.r.t. y is

$$\begin{aligned}\frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y}\left(3u_\xi - 2u_\eta\right) = 3\frac{\partial u_\xi}{\partial y} - 2\frac{\partial u_\eta}{\partial y}\\ &= 3\left(\frac{\partial u_\xi}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial u_\xi}{\partial \eta}\frac{\partial \eta}{\partial y}\right) - 2\left(\frac{\partial u_\eta}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial u_\eta}{\partial \eta}\frac{\partial \eta}{\partial y}\right)\\ &= 3\frac{\partial u_\xi}{\partial \xi}\cdot 3 + 3\frac{\partial u_\xi}{\partial \eta}\cdot(-2) - 2\frac{\partial u_\eta}{\partial \xi}\cdot 3 - 2\frac{\partial u_\eta}{\partial \eta}\cdot(-2)\\ &= 9u_{\xi\xi} - 12u_{\xi\eta} + 4u_{\eta\eta}.\end{aligned}$$

Convince yourself that the mixed second-order derivative is

$$u_{xy} = 3u_{\xi\xi} + u_{\xi\eta} - 2u_{\eta\eta}.$$

This allows to write out (2.4):

$$\begin{aligned}&3\frac{\partial^2 u}{\partial x^2} + \frac{1}{2}\frac{\partial^2 u}{\partial x \partial y} - \frac{1}{2}\frac{\partial^2 u}{\partial y^2}\\ &\quad= 3\left[u_{\xi\xi} + 2u_{\xi\eta} + u_{\eta\eta}\right] + \frac{1}{2}\left[3u_{\xi\xi} + u_{\xi\eta} - 2u_{\eta\eta}\right] - \frac{1}{2}\left[9u_{\xi\xi} - 12u_{\xi\eta} + 4u_{\eta\eta}\right]\\ &\quad= 0\cdot u_{\xi\xi} + \left(6 + \frac{1}{2} + 6\right)\cdot u_{\xi\eta} + 0\cdot u_{\eta\eta} = \frac{25}{2}u_{\xi\eta}.\end{aligned}$$

Later, in Sec. 4.8, we will use computations of this kind to transform combinations of second-order partial derivatives (as in (2.4)) into single second-order derivatives with respect to new variables (here: ${}^{25}/_{2}\cdot u_{\xi\eta}$).

Definition 2.22 — Level Sets. Given a function f of n variables and a constant c in the range of f, the set of points in the domain of f with

$$f = c$$

is called a ***level set*** of f.

Remark 2.23

(i) Level sets usually have one dimension less than the domain of the function. For example, the function

$f(x,y) = x^2 + y^2$ is defined on the xy-plane, $\mathbb{R}^2$, which is two-dimensional. We have seen in example 2.4 that the condition

$$f(x,y) = x^2 + y^2 = 1$$

describes a curve, i.e. a one-dimensional object, namely the circle of radius 1. In fact, $f(x,y) = x^2 + y^2 = c$ describes a one-dimensional object for any $c > 0$. However, the only solution to $f = 0$ is the point $(x,y) = (0,0)$, and points are considered zero-dimensional. A constant $c < 0$ would not be in the range of f and therefore $f = c$ has no solutions at all.

Similarly, consider the function $f(x,y,z) = x^2 + y^2 + z^2$. It is defined on the three-dimensional space $\mathbb{R}^3$, and its level sets $f = c$ for $c > 0$ are two-dimensional objects: spheres of radius $\sqrt{c}$ (the surfaces of the balls of radius $\sqrt{c}$ centred at the origin of the coordinate system).

We had already discussed a level set at the very beginning of this chapter, namely the set S in example 2.4 (v) (also compare to the figure in (i)).

We now study some basic theory surrounding level sets and then use it for differentiation. You will encounter this topic again later in your studies – keyword, e.g., "Implicit Function Theorem".

(ii) In the xy-plane, where x and y are independent variables, we have

$$\frac{\partial y}{\partial x} = 0$$

(the x-derivative of the function $g(x,y) = y$ is zero, since g does not depend on x).

Now consider a level set $f(x,y) = c$. Under this constraint, the variables cannot both move freely any more. Such a level set will be one-dimensional, i.e. a curve, and we can use x as a parameter for it. That is, we can let x vary freely (independent variable) and then y will be determined (not independent any more) by the choice of x.

E.g., requiring $f_1 = 0$ in example 2.16 gives $y(x) = x$, and $f_3 = \ln(2)$ gives $y(x) = 1 + x^2$.

Theorem 2.24 — Implicit Differentiation. If the independent variable y is turned into a dependent variable $y(x)$ via a condition $f(x,y) = c$ (as in (ii) of the previous remark), then

$$\frac{dy}{dx} = -\frac{f_x}{f_y}.$$

Similarly in $\mathbb{R}^3$: If the independent variable z is turned into a dependent variable $z(x,y)$ via a condition $f(x,y,z) = c$, then

$$\frac{\partial z}{\partial x} = -\frac{f_x}{f_z},$$
$$\frac{\partial z}{\partial y} = -\frac{f_y}{f_z}.$$

Proof. We differentiate the condition $f(x,y(x)) = c$ with respect to x. On the right-hand side, we have

$$\frac{dc}{dx} = 0.$$

For the other side, we use chain rule I, theorem 2.17:

$$\frac{df(x,y(x))}{dx} = \frac{\partial f(x,y)}{\partial x}\frac{dx}{dx} + \frac{\partial f(x,y)}{\partial y}\frac{dy}{dx}.$$

Note that here, f appears as a function of one independent variable on the left, and as a two-variable function on the right. Since ${}^{dx}/_{dx} = 1$, combining the two leads to the

claimed formula for $^{dy}/_{dx}$.

Similarly, we differentiate the condition $f(x,y,z(x,y)) = c$ with respect to x and y,

$$\frac{\partial f(x,y,z(x,y))}{\partial x} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial x} = f_x \cdot 1 + f_y \cdot 0 + f_z \cdot \frac{\partial z}{\partial x} = 0,$$
$$\frac{\partial f(x,y,z(x,y))}{\partial y} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial y} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial y} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial y} = f_x \cdot 0 + f_y \cdot 1 + f_z \cdot \frac{\partial z}{\partial y} = 0,$$

to derive the other formulas stated in the theorem. Here, f appears as a two-variable function on the left, and as a three-variable function in the other expressions. It was mentioned in the previous remark that $^{\partial y}/_{\partial x} = 0$. ■

Example 2.25 (i) Find $^{dy}/_{dx}$ if $x^3 + y^3 = 6xy$.

Sol.:

(ii) Find $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$, $\frac{\partial^2 z}{\partial x \partial y}$, where $z = z(x,y)$ is defined by the equation

$$x + y - z = e^z.$$

Sol.:

(iii) For $z = u^v$, where $u = \sin x$ and $v = \cos x$, find $^{\mathrm{d}z}/_{\mathrm{d}x}$.

Sol.: The chain rule for $z = z(x)$ reads

$$\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x}.$$

The partials of $z = z(u, v)$ are

$$\frac{\partial z}{\partial u} = vu^{v-1}, \quad \frac{\partial z}{\partial v} = u^v \ln u,$$

and therefore we find

$$\frac{\mathrm{d}z}{\mathrm{d}x} = vu^{v-1}\cos x - u^v \ln u \cdot \sin x = (\sin x)^{\cos x - 1}\cos^2 x - \ln(\sin x)\cdot(\sin x)^{\cos x + 1}.$$

(iv) (Logarithmic Differentiation) Find $^{\partial z}/_{\partial x}$, $^{\partial z}/_{\partial y}$ for

$$z = (x^2y + xy^2)^{\cos(xy)}. \tag{2.5}$$

Sol.: The idea for dealing with the function in the power on the right is to use the rule $\ln(a^p) = p\ln(a)$ for logarithms. If $a = b$, then $\ln(a) = \ln(b)$ and therefore

$$\ln z = \ln\left((x^2y + xy^2)^{\cos(xy)}\right) = \cos(xy)\ln(x^2y + xy^2).$$

Differentiating both sides with respect to x, we obtain

$$\frac{1}{z}\frac{\partial z}{\partial x} = -y\sin(xy)\ln(x^2y + xy^2) + \cos(xy)\,\frac{2xy + y^2}{x^2y + xy^2}$$

and

$$\frac{\partial z}{\partial x} = (x^2y + xy^2)^{\cos(xy)} \left[-y \sin(xy) \ln(x^2y + xy^2) + \cos(xy) \frac{2x + y}{x^2 + xy} \right].$$

One could now repeat this computation using the operator $\partial/\partial y$, or, better, one could point to the symmetry in (2.5) and immediately write down $\partial z/\partial y$.

Exercise 2.26 (i) Practise advanced multivariate differentiation – chain rule I, chain rule II, implicit differentiation, logarithmic differentiation by working through additional examples from the lectures and by doing the practice problems on the tutorial sheet. You should also try to gain familiarity with curves, changes of variables, level sets, and 3D graphing. Experimenting with maths software would be a good approach to the latter. Ask the lecturer or your tutor, or stop by the the drop-in surgeries, in case you have questions.

(ii) Find functions f_i and corresponding constants c_i such that the level set $f_i = c_i$ are: a straight line of slope $+1$, a straight line of slope different from ± 1, a circle, a parabola, an ellipse, two concentric circles, a sketch of your favourite animal. Check your examples using software[a].

(iii) Write out the function $f(x,y) = x + y^2$ in polar coordinates, i.e. find the formula for $F(r,\theta) = f(x,y)$ in terms of r and θ. Then differentiate these expressions directly and compare to the r and θ derivatives of F obtained via chain rule II (as in example 2.21 (ii)). Again, you can have a look at the MTH1000 worksheet "Polar coordinates" for more practice on this important change of variables.

(iv) Find the first-order partial derivatives of[a]

$$z(x,y) = (\cos(2x))^{\ln y}.$$

(v) In example 2.25 (iv), the x-derivative of z in (2.5) is found by applying $\partial/\partial x$ to the $\ln(\dots)$ of (2.5). Now, find the derivative with respect to y by repeating those steps. Looking at this result, try to interpret the reference to "symmetry" at the end of the example[b].

[a] Hint: For WolframAlpha, the syntax would be, for example, `plot sin(x)*sin(y) = 0.1 .`
[a] Hint: You can *check* (not "find"!) your result with `differentiate z(x,y) = ....`
[b] Hint: In (2.5), the variable names x and y can be swapped, can't they? Having that in mind, what is the difference between the given z_x and the z_y you have found (they should be very similar – if not, you've made a mistake computing z_y)?

2.3 Directional Derivatives and the Gradient Vector

Definition 2.27 — Gradient Vectors. The ***gradient vector*** ∇f of a function $f(x,y)$ is the row vector containing its partial derivatives,

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix}.$$

Similarly for $f = f(x,y,z)$,

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{bmatrix}.$$

Definition 2.28 — Directional Derivatives. For a vector $\vec{v} = \begin{bmatrix} a & b \end{bmatrix}^\top$ and a function f of two variables, the ***directional derivative*** of f along $\vec{v}$ is

$$\mathrm{D}_v f = a\frac{\partial f}{\partial x} + b\frac{\partial f}{\partial y} = \nabla f \begin{bmatrix} a \\ b \end{bmatrix},$$

and similarly for three-variable functions and directions $\vec{v} = \begin{bmatrix} a & b & c \end{bmatrix}^\top$ (recall our conventions regarding vector notation from Chapter 1: "vector" (without specification of "column" or "row") means "column vector", and such column vectors may be written as the transposes of row vectors to save white space).

Example 2.29 Find the gradient of the function $f(x,y) = x\cos(y) - \sin(xy)$, and further its directional derivative along $\vec{v} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \end{bmatrix}^\top$ at the point $(x_0, y_0) = (\pi, 0)$.
Sol.:

Remark 2.30

(i) Consider a function $f = f(x,y)$ and a curve

$$\gamma(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

in the xy-plane (in the previous section, we had written "$(x(t), y(t))$" for curves – we shall see in the following that it is also useful to write this pair, (x,y), as a vector). Define the function $F(t) = f(\gamma(t)) = f(x(t), y(t))$. Then, comparing the definition of directional derivatives to chain rule I, we

find

$$\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} = \begin{bmatrix} f_x & f_y \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = \mathrm{D}_{\gamma'} f,$$

where $\gamma' = \begin{bmatrix} x' & y' \end{bmatrix}^\top$.

Definition 2.31 — Hessian Matrix. The ***Hessian matrix*** or *Hessian* of $f = f(x,y)$ is the matrix

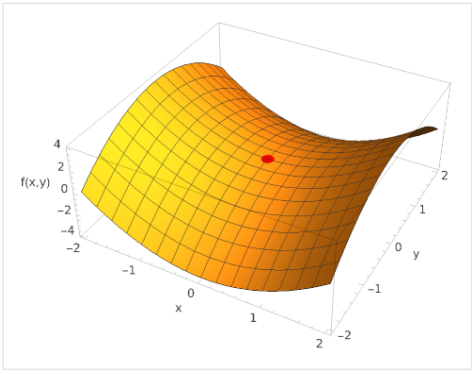
$$\mathrm{Hess} f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

Similarly for functions of three or more variables.

Example 2.32 Find the Hessian of $f(x,y) = x^2 - y^2$.

Sol.:

$$\mathrm{Hess} f = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$



Example 2.33 Find the gradient and the Hessian of $f(x,y,z) = xy^2 + z^3$.

Sol.:

Exercise 2.34 (i) Work through the practice problems on the tutorial sheet. If necessary, review some of the topics from Chapter 1 to be able to comfortably work with the gradient vector and the Hessian matrix.

(ii) For $f(x,y) = \ln(x+y^2)$, find the gradient, the Hessian matrix, and the directional derivative in the direction $v = {}^1/_{\sqrt{5}}\begin{bmatrix}2 & 1\end{bmatrix}^\top$ at[a] $(x,y) = (3,1)$.

(iii) For $f(x,y) = \left(e^{3x} + \sin(4y)\right)^5$, find the unit vector v (i.e., we need $|v| = 1$) of steepest descent at[b] $(x,y) = (0, {}^\pi/_4)$.

(iv) In remark 2.30, we have written out chain rule I as a matrix multiplication of a row vector and a column vector. Do the same for chain rule II: Find the matrix M such that for the functions f and F in theorem 2.20 we have

$$\nabla F = \nabla f \cdot M.$$

Note that the gradient of F is with respect to its variables, s and t, while ∇f is w.r.t. x and y.

---

[a] Answer:

$$D_v f(3,1) = \frac{1}{\sqrt{5}}, \quad \text{Hess} f(3,1) = \frac{1}{16}\begin{bmatrix}-1 & -2\\ -2 & +4\end{bmatrix}.$$

[b] Hint: The steps for solving this are: Compare to theorem **??**, find the gradient of f at the given point, normalise, choose one of the two possible directions.
Answer: $v = \begin{bmatrix}-0.6 & 0.8\end{bmatrix}^\top$.

2.4 Taylor Approximations

Remark 2.35

Recall the single-variable theory of Taylor series and Taylor approximations from last term: For $F = F(t)$, we have the Taylor series

$$F(t) = F(a) + F'(a)(t-a) + \frac{1}{2}F''(a)(t-a)^2 + \frac{1}{6}F'''(a)(t-a)^3 + \dots$$

in some neighbourhood of a, provided certain technical conditions on F are satisfied. Truncating this series, say after the term of order 2, then gives an approximation,

$$F(t) \approx F(a) + F'(a)\cdot(t-a) + \frac{1}{2}F''(a)\cdot(t-a)^2,$$

which is valid close to the point $t = a$ about which the Taylor series/approximation has been developed. In this section, we will work out the corresponding formulas for functions of several variables.

Theorem 2.36 — Taylor Approximation. Let $f(x,y)$ have continuous partial derivatives of order up to 2, and let the point (a,b) be in its domain. Then the *first- and second-order Taylor approximations* of f at (a,b) are

$$T^{(1)}_{(a,b)}f(x,y) = f(a,b) + \nabla f(a,b)\begin{bmatrix} x-a \\ y-b \end{bmatrix}$$

$$T^{(2)}_{(a,b)}f(x,y) = f(a,b) + \nabla f(a,b)\begin{bmatrix} x-a \\ y-b \end{bmatrix} + \frac{1}{2}\begin{bmatrix} x-a & y-b \end{bmatrix}\operatorname{Hess} f(a,b)\begin{bmatrix} x-a \\ y-b \end{bmatrix}, \tag{2.6}$$

where we omit the subscript (a,b) if the point, from which the approximation is carried out, is clear. Close

to that point, we have

$$T^{(1)}f \approx f, \qquad T^{(2)}f \approx f,$$

where the second-order approximation is better in the sense that the error $\left|T^{(2)}f(x,y) - f(x,y)\right|$ decays faster than $\left|T^{(1)}f(x,y) - f(x,y)\right|$ as $(x,y) \to (a,b)$. Similarly for $f = f(x,y,z)$.

Proof. Let (x,y) be close to the point (a,b), so that both

$$h = x - a \quad \text{and} \quad k = y - b$$

are small. Define the curve

$$\gamma(t) = (a + ht, b + kt),$$

and let $F(t) = f(\gamma(t))$. Note that $\gamma(0) = (a,b)$ and $\gamma(1) = (x,y)$. We now find the second-order Taylor approximation of F (single-variable!) at $t = 0$:

$$\begin{aligned} F(t) \approx T_0^{(2)}F(t) &= F(0) + F'(0)\cdot t + \frac{1}{2}F''(0)\cdot t^2 \\ &= F(0) + t\cdot\frac{\mathrm{d}F}{\mathrm{d}t}(0) + \frac{1}{2}t^2\cdot\frac{\mathrm{d}^2F}{\mathrm{d}t^2}(0) \\ &= f(a,b) + t\cdot\left[\frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}\right]_{t=0} + \frac{1}{2}t^2\cdot\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}\right]_{t=0} \\ &= f(a,b) + t\cdot\nabla f(a,b)\begin{bmatrix}h\\k\end{bmatrix} + \frac{1}{2}t^2\cdot\frac{\mathrm{d}}{\mathrm{d}t}\left[f_x h + f_y k\right]_{t=0}. \end{aligned}$$

The remaining t-derivative is that of a composition with γ (after all, the evaluation at $t = 0$, indicated earlier as $[\ldots]_{t=0}$, does not mean that f_x and f_y are evaluated at $t = 0$ – they are to be evaluated at $\gamma(0) = (a,b)$), and therefore, chain rule I needs to be applied one more time – note that h and k do not depend on t.

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\left[f_x h + f_y k\right]_{t=0} &= h\cdot\left[\frac{\partial f_x}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f_x}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}\right] + k\cdot\left[\frac{\partial f_y}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f_y}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}\right] \\ &= h^2 f_{xx} + hk f_{yx} + kh f_{xy} + k^2 f_{yy}, \end{aligned}$$

where all the partials are evaluated at (a,b). The last expression can be written as a matrix multiplication of the

row vector containing h and k, a matrix containing the second-order derivatives of f, and the column vector containing h and k. This gives

$$F(t) \approx f(a,b) + t\,\nabla f(a,b) \begin{bmatrix} h \\ k \end{bmatrix} + \frac{t^2}{2} \begin{bmatrix} h & k \end{bmatrix} \mathrm{Hess} f(a,b) \begin{bmatrix} h \\ k \end{bmatrix}.$$

Evaluating this approximation at $t = 1$, we obtain the stated second-order approximation of f,

$$f(x,y) \approx f(a,b) + \nabla f(a,b) \begin{bmatrix} h \\ k \end{bmatrix} + \frac{1}{2} \begin{bmatrix} h & k \end{bmatrix} \mathrm{Hess} f(a,b) \begin{bmatrix} h \\ k \end{bmatrix}.$$

Note that $t = 1$ is not small, i.e. that it is not close to the point $t = 0$ about which F was developed, and that in general it might be too big for a good approximation $T^{(2)}G(1) \approx G(1)$ of a single-variable function $G = G(t)$. However, in the above situation, this can be controlled by requiring h and k to be small, that is, by letting (x,y) be close to the point (a,b) from which the approximation of f is carried out. ■

Example 2.37 (i) Find the first- and second-order Taylor approximation of $f(x,y) = e^x \ln(1+y)$ about the point $(a,b) = (0,0)$.

Sol.:

(ii) Find the first- and second-order Taylor approximation of

$f(x,y) = 2x^2 + y^2$ about the point $(a,b) = (1,1)$.

Sol.:

$$\begin{aligned}
f(x,y) &= 2x^2 + y^2, \\
f(1,1) &= 3, \\
\nabla f(x,y) &= \begin{bmatrix} 4x & 2y \end{bmatrix}, \\
\nabla f(1,1) &= \begin{bmatrix} 4 & 2 \end{bmatrix}, \\
\mathrm{Hess} f(x,y) &= \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}, \\
\mathrm{Hess} f(1,1) &= \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix},
\end{aligned}$$

$$\begin{aligned}
T^{(2)}_{(1,1)} f(x,y) &= 3 + \begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} x-1 \\ y-1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} x-1 & y-1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x-1 \\ y-1 \end{bmatrix} \\
&= 3 + 4x - 4 + 2y - 2 + \frac{1}{2} \begin{bmatrix} x-1 & y-1 \end{bmatrix} \begin{bmatrix} 4x-4 \\ 2y-2 \end{bmatrix} \\
&= -3 + 4x + 2y + \begin{bmatrix} x-1 & y-1 \end{bmatrix} \begin{bmatrix} 2x-2 \\ y-1 \end{bmatrix} \\
&= -3 + 4x + 2y + 2x^2 - 2x - 2x + 2 + y^2 - y - y + 1 \\
&= 2x^2 + y^2.
\end{aligned}$$

Note that here, we have obtained $T^{(2)} f = f$, because $f(x,y)$ is already a very simple function – an expression of x and y containing terms of order at most 2. The analogous situation in the single-variable theory from last term is: The m-th-order Taylor approximation of a polynomial of order m is the original polynomial itself, and any higher-order approximation will be the same.

It is important not to forget about one of the parts we were asked about – the first-order approximation! It is contained in the above computation:

$$T^{(1)}_{(1,1)} f(x,y) = -3 + 4x + 2y.$$

$T^{(1)}_{(1,1)} f$ is again a function of two variables, and we plot

such functions in $\mathbb{R}^3$ by drawing its function values as the z-coordinate. Hence we could write

$$z = -3 + 4x + 2y,$$

which is the equation of a plane in $\mathbb{R}^3$, namely the tangent plane of f at the point $(x_0, y_0, z_0) = (1, 1, f(1,1))$

(iii) Find the second-order Taylor approximation of

$$f(x,y) = \sin(x)\sin(y)$$

about the point $(a,b) = (\pi/4, \pi/4)$.

Sol.: Note that $\sin(\pi/4) = \cos(\pi/4) = \sqrt{2}/2$. The required quantities at (a,b) are

$$\begin{aligned}
f(x,y) &= \sin(x)\sin(y), \\
f(\pi/4,\pi/4) &= \frac{1}{2}, \\
\nabla f(x,y) &= \begin{bmatrix} \cos(x)\sin(y) & \sin(x)\cos(y) \end{bmatrix}, \\
\nabla f(\pi/4,\pi/4) &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \\
\mathrm{Hess} f(x,y) &= \begin{bmatrix} -\sin(x)\sin(y) & \cos(x)\cos(y) \\ \cos(x)\cos(y) & -\sin(x)\sin(y) \end{bmatrix}, \\
\mathrm{Hess} f(\pi/4,\pi/4) &= \frac{1}{2}\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}.
\end{aligned}$$

For simplicity, we write out the approximation in terms of $h = x - \pi/4$ and $k = y - \pi/4$:

$$\begin{aligned}
T^{(2)}_{(\pi/4,\pi/4)} f(x,y) &= \frac{1}{2} + \frac{1}{2}\begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} h \\ k \end{bmatrix} + \frac{1}{4}\begin{bmatrix} h & k \end{bmatrix}\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} h \\ k \end{bmatrix} \\
&= \frac{1}{2} + \frac{h}{2} + \frac{k}{2} + \frac{1}{4}\begin{bmatrix} h & k \end{bmatrix}\begin{bmatrix} -h+k \\ h-k \end{bmatrix} \\
&= \frac{1}{2} + \frac{h}{2} + \frac{k}{2} - \frac{h^2}{4} + \frac{hk}{4} + \frac{kh}{4} - \frac{k^2}{4} \\
&= \frac{1}{2}\left[1 + h + k + hk - \frac{h^2}{2} - \frac{k^2}{2}\right],
\end{aligned}$$

which can then be translated back to an expression of x and y by replacing h and k with $x - \pi/4$ and $y - \pi/4$ respectively.

Exercise 2.38 (i) Work through the tutorial sheet and your material from the lectures, and review Taylor series from last term and related theory if necessary.

(ii) Find the second-order Taylor approximation of $f(x,y,z) = xy^2 + z^3$ at the point[a] $(a,b,c) = (1,1,1)$.

(iii) Re-do example (ii) above, $f(x,y) = 2x^2 + y^2$, but now at a general point $(a,b) \in \mathbb{R}^2$ (i.e., use parameters a and b for the point at which the approximation is carried out, rather than plugging in specific numbers)[b].

(iv) Explore the functions of the form

$$h(x,y) = \begin{bmatrix} x & y \end{bmatrix} M \begin{bmatrix} x \\ y \end{bmatrix},$$

where M is a symmetric 2×2 matrix (i.e. $a_{12} = a_{21}$), by doing the following: First write out $h(x,y)$ in a form that you can enter in WolframAlpha or other graphing software[c]; then, for different choices of the entries of M, plot h. How many different shapes can you find[d]?

---

[a] Hint: You can compare to the example at the end of the previous section to see whether the gradient vector and Hessian matrix you have found are correct. You can quickly check two conditions that your answer needs to satisfy: (i)

$$T^{(2)}_{(a,b,c)}(a,b,c) = f(a,b,c)$$

– your approximation should agree with the function you are approximating at the point from which you are developing your approximation (same in two variables, at (a,b) in this case), and (ii) your approximation should only contain terms of order at most 2 (for mixed terms, one adds the powers, hence terms like x, y^2, xz may appear in your approximation – x^3, y^2z, x^3yz^2 may not). If you further check the following two conditions, you will even have certainty that your answer is correct: (iii) the first derivative (gradient) and (iv) the second derivative (Hessian) of f and $T^{(2)}$ at the point (a,b,c) ((a,b) in two variables) need to agree.

[b] Answer: For the second-order approximation, you should obtain the same result as before – for the same reason.

[c] Hint: E.g., for $M = I$, $h(x,y) = x^2 + y^2$.

[d] Hint: These shapes will be important in the next section, and it will be answered then how many qualitatively different shapes there are.

2.5 Local Extrema and Saddle Points

Definition 2.39 — Local Extrema and Critical Points. For a function $f(x,y)$ and a point (a,b) in its domain (similarly in three variables):

(i) (a,b) is a ***local maximum*** of f if

$$f(x,y) \leq f(a,b)$$

in some neighbourhood of (a,b).

(ii) (a,b) is a ***local minimum*** of f if there exists an $\delta > 0$ such that

$$f(x,y) \geq f(a,b)$$

for all (x,y) with $\left\| \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} a \\ b \end{bmatrix} \right\| < \delta$.

(iii) (a,b) is a ***critical point*** of f if

$$\nabla f(a,b) = \begin{bmatrix} 0 & 0 \end{bmatrix},$$

or if $f_x(a,b)$ and/or $f_y(a,b)$ doesn't exist.

Remark 2.40

(i) Neighbourhoods are small sets surrounding the point in question. For example, in one dimension: For $t_0 = 0.001$, we can find a small neighbourhood that contains only positive numbers, e.g. the open interval $(t_0 - 0.0005, t_0 + 0.0005)$. For $t_0 = 0$, however, this is not possible: every neighbourhood – no matter how small – contains both positive and negative numbers. In the higher dimensional domains of multivariate functions, one can work with disks (in 2D) or balls (in 3D) around the points in

question. This notion is very important in Mathematics and used informally in part (i) of the definition above. In part (ii), the same condition is written out more formally using a parameter δ and the corresponding disk of radius δ around the point (a,b). Note that it does not matter how small δ is, as long as it is positive.

(ii) A local extremum is either a local minimum or a local maximum. Local maxima are not surpassed in some small neighbourhood, while for local minima there exists a neighbourhood in which none of the other function values is smaller. Note that there is no requirement for surrounding function values to be strictly smaller, respectively strictly larger – that means, for example, that for a constant function $f(x,y) = c$, every point in its domain is both a local maximum and a local minimum.

(iii) For the material in this section, it is useful to recall the corresponding single-variable theory from last term. The following result should look familiar, and its proof will be left as an exercise.

(iv) Terminology: Minima, maxima, extrema are points in the domain at which extreme values are taken. So, make sure to always state those points in your answer – not just the corresponding value. You do not have to state those values, unless you are explicitly asked to do so. However, stating the values is good practice – especially for worded optimisation problems – and in other modules you may be required to state the values as well.

Theorem 2.41 If (a,b) is a local extremum of f, then (a,b) is a critical point of f.

Example 2.42 (i) Find the critical points of $f(x,y) = x^2y - y + 7$.

Sol.:

(ii) Find the critical points of $f(x,y) = xy$.

Sol.:

(iii) Find the critical points of $f(x,y) = x^3 - 3x + 3xy^2$.

Sol.:

Theorem 2.43 — Second-Derivative Test. Let (a,b) be a critical point of $f(x,y)$ and let

$$\Delta = \det\left(\text{Hess} f(a,b)\right) = \left(f_{xx}f_{yy} - f_{xy}^2\right)_{|(a,b)}.$$

Then (a,b) can be classified as follows:

(i) If $\Delta > 0$ and $f_{xx}(a,b) > 0$, then (a,b) is a local minimum.

(ii) If $\Delta > 0$ and $f_{xx}(a,b) < 0$, then (a,b) is a local maximum.

(iii) If $\Delta < 0$, then (a,b) is not a local extremum.

(iv) If $\Delta = 0$, then the test is inconclusive and (a,b) could be any of the three possibilities above (local maximum, local minimum, or neither).

Example 2.44 (i) Find and classify all critical points of $f(x,y) = x^2y - y + 7$.

Sol.:

(ii) Find and classify all critical points of $f(x,y) = x^3 - 3x + 3xy^2$.

Sol.:

(iii) Find all critical points of $f(x,y) = x - \ln x + 2y^2 + xy$ and classify them using the second-derivative test.

Sol.:

$$\begin{aligned} f(x,y) &= x - \ln x + 2y^2 + xy, \\ \nabla f(x,y) &= \begin{bmatrix} 1 - \frac{1}{x} + y & 4y + x \end{bmatrix}, \\ \mathrm{Hess} f(x,y) &= \begin{bmatrix} {}^{1}/_{x^2} & 1 \\ 1 & 4 \end{bmatrix}, \\ P &= (2, -\tfrac{1}{2}), \\ \Delta &= 0. \end{aligned}$$

Therefore, the second-derivative test is inconclusive, and more work (e.g. analyse the graph or manually check first derivatives around the critical point) is required to find out what the critical point $P = (2, -\frac{1}{2})$ of f is.

Exercise 2.45 (i) Work through the problems on the tutorial sheet and your material from the lectures, and review the 1D second-derivative test from last term if necessary.

(ii) Make sure to properly work through the computations (iii) of examples 2.42 and 2.44 that are only sketched. For 2.44 (iii), use graphing software to try to find out what the point P is[a].

(iii) Convince yourself that theorem 2.41 is true[b]; or, if you like, try to prove it formally. Ask your tutor or lecturer for help if necessary.

(iv) Give examples of (1) a local minimum, (2) a local maximum, and (3) a critical point that is neither, that cannot be classified with the second-derivative test[c].

[a] Hint: This part (second sentence of (ii)) is quite difficult and not core material. Zooming in by appending `for x from 1.99 to 2.01 and y from -0.51 to -0.49` to your WolframAlpha plot command provides some insight, but unfortunately it does not quite give an answer.

[b] Hint: You could contrapose that statement and/or look at theorem **??** for inspiration.

[c] Hint: $f(x,y) = x^6 + y^4$ covers one of those cases.

2.6 Extrema under Constraints: Lagrange Multipliers

Example 2.46 Find the maximum of the function $z = f(x,y) = 2x + y$ on the circle $S = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$.

Sol.:

Theorem 2.47 — Lagrange Multipliers. To maximise or minimise a function $f(x,y)$ on a set $S = \{(x,y) \in \mathbb{R}^2 \mid g(x,y) = 0\}$, define the function

$$F(\lambda, x, y) = f(x,y) - \lambda g(x,y),$$

and then solve

$$\nabla F = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}.$$

The function f can then be evaluated at the solutions (x,y) of that system to identify the extremum that is sought. The parameter λ is called *Lagrange multiplier* (its value is not going to be directly relevant for our purposes).

Example 2.48 (i) Find the point on the surface $(x-y)^2 - z^2 = 1$ that is closest to the origin $(x,y,z) = (0,0,0)$ of $\mathbb{R}^3$.

Sol.:

(ii) Bob wants to sell free-range eggs and needs to decide

how many chicken to keep ("x") and how much land to lease ("y"). In order to be able to label his eggs as "free-range", every chicken needs to have at least $4\,m^2$ of space. Leasing land costs $f_2(y) = {}^{y}/_{10}$ and selling eggs brings in an income $f_1(x) = 8\sqrt{x}$. Help Bob optimise his profits. (The numbers in this problem are fictional.)

Sol.:

Exercise 2.49 (i) Work through the material – in particular, the examples – from the lectures and solve the optimisation problems on the tutorial sheet.

(ii) A single-variable problem similar to example 2.48 (i): For $c \in \{1, -4\}$, consider the curve $y = x^2 + c$ and the distances of its points to the origin $(0,0)$ of the xy-plane. Sketch the curves and think about what kind of extrema exist in each case. Then compute all local and global extrema as well as the corresponding distances to the origin[a].

(iii) Just to make sure we are not sending Bob down the wrong path: also solve his optimisation problem, example 2.48 (ii), with single-variable theory, as in example 2.46.

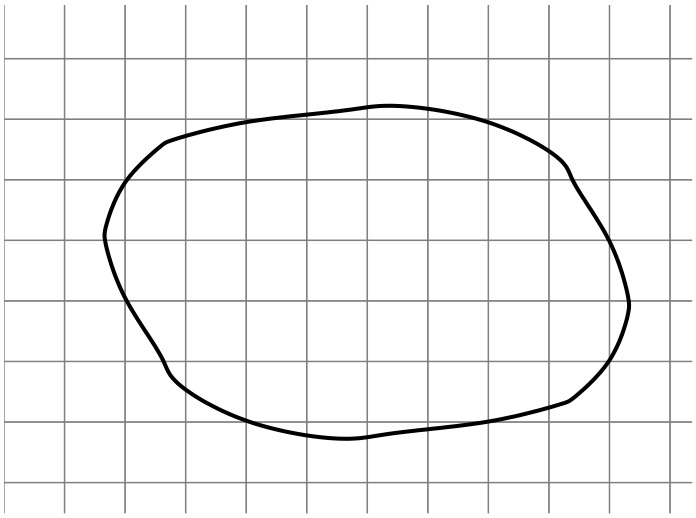
(iv) Find the point on $S : z = xy$ that is closest to the sphere of radius 1 centred at[b] $(x, y, z) = (0, 0, 5)$.

---

[a] Answer: You should obtain four extrema with distances 1, 1.936, and 4 to the origin. The minima are both local and global, the maximum is only local. Make sure to state the extrema (the points at which the minimal/maximal distances are realised), as they are part of what you were asked for – we are only giving you the corresponding distances as they are enough information to verify your answers.

[b] Hint: A point on S, that is closest to the sphere, is also closest to its centre (!!); considering different cases helps solving $\nabla F = 0$; to check your answer: you should get a minimum distance of 2.

3 Integration



Suppose you have an oddly shaped room with square tiles, and you want to estimate its square footage. Let n be the number of full tiles you count, and let m stand for the number of broken tiles. Denote the area of a single tile – which can be computed: it is the square of the side length – by A_0. Then we have the lower and upper bounds

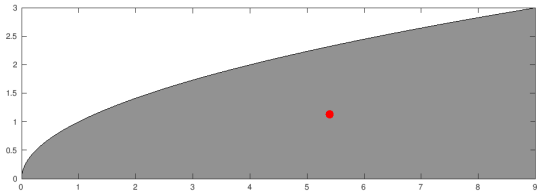
$$n \cdot A_0 \leq A \leq (n+m) \cdot A_0$$

for the area A of the room. Now, convince yourself that this estimate would be better (tighter), if the tiles were smaller*!

Finding areas under curves or volumes under surfaces is called *integration*. It is one of the most fundamental and important concepts of mathematics, and it has applications in every scientific discipline. The above approach of approximating areas with large numbers of simple pieces can be quite cumbersome. Fortunately, there is very powerful connection to differentiation, which allows to compute integrals in a more efficient way.

Example 3.1 A large metal plate of uniform thickness and the shape in the sketch (let us call it S; the area under $f(x) = \sqrt{x}$ for x from 0 to 9) needs to be balanced on a single point.

*Hint: Mark the areas $n \cdot A_0$ and $(n+m) \cdot A_0$ in the sketch above, and then think about what would happen if we had tiles of half the side length



That is, the centre of mass of S needs to be found.

This can be solved with integration: The x and y coordinates of the centre of mass of S are

$$x_c = \frac{\iint_S x \, dA}{\iint_S 1 \, dA} = 5.4,$$
$$y_c = \frac{\iint_S y \, dA}{\iint_S 1 \, dA} = 1.125.$$

We will learn how to compute those integrals in this chapter.

3.1 Theory of Integration in One Dimension

Definition 3.2 — Definite Integral, Riemann Sum. Let $f(x)$ be continuous on $a \leq x \leq b$ and divide the interval $[a,b]$ into n subintervals of equal width $\Delta x = {}^{(b-a)}/_{n}$. Let

$$x_0 = a, \; x_1 = a + \Delta x, \; x_2 = x_1 + \Delta x = x_0 + 2\Delta x, \; \ldots,$$
$$x_n = x_0 + n\Delta x = b$$

be the endpoints of these subintervals.

(i) Then the ***definite integral*** of f from a to b is defined as

$$\int_a^b f(x)\,dx := \lim_{n\to\infty} \sum_{j=1}^{n} f(c_j)\Delta x, \tag{3.1}$$

where $x_{j-1} \leq c_j \leq x_j$. Here $f(x)$ is called the ***integrand***, a the lower boundary, and b the upper boundary.

(ii) The sum appearing in the above definition is called a ***Riemann sum*** of f over $[a,b]$. If the points c_j are always chosen to be the right-hand end point

of their subinterval, that is $c_j = x_j$, we obtain the right-hand Riemann sum

$$\sum_{j=1}^{n} f(x_j)\Delta x.$$

Using the left end points of the subintervals, $c_j = x_{j-1}$, gives the left-hand Riemann sum

$$\sum_{j=1}^{n} f(x_{j-1})\Delta x = \sum_{j=0}^{n-1} f(x_j)\Delta x.$$

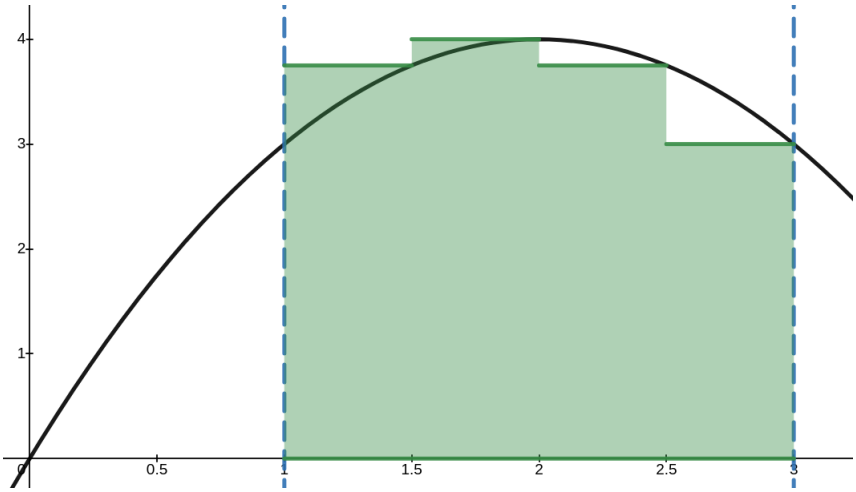
Remark 3.3

(i) Equation (3.1) introduces a new quantity – the definite integral of f over $[a,b]$ – and *defines* (that is what the ":=" means) it to be equal to the limit for $n \to \infty$ of the Riemann sums on the right. Note that on the right-hand side of (3.1), different choices could be made for the points c_j. One should therefore show now that all these possible different choices lead to the same limit. That is, one should show that $\int_a^b f(x)\,\mathrm{d}x$ as defined above is *well-defined*! This will be given as an exercise at the end of this section.

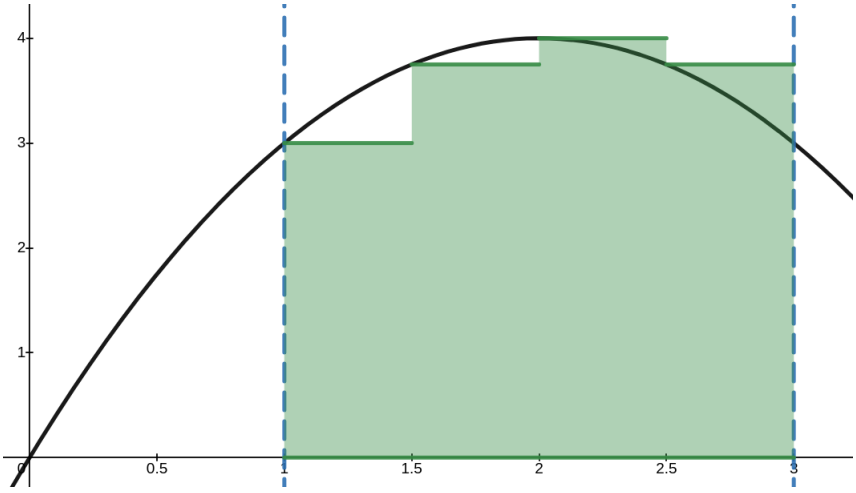
(ii) The following figure illustrates the right-hand Riemann sum for $n = 4$,

$$\sum_{j=1}^{4} f(x_j)\Delta x,$$

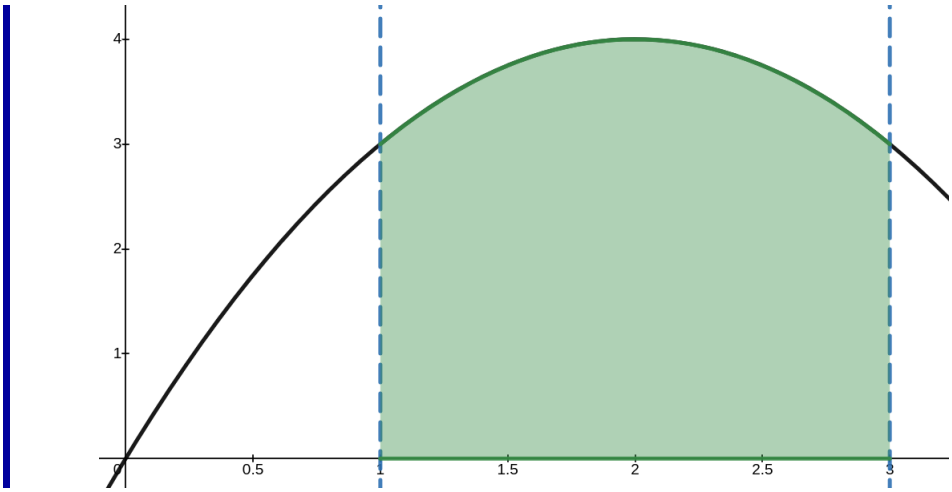
of $f(x) = x(4-x)$



The rectangles in this figure have areas "height times width" $= f(x_j) \cdot \Delta x$, and they are added up to obtain the Riemann sum. For comparison, the left-hand Riemann sum is



We see that in both cases, the Riemann sums approximate the area under the curve $y = f(x) = x(4-x)$, and as in the example of the oddly-shaped room in the introduction, those approximations will get better if we choose n to be larger. We can therefore interpret their limit, i.e., $\int_a^b f(x)\,\mathrm{d}x = \int_1^3 x(4-x)\,\mathrm{d}x$, as the signed area under the curve $y = f(x) = x(4-x)$:



Example 3.4 Consider $f(x) = {}^{3}/_{5}\,x$, $a = 0$, $b = 5$. First, let us find the left-hand Riemann sum for $n = 5$ of f. The partition of $[a,b] = [0,5]$ is

$$\Delta x = \frac{b-a}{n} = \frac{5-0}{5} = 1,$$
$$x_0 = 0,\ x_1 = 1,\ x_2 = 2,\ \ldots, x_5 = 5,$$

and therefore

$$\int_0^5 \frac{3}{5}\,x\,\mathrm{d}x \approx \sum_{j=0}^{4} f(x_j)\Delta x = [f(0)+f(1)+f(2)+f(3)+f(4)]\cdot 1$$
$$= \frac{3}{5}\,[0+1+2+3+4] = 6.$$

We now generalise this computation to find the definite integral f over the interval $[a,b]$. The partition of the interval is

$$\Delta x = \frac{b-a}{n} = \frac{5-0}{n} = \frac{5}{n},$$
$$x_0 = 0,\ x_1 = \frac{5}{n},\ x_2 = 2\frac{5}{n},\ \ldots, x_n = n\frac{5}{n} = 5,$$

and therefore

$$\int_0^5 \frac{3}{5}\,x\,\mathrm{d}x = \lim_{n\to\infty}\sum_{j=0}^{n-1} f(x_j)\Delta x = \lim_{n\to\infty}\frac{5}{n}\sum_{j=0}^{n-1}\frac{3}{5}x_j$$
$$= \lim_{n\to\infty}\frac{5}{n}\sum_{j=0}^{n-1}\frac{3}{5}\frac{5}{n}j = \lim_{n\to\infty}\frac{15}{n^2}\sum_{j=0}^{n-1} j$$
$$= \lim_{n\to\infty}\frac{15}{n^2}\frac{(n-1)n}{2} = \lim_{n\to\infty}\frac{15}{2}\frac{n-1}{n} = 7.5,$$

Note that this result agrees with the area

$$A = \frac{3\cdot 5}{2},$$

found by noticing that the region under the curve $y = f(x)$ is a right-angled triangle.

Properties

Let $a, b, \alpha \in \mathbb{R}$, $a < b$, and let $f(x), g(x)$ be continuous functions on $[a, b]$. Then:

(1) $\displaystyle\int_a^a f(x)\,\mathrm{d}x = 0,$

(2) $\displaystyle\int_a^b f(x)\,\mathrm{d}x = -\int_b^a f(x)\,\mathrm{d}x,$

(3) $\displaystyle\int_a^b [f(x) + g(x)]\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + \int_a^b g(x)\,\mathrm{d}x,$

(4) $\displaystyle\int_a^b \alpha \cdot f(x)\,\mathrm{d}x = \alpha \cdot \int_a^b f(x)\,\mathrm{d}x,$

(5) $\displaystyle\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x$

(6) If $f(x) \geq 0$ on $[a, b]$, then $\displaystyle\int_a^b f(x)\,\mathrm{d}x \geq 0.$

Now suppose that $f(x)$ is continuous on the interval $[-a, a]$, where $a > 0$. Then we have

(7) If $f(x)$ is even, i.e. $f(-x) = f(x)$, then $\displaystyle\int_{-a}^a f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x,$

(8) If $f(x)$ is odd, i.e. $f(-x) = -f(x)$, then $\displaystyle\int_{-a}^a f(x)\,\mathrm{d}x = 0.$

(i) All those properties are proven starting from the definition 3.1 of the definite integral. (There is nothing else to start from, is there?) Only the proof of the last property is written out below.

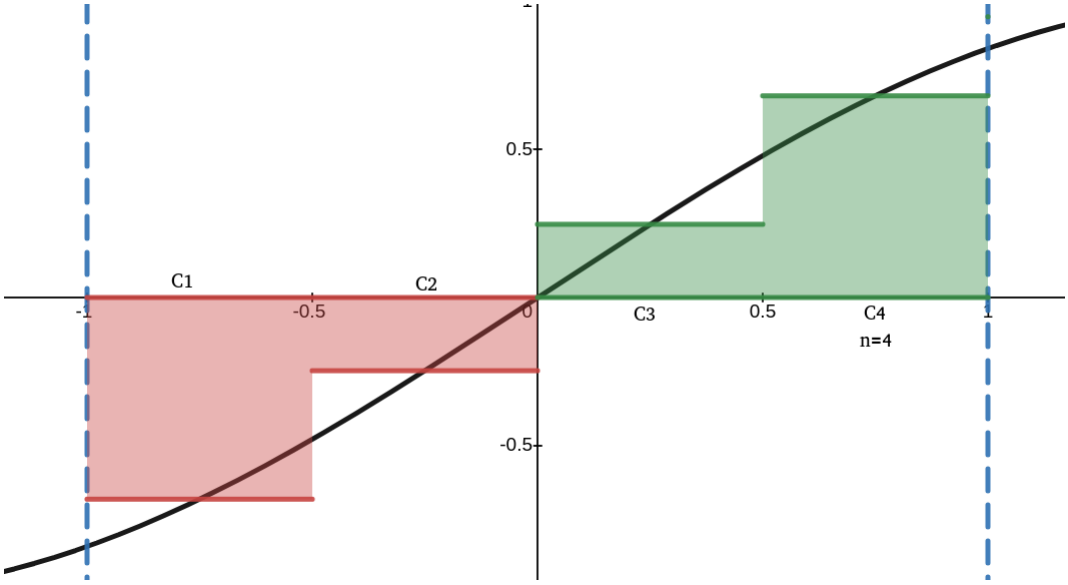
(ii) Properties (3) and (4) state that integration is *linear*, and they can by proven by using the corresponding laws for sums and limits (e.g., pull the constant α out of the sum and then out of the limit in definition 3.1).

Proof. For the proof of (8), we consider an odd function and we use only even n for the limit $n \to \infty$. Since all choices for the points c_j lead to the same limit, we make a choice that suits our purpose well: let the c_j be the mid-

points of their subintervals. The Riemann sums are

$$\sum_{j=1}^{n} f(c_j)\Delta x = \Delta x \cdot [f(c_1) + f(c_2) + \dots f(c_{n-1}) + f(c_n)]\,,$$

and comparing to



we see that this Riemann sum is zero since $f(c_1) + f(c_n) = 0$, $f(c_2) + f(c_{n-1}) = 0$, ... This gives

$$\int_{-a}^{a} f(x)\,\mathrm{d}x = \lim_{n\to\infty} 0 = 0.$$

Now, the above assertion that $f(c_j) + f(c_{n-j+1}) = 0$ seems to be confirmed by the sketch, but we should prove it properly: After working out formulas for the partition points x_j, we find

$$c_j = -a + \Delta x\left(j - \frac{1}{2}\right) = -a + \frac{2a}{n}\left(j - \frac{1}{2}\right)$$

for the midpoints c_j of the subintervals. This gives

$$\begin{aligned} c_{n-j+1} &= -a + \frac{2a}{n}\left(n - j + 1 - \frac{1}{2}\right) \\ &= -a + 2a + \frac{2a}{n}\left(-j + \frac{1}{2}\right) = -c_j, \end{aligned}$$

and therefore, since f is odd,

$$f(c_j) + f(c_{n-j+1}) = f(c_j) + f(-c_j) = f(c_j) - f(c_j) = 0.$$

■

Definition 3.5 — Mean. For continuous $f : [a,b] \to \mathbb{R}$, the real number

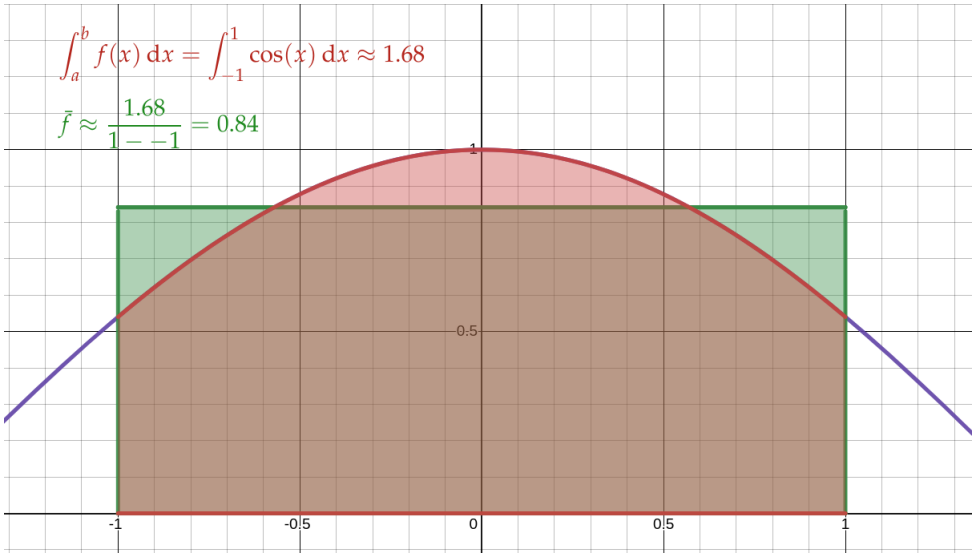
$$\bar{f} := \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x$$

is called the ***mean*** or ***average*** of f over the interval $[a,b]$.

We have

$$\bar{f}\cdot(b-a) = \int_a^b f(x)\,\mathrm{d}x,$$

that is, the area under the constant function $\bar{f}$ over $[a,b]$ is the same as the area under the graph of f (e.g $f(x) = \cos(x)$):



Note that this is similar for the average of numbers – if Alice has an average of $\bar{m}$ on all the tests in a module, and Bob scored exactly $\bar{m}$ each time, then they have the same total mark.

Theorem 3.6 — Mean Value Theorem for Integrals (MVTI). Let $f : [a,b] \to \mathbb{R}$ be continuous. Then there exists a point $c \in (a,b)$ such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x.$$

That is, f attains its mean value at some point in (a,b).

Proof. Let

$$M := \max_{x\in[a,b]} f(x), \qquad m := \min_{x\in[a,b]} f(x).$$

Since f is continuous, it takes every value between m and M. Therefore, we need to show that

$$m \leq \bar{f} \leq M. \tag{3.2}$$

We have

$$\begin{aligned} & M - f(x) \geq 0 \qquad \text{on } [a,b] \\ \overset{\text{prop. (6)}}{\Longrightarrow} \quad & \int_a^b M - f(x)\,\mathrm{d}x \geq 0 \\ \overset{\text{prop. (3)}}{\Longrightarrow} \quad & \int_a^b M\,\mathrm{d}x - \int_a^b f(x)\,\mathrm{d}x \geq 0 \\ \Longrightarrow \quad & \int_a^b f(x)\,\mathrm{d}x \leq \int_a^b M\,\mathrm{d}x = M(b-a). \end{aligned}$$

Combining this with the outcome of the analogous computation for $f(x) - m$ gives

$$m(b-a) \leq \int_a^b f(x)\,\mathrm{d}x \leq M(b-a).$$

Division by $(b-a)$ gives the inequality 3.2, and hence completes the proof. ■

Definition 3.7 — Area Function. For continuous $f : [a,b] \to \mathbb{R}$, the ***area function***

$$A : [a,b] \to \mathbb{R}$$

of f is

$$A(x) := \int_a^x f(t)\,\mathrm{d}t.$$

Remark 3.8

We use the variable t in the integrand to avoid confusion with the variable of A.

Theorem 3.9 — Fundamental Theorem of Calculus (FTC). Let $f : [a,b] \to \mathbb{R}$ be continuous.

(i) The derivative of the area function is

$$A'(x) = \frac{\mathrm{d}}{\mathrm{d}x} A(x) = f(x).$$

That is,

$$\frac{\mathrm{d}}{\mathrm{d}x} \int_a^x f(t)\,\mathrm{d}t = f(x).$$

(ii) If F is any *antiderivative* of f, i.e. a function with $F'(x) = f(x)$, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Proof. (i) Using the definition of the derivative, the definition of the area function, and property (5) of the definite integral, we obtain

$$A'(x) = \lim_{h\to 0} \frac{A(x+h) - A(x)}{h} = \lim_{h\to 0} \frac{1}{h} \int_x^{x+h} f(t)\,\mathrm{d}t.$$

The expression within the limit on the right is the average of f over the interval $[x, x+h]$. By the MVTI, there exist $c_h \in [x, x+h]$ such that

$$f(c_h) = \frac{1}{h} \int_x^{x+h} f(t)\,\mathrm{d}t.$$

This allows to continue the computation of $A'(x)$ above as follows:

$$\begin{aligned} A'(x) &= \lim_{h\to 0} f(c_h) \qquad \text{where } c_h \in [x, x+h] \\ &= f\left(\lim_{h\to 0} c_h\right) = f(x). \end{aligned}$$

(ii) The area function $A(x)$ is an antiderivative of f – by (i) – and for it, the claim is true by definition:

$$A(b) - A(a) = A(b) = \int_a^b f(t)\,\mathrm{d}t.$$

Now let $F(t)$ be a different antiderivative of f. Then

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(F(x) - A(x)\right) = f(x) - f(x) = 0,$$

and therefore F and A differ only by a constant,

$$F(x) = A(x) + c.$$

This gives

$$F(b) - F(a) = (A(b) + c) - (A(a) + c) = A(b) - A(a) = \int_a^b f(t)\,\mathrm{d}t.$$

■

Example 3.10 We solve the following definite integral by guessing a function whose derivative is the integrand. Once this guessed antiderivative is written down, one should always check whether it really differentiates to the original integrand.

$$\int_1^3 (x^3 - 6x)\,\mathrm{d}x = \left(\frac{x^4}{4} - 3x^2\Big|_1^3\right) \qquad \left[\text{check}: \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^4}{4} - 3x^2\right) = x^3 - 6x \quad \checkmark\right]$$
$$= \left(\frac{3^4}{4} - 3\cdot 3^2\right) - \left(\frac{1^4}{4} - 3\cdot 1^2\right) = \frac{81}{4} - 27 - \frac{1}{4} + 3 = -4$$

The vertical line in the second expression stands for "evaluate at $x = 3$ and then subtract the evaluation at $x = 1$" – as on the right-hand side of FTC (ii). To foreshadow the proof of the FTC, note that the choice of antiderivative is not unique, but different choices seem to be leading to the same result; e.g.,

$$\int_1^3 x^3 - 6x\,\mathrm{d}x = \left(\frac{x^4}{4} - 3x^2 + 42\Big|_1^3\right) = \frac{81}{4} - 27 + 42 - \frac{1}{4} + 3 - 42 = -4.$$

Corollary 3.11 (i)

$$\int_a^b F'(t)\,\mathrm{d}t = F(b) - F(a).$$

(ii)

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_{a(x)}^{b(x)} f(t)\,\mathrm{d}t = f(b(x))\,b'(x) - f(a(x))\,a'(x).$$

Some notes:

(i) The word "corollary" is usually used for results that follow from important theorems.

(ii) The first identity is also called the "Total Change Theorem", and it follows immediately from FTC (ii). It reads: "Integrating the rate of change of a function over an interval gives the total change over that interval."

(iii) Make sure to take note of the second identity – it is important for differentiating more complicated integral expressions. It follows from the chain rule:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\int_c^{b(x)} f(t)\,\mathrm{d}t &= \frac{\mathrm{d}}{\mathrm{d}x}\int_c^{y} f(t)\,\mathrm{d}t && (y = b(x))\\ &= \frac{\mathrm{d}}{\mathrm{d}y}\int_c^{y} f(t)\,\mathrm{d}t \cdot \frac{\mathrm{d}y}{\mathrm{d}x} && (\text{chain rule})\\ &= f(y)\cdot\frac{\mathrm{d}y}{\mathrm{d}x} = f(b(x))\,b'(x),\end{aligned}$$

and using property (5) of the definite integral, we can prove the formula in full generality, that is, for the case when the lower bound is a function of x as well.

Definition 3.12 — Indefinite Integral. The ***indefinite integral***

$$\int f(x)\,\mathrm{d}x$$

of $f(x)$ is the collection of all antiderivatives of the function f.

Example 3.13 (i) For

$$F(x) = \int_1^{x^3} t^5\,\mathrm{d}t,$$

find $F'(x)$. identifying $f(t) = t^5$ and using the formula from theorem 3.11,

$$\begin{aligned}F'(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\int_1^{x^3} f(t)\,\mathrm{d}t\\ &= f(x^3)\cdot\frac{\mathrm{d}}{\mathrm{d}x}\left(x^3\right) - f(1)\cdot\frac{\mathrm{d}}{\mathrm{d}x}(1)\\ &= \left(x^3\right)^5\cdot 3x^2 + 0 = 3x^{17}.\end{aligned}$$

We can check this by finding $F(x)$ explicitly,

$$F(x) = \int_1^{x^3} t^5 \, \mathrm{d}t = \left(\frac{t^6}{6} \Big|_1^{x^3} \right) = \frac{(x^3)^6}{6} - \frac{1}{6},$$

and then differentiating:

$$F'(x) = \frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{x^{18}}{6} - \frac{1}{6} \right) = 18 \cdot \frac{x^{17}}{6} - 0 = 3x^{17} \quad \checkmark$$

(ii)

$$\int \mathrm{e}^{3x} \, \mathrm{d}x = \frac{1}{3}\mathrm{e}^{3x} + c.$$

Here, we use the constant c as a place holder to cover all the infinitely many possible antiderivatives, e.g.

$$\frac{1}{3}\mathrm{e}^{3x}, \quad \frac{1}{3}\mathrm{e}^{3x} - 5, \quad \frac{1}{3}\mathrm{e}^{3x} + 117.38, \quad \ldots,$$

of $f(x) = \mathrm{e}^{3x}$.

Remark 3.14

We conclude this section with the following remarks:

(i) Integration is "the opposite of differentiation".

(ii) Integrals with boundaries are definite integrals, and then the result is a real number.

(iii) Integrals without boundaries are indefinite integrals. Make sure not to forget the constant of integration.

Exercise 3.15 (i) Work through all the material from the lectures and the practice problems on the tutorial sheet that corresponding to this section. Interpret the properties 3.1 – perhaps add a few sketches to your notes – and prove one or two of them.

(ii) Find the area function A_0 of $f(x) = x^3 - 6x$ with $a = 0$. Then set $a = 1$ and find the corresponding area function A_1. Compare A_0 and A_1 to each

other and to the indefinite integral of[a] $f(x)$.

(iii) Find the definite integral

$$\int_0^1 e^x \, dx$$

using Riemann sums[b], and then compare to the value found via direct integration.

(iv) Show that definite integral (3.1) is well-defined[c].

(v) Find the derivative of[d]

$$F(x) = \int_0^{x^2} x \, dt.$$

[a] Answer: $A_0(x)$ and $A_1(x)$ should differ by the constant $-11/4$ – where does this number come from? By choosing the constant of integration suitably, you should be able to obtain both A_0 and A_1 from the indefinite integral.

[b] Hint: You will need the formula for the sum of the first n terms of the geometric series.

[c] Hint: Cf. remark 3.3 for the meaning of 'well-defined" in this context. For fixed n, we obtain the largest possible Riemann sum by letting the c_j be the points at which f takes its maximal value on $[x_{j-1}, x_j]$. Similarly for the smallest possible Riemann sum. What is the difference between those two extremes, and what happens for $n \to \infty$?

[d] Hint: Careful, the FTC can not immediately by applied, because the x appears in the integrand as well. Hence pull out the x and then differentiate the product $x \cdot \int_0^{x^2} \ldots$. Note that there is a different way to solve this: find an explicit formula for F in terms of x – without an integral – and then differentiate. Try this approach as well and make sure your answers agree!

3.2 Basic Integrals

(1) $\displaystyle\int x^r \,\mathrm{d}x = \frac{x^{r+1}}{r+1} + c \qquad \text{for } r \neq -1,$

(2) $\displaystyle\int \frac{1}{x} \,\mathrm{d}x = \ln|x| + c,$

(3) $\displaystyle\int \mathrm{e}^x \,\mathrm{d}x = \mathrm{e}^x + c,$

(4) $\displaystyle\int a^x \,\mathrm{d}x = \frac{a^x}{\ln a} + c \qquad \text{for } a > 0,$

(5) $\displaystyle\int \cos x \,\mathrm{d}x = \sin x + c,$

(6) $\displaystyle\int \sin x \,\mathrm{d}x = -\cos x + c,$

(7) $\displaystyle\int \frac{1}{\cos^2 x} \,\mathrm{d}x = \tan x + c,$

(8) $\displaystyle\int \frac{1}{\sin^2 x} \,\mathrm{d}x = -\cot x + c,$

(9) $\displaystyle\int \frac{1}{1+x^2} \,\mathrm{d}x = \arctan x + c,$

(10) $\displaystyle\int \frac{1}{\sqrt{1-x^2}} \,\mathrm{d}x = \arcsin x + c,$

(11) $\displaystyle\int \cosh x \,\mathrm{d}x = \sinh x + c,$

(12) $\displaystyle\int \sinh x \,\mathrm{d}x = \cosh x + c.$

Remark 3.16

(i) The function $\arctan x$ is the inverse function of $\tan x$. It is sometimes denoted $\tan^{-1}$, but this carries potential for confusion, as $\arctan x \neq {}^1/_{\tan x}$! Similarly, for $\cos x$ and $\sin x$ and their inverse functions. The expressions appearing in (7) and (8) could be rewritten using the conventions

$$\sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}, \quad \cot x = \frac{\cos x}{\sin x}.$$

(ii) The linearity properties (3) and (4) of 3.1 remain true for indefinite integrals. Therefore, we can now integrate all linear combinations of functions

appearing in the integrands of properties 3.2.

(iii) All the integrals above are verified by differentiating the antiderivative on the right and comparing to the integrand on the left. For example,

$$\frac{\mathrm{d}}{\mathrm{d}x}(-\cos x + c) = \sin x,$$

and hence (6) is correct.

(iv) Formula (2) can be used to evaluate definite integrals such as $\int_1^2 {}^1\!/_x \,\mathrm{d}x$ or $\int_{-23}^{-5} {}^1\!/_x \,\mathrm{d}x$, but definite integrals across $x = 0$ are not permitted since $f(x) = {}^1\!/_x$ is not defined at $x = 0$. This means that in this context, x is either always positive or always negative. This allows to verify (2) by considering two cases:

Case (a): $x > 0$

If x is positive, then $|x| = x$ and

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln|x| + c) = \frac{\mathrm{d}}{\mathrm{d}x}\ln x = \frac{1}{x} \qquad \checkmark$$

Case (b): $x < 0$

If x is negative, then $|x| = -x$ and we obtain

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln|x| + c) = \frac{\mathrm{d}}{\mathrm{d}x}\ln(-x) = \frac{1}{-x}\cdot(-1) = \frac{1}{x}$$

as well.

Example 3.17 (i) $\int x^{11} + 5\cos x - 7^x \,\mathrm{d}x$

(ii) $\int_1^{\mathrm{e}} \frac{1}{x}\,\mathrm{d}x$

Remark 3.18

In the following example, recognising an expression coming from the chain rule allows us to "guess" an integral:

$$\int 2x\,\cosh(x^2)\,\mathrm{d}x = \sinh(x^2) + c.$$

The next theorem makes the application of this idea more systematic.

3.3 Substitution

Theorem 3.19 — Substitution. If $u = u(x)$ is differentiable and f continuous, then

(i)
$$\int f(u(x)) \cdot u'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u.$$

(ii)
$$\int_a^b f(u(x)) \cdot u'(x)\,\mathrm{d}x = \int_{u(a)}^{u(b)} f(u)\,\mathrm{d}u.$$

Example 3.20 (i) $\int x\left(1+x^2\right)^{13}\,\mathrm{d}x$

(ii) $\int_0^{\pi/2} \cos(\cos x)\sin x\,\mathrm{d}x$

(iii)

$$I = \int_0^{\sqrt{3}} \mathrm{e}^{\sqrt{1+x^2}} \frac{x}{\sqrt{1+x^2}}\,\mathrm{d}x \qquad \left[\begin{array}{rcl} u &=& \sqrt{1+x^2} \\ \mathrm{d}u &=& \dfrac{x}{\sqrt{1+x^2}}\mathrm{d}x \\ u=2 &\leftrightarrow& x=\sqrt{3} \\ u=1 &\leftrightarrow& x=0 \end{array}\right]$$

$$= \int_1^2 \mathrm{e}^u\,\mathrm{d}u$$

$$= \mathrm{e}^u\,|_1^2 = \mathrm{e}^2 - \mathrm{e}^1 = \mathrm{e}(\mathrm{e}-1).$$

Alternatively, one could leave the boundaries in terms of x and evaluate the antiderivative after substituting back to x:

$$I = \mathrm{e}^u\,|_{\cdot}^{\cdot} = \mathrm{e}^{\sqrt{1+x^2}}\Big|_0^{\sqrt{3}} = \mathrm{e}^{\sqrt{1+3}} - \mathrm{e}^{\sqrt{1}} = \mathrm{e}(\mathrm{e}-1).$$

In the integrands of the previous examples, we have seen the inner functions

$$1+x^2, \quad \cos x, \quad \sqrt{1+x^2},$$

to which the outer functions

$$x^{13}, \quad \cos x, \quad \mathrm{e}^x,$$

respectively, are applied. A first attempt for solving integrals of that kind should always be to substitute for the identified inner function. The question is then whether the other factors in the integrand are absorbed by the transformation of the differential $\mathrm{d}\cdot$. If not, then more work or a different approach is required.

Example 3.21 Compute the indefinite integral

$$I = \int \frac{1}{4+x^2}\,\mathrm{d}x.$$

Note that we could read off the answer immediately from the list in properties 3.2 if the 4 in the denominator was a 1. We

therefore bring the integral in that form using substitution:

$$\begin{aligned} I = \int \frac{1}{4+x^2}\,\mathrm{d}x &= \int \frac{1}{4\left(1+{}^{x^2}/_4\right)}\,\mathrm{d}x \\ &= \frac{1}{4}\int \frac{1}{1+\left({}^{x}/_2\right)^2}\,\mathrm{d}x \qquad \left[\begin{array}{rcl} u &=& {}^{x}/_2 \\ \mathrm{d}u &=& {}^{1}/_2\,\mathrm{d}x \end{array}\right] \\ &= \frac{1}{2}\int \frac{1}{1+u^2}\,\mathrm{d}u \\ &= \frac{1}{2}\arctan u + c = \frac{1}{2}\arctan\left(\frac{x}{2}\right) + c. \end{aligned}$$

3.4 Trigonometric Identities

Reminder: Properties of Trigonometric Identities

(1) $\cos^2 x + \sin^2 x = 1,$

(2) $\cos(x+y) = \cos x \cos y - \sin x \sin y,$

(3) $\sin(x+y) = \sin x \cos y + \cos x \sin y,$

(4) $\cos(2x) = 2\cos^2 x - 1 = 1 - 2\sin^2 x,$

(5) $\sin(2x) = 2\sin x \cos x,$

(6) $2\sin x \cos y = \sin(x-y) + \sin(x+y),$

(7) $2\cos x \cos y = \cos(x-y) + \cos(x+y),$

(8) $2\sin x \sin y = \cos(x-y) - \cos(x+y),$

(9) $\cosh^2 x - \sinh^2 x = 1,$

Example 3.22 (i)

$$\begin{aligned} \int_0^\pi \sin^2 x\,\mathrm{d}x = \int_0^\pi \sin x \cdot \sin x\,\mathrm{d}x &\overset{(8)}{=} \int_0^\pi \frac{1}{2}\left[\cos(x-x) - \cos(x+x)\right]\,\mathrm{d}x \\ &= \int_0^\pi \frac{1}{2}\left[\cos(0) - \cos(2x)\right]\,\mathrm{d}x = \frac{1}{2}\int_0^\pi 1 - \cos(2x)\,\mathrm{d}x \\ &= \frac{1}{2}\left(x - {}^{1}/_2\sin(2x)\Big|_0^\pi\right) = \frac{\pi}{2}. \end{aligned}$$

A better way to compute this integral would be to use the identity (4) – this allows to go directly from the first expression to the fifth.

(ii)

$$\begin{aligned}\int \cos^4 x \, \mathrm{d}x &= \int \left(\cos^2 x\right)^2 \, \mathrm{d}x \overset{(4)}{=} \int \left(\frac{\cos(2x)+1}{2}\right)^2 \, \mathrm{d}x \\ &= \int \frac{1}{4}\left(\cos^2(2x) + 2\cos(2x) + 1\right) \, \mathrm{d}x \\ &\overset{(4)}{=} \int \frac{1}{4}\left(\frac{\cos(4x)+1}{2} + 2\cos(2x) + 1\right) \, \mathrm{d}x \\ &= \frac{1}{32}\sin(4x) + \frac{1}{4}\sin(2x) + \frac{3}{8}x + c.\end{aligned}$$

(iii)

$$\int \sin(2x)\sin(7x) \, \mathrm{d}x \overset{(8)}{=} \int \frac{1}{2}\left[\cos(-5x) - \cos(9x)\right] \, \mathrm{d}x = \frac{\sin(5x)}{10} - \frac{\sin(9x)}{18} + c.$$

(iv) The following example contains a typical substitution with trigonometric functions, and some comments on it will be made in the next remark.

$$\int_{2.5}^{5} \frac{\sqrt{25-x^2}}{x^2} \, \mathrm{d}x \qquad \left[\begin{array}{rcl} x &=& 5\sin\theta \\ \mathrm{d}x &=& 5\cos\theta \, \mathrm{d}\theta \\ x = 5 &\leftrightarrow& \theta = {}^{\pi}/_{2} \\ x = 2.5 &\leftrightarrow& \theta = {}^{\pi}/_{6} \end{array}\right]$$

$$\begin{aligned} &= \int_{{}^{\pi}/_{6}}^{{}^{\pi}/_{2}} \frac{\sqrt{25-(5\sin\theta)^2}}{(5\sin\theta)^2} 5\cos\theta \, \mathrm{d}\theta \\ &= \int_{{}^{\pi}/_{6}}^{{}^{\pi}/_{2}} \frac{\sqrt{1-\sin^2\theta}}{\sin^2\theta} \cos\theta \, \mathrm{d}\theta \overset{(1)}{=} \int_{{}^{\pi}/_{6}}^{{}^{\pi}/_{2}} \frac{\cos^2\theta}{\sin^2\theta} \, \mathrm{d}\theta \\ &\overset{(1)}{=} \int_{{}^{\pi}/_{6}}^{{}^{\pi}/_{2}} \frac{1-\sin^2\theta}{\sin^2\theta} \, \mathrm{d}\theta = \int_{{}^{\pi}/_{6}}^{{}^{\pi}/_{2}} \frac{1}{\sin^2\theta} - 1 \, \mathrm{d}\theta \\ &= -\cot\theta - \theta \,\big|_{{}^{\pi}/_{6}}^{{}^{\pi}/_{2}} = -0 - \frac{\pi}{2} + \frac{{}^{\sqrt{3}}/_{2}}{{}^{1}/_{2}} + \frac{\pi}{6} = \sqrt{3} - \frac{\pi}{3}. \end{aligned}$$

(v)

$$\int \frac{1}{\sqrt{x^2-a^2}} \, \mathrm{d}x \qquad \left[\begin{array}{rcl} x &=& a\cosh t \\ \mathrm{d}x &=& a\sinh t \, \mathrm{d}t \end{array}\right]$$

$$\begin{aligned} &= \int \frac{a\sinh t}{\sqrt{(a\cosh t)^2 - a^2}} \, \mathrm{d}t \\ &\overset{(9)}{=} \int 1 \, \mathrm{d}t = t + c = \operatorname{arccosh}\left(\frac{x}{a}\right) + c. \end{aligned}$$

Remark 3.23

(i) Note the similarities in the approaches for (iv) and (v) above, and also note the differences: In both cases, we have square roots of expressions $\pm(a^2 - x^2)$ in the integrand, and we use the trigonometric Pythagoras formula (1) of properties 3.4 and the corresponding hyperbolic version (9) to simplify them. The following thoughts might help you to avoid confusion of the two. For $\sqrt{a^2 - x^2}$, we need x to be small, namely in the range $[0, a]$, and the trigonometric functions have a bounded range. For $\sqrt{x^2 - a^2}$, we need x to be large, and the function $g(t) = a\cosh t$ has the correct range, $R(g) = [a, \infty)$.

(ii) For our next integration method, we apply the product rule to differentiate the product fg, and then bring one of the terms on the other side,

$$f'g = (fg)' - fg'.$$

Integrating this expression with respect to x, we obtain the following result.

3.5 Integration by Parts

Theorem 3.24 — Integration by Parts.

$$\int f'(x)g(x)\,\mathrm{d}x = f(x)g(x) - \int f(x)g'(x)\,\mathrm{d}x.$$

Example 3.25 (i) The integration by parts formula allows us to find an important integral that was missing in properties 3.2 – the integral of the logarithm: $\int \ln x\,\mathrm{d}x$

(ii)

$$\int x^2 e^{3x}\,dx$$

(iii)

$$\int x(1+x)^{17}\,dx$$

(iv)

$$I = \int e^x \cos x\,dx$$

Remark 3.26

(i) When faced with an integrand that contains a power of x and other, more complicated terms, one should first try substitution. If this does not work, integration by parts is the next best option. In this

case, one would usually choose the power of x as the term that is to be differentiated – the reason for this is that differentiation makes powers simpler and repeated integration by parts will eventually dispose of them. However, there are no firm rules for integration and one should remain open-minded to non-standard approaches.

(ii) Many of the examples above are specifically designed so that the computations work out relatively smoothly. In general, integration is quite hard and sometimes even impossible. It is therefore important to keep in mind that differentiation provides a straight-forward way to verify the integrals you have found. Here is an integral that cannot be solved:

$$\int \mathrm{e}^{-x^2}\,\mathrm{d}x.$$

(iii) Consider the computation

$$\frac{2}{x+4}-\frac{3}{x-1}=\frac{2(x-1)-3(x+4)}{(x+4)(x-1)}=\frac{-(x+14)}{x^2+3x-4},$$

and note that, while we can integrate the expression on the left-hand side – e.g.,

$$\int \frac{2}{x+4}\,\mathrm{d}x = 2\ln(x+4)+c$$

– the expression on the right is not covered by the integration methods we have developed so far. Reversing finding the common denominator therefore extends the set of functions we can integrate. This is called integration by partial fractions.

3.6 Partial Fractions

Example 3.27 (i) Find the integral

$$I=\int \frac{2x+16}{x^2+2x-35}\,\mathrm{d}x.$$

Sol.:

$$\frac{2x+16}{x^2+2x-35} = \frac{2x+16}{(x-5)(x+7)} = \frac{A}{x-5} + \frac{B}{x+7}$$
$$= \frac{A(x+7)+B(x-5)}{(x-5)(x+7)} = \frac{(A+B)x+(7A-5B)}{(x-5)(x+7)}$$

$$\implies \begin{cases} A+B & = 2 \\ 7A-5B & = 16 \end{cases} \implies \begin{cases} A & = {}^{13}/_{6} \\ B & = {}^{-1}/_{6} \end{cases}$$

$$\implies I = \int \frac{2x+16}{x^2+2x-35}\,\mathrm{d}x = \int \frac{{}^{13}/_{6}}{x-5} + \frac{{}^{-1}/_{6}}{x+7}\,\mathrm{d}x$$
$$= \frac{13}{6}\int \frac{1}{x-5}\,\mathrm{d}x - \frac{1}{6}\int \frac{1}{x+7}\,\mathrm{d}x = \frac{13}{6}\ln|x-5| - \frac{1}{6}\ln|x+7| + c.$$

(ii) Integrate

$$f(x) = \frac{x^4-x^3-3x^2+x-2}{x^3+9x}$$

over the interval $[1,2]$.

Sol.: Before we can "split up" the fraction as above, we have to bring it in a form in which the degree of the numerator is smaller than the degree of the denominator. The idea for doing that is

$$\frac{x^4-x^3-3x^2+x-2}{x^3+9x} = \frac{x^4 \,+9x^2-9x^2\, -x^3-3x^2+x-2}{x^3+9x}$$
$$= \frac{x^4+9x^2}{x^3+9x} + \frac{-9x^2-x^3-3x^2+x-2}{x^3+9x}$$
$$= x + \frac{-x^3-12x^2+x-2}{x^3+9x} = \ldots$$

This is long division of polynomials, and it is usually written out more systematically as

$$\begin{array}{rrrrrrcrcl} (& x^4 & -x^3 & -3x^2 & +x & -2 &) & : & (x^3+9x) & = \\ -(& x^4 & & +9x^2 & &) & & & & \\ & (& -x^3 & -12x^2 & +x & -2 &) & & & \\ & -(& -x^3 & & -9x &) & & & & \\ & & (& -12x^2 & +10x & -2 &) & & & \end{array} \quad x-1+\frac{-12x^2+10x-2}{x^3+9x}.$$

Next, we split up the remaining fraction, as in the example

above:

$$\frac{-12x^2+10x-2}{x^3+9x} = \frac{-12x^2+10x-2}{x(x^2+9)} = \frac{A}{x}+\frac{Bx+C}{x^2+9}$$
$$= \frac{A(x^2+9)+(Bx+C)x}{x(x^2+9)} = \frac{(A+B)x^2+Cx+9A}{x(x^2+9)}.$$

General guidelines for how to make the ansatz involving the constants A, B, C in the middle expression will be given in the remark below. The next step is to compare coefficients in the numerator on the left and on the right:

$$\begin{cases} A+B &= -12 \\ C &= 10 \\ 9A &= -2 \end{cases} \implies \begin{cases} A &= {}^{-2}/_{9} \\ B &= {}^{-106}/_{9} \\ C &= 10. \end{cases}$$

We can now write the definite integral as a sum of simpler integrals:

$$I = \int_1^2 \frac{x^4-x^3-3x^2+x-2}{x^3+9x}\,\mathrm{d}x = \int_1^2 x-1+\frac{{}^{-2}/_{9}}{x}+\frac{{}^{-106}/_{9}\,x+10}{x^2+9}\,\mathrm{d}x$$
$$= \int_1^2 x\,\mathrm{d}x - \int_1^2 1\,\mathrm{d}x - \frac{2}{9}\int_1^2 \frac{1}{x}\,\mathrm{d}x + \int_1^2 \frac{{}^{-106}/_{9}\,x+10}{x^2+9}\,\mathrm{d}x.$$

The first integrals are basic integrals, denote them I_1, I_2, I_3.

The fourth we split up further as

$$I = I_1+I_2+I_3 - \frac{53}{9}\int_1^2 \frac{2x}{x^2+9}\,\mathrm{d}x + 10\int_1^2 \frac{1}{x^2+9}\,\mathrm{d}x,$$

so that the fourth integral can be computed with a straightforward substitution, and the last one as in example 3.21.

Computing the five definite integrals and adding them together gives the answer,

$$I = \frac{1}{2} - \frac{2}{9}\ln(2) - \frac{53}{9}\ln(1.3) + \frac{10}{3}\left(\arctan({}^{2}/_{3}) - \arctan({}^{1}/_{3})\right) \approx -0.312.$$

(iii) Find the integral

$$I = \int \frac{x-{}^{3}/_{2}}{x^2-3x+7}\,\mathrm{d}x.$$

Sol.: Flexibility is key for integration – this is a substitu-

tion problem:

$$I = \frac{1}{2}\int \frac{2x-3}{x^2-3x+7}\,\mathrm{d}x = \frac{1}{2}\int \frac{1}{u}\,\mathrm{d}u$$
$$= \frac{1}{2}\ln|u| + c = \frac{1}{2}\ln\left|x^2-3x+7\right| + c.$$

There are examples where both approaches – the substitution here in (iii), and the partial fraction approach in (i) – can be used, cf. the exercise below. However, for the problem at hand, the partial fraction approach would not work, as the denominator x^2-3x+7 can not be factored.

The steps for integration by partial fractions are:

(1) If necessary, use polynomial division to transform to a polynomial plus a fraction in which the degree of the numerator is smaller than degree of the denominator.

(2) Write the denominator as a product of irreducible factors. Each of them will have degree at most two. In this module, we only consider the case where none of these factors is repeated.

(3) For each factor, define a fraction that has that factor in the denominator and a general polynomial of degree one less in the numerator. That is, if the denominator is a linear expression (polynomial of degree 1), then the numerator is just a constant c (polynomial of degree 0). If the denominator is a quadratic expression (polynomial of degree 2), then the numerator should be of the form $c_1x + c_2$ (polynomial of degree 1).

(4) Make the ansatz that the sum of the fractions from the previous step is equal to the original fraction. This will lead to a system of equations for all the constants appearing in the numerators defined in the previous step. Solve it.

(5) In the computation of the integral, replace the original integrand with the sum of partial fractions you have found, plus

possibly the polynomial that was obtained in step (1). Now use the linearity of the integral to obtain a combination of simple integrals that can be solved with the methods we have seen in this section. (In general, it is possible to obtain integrals at this point that can not be solved with the methods we have learned so far – but there will not be any such examples in the exam.)

Exercise 3.28 (i) We recommend to practise integration intensively by working through a large number of examples. This is important as integration is a very fundamental skill for all branches of mathematics, engineering, and other sciences. Besides the material from the lectures and the tutorial sheet, you can look for more practice examples in books and online. You can even make up you own examples and check your results using differentiation. Mathematics software such as WolframAlpha can compute many integrals, but *do not rely on this*.

(ii) Make sure to complete all the steps in this section that were only sketched, e.g. the last lines of 3.27 (ii). Following the comment at the end of 3.27 (iii), compute

$$\int \frac{x+1}{x^2+2x-3}\,\mathrm{d}x$$

in two different ways.

(iii) Using integration, find the area inside the circle of radius[a] R.

(iv) Find[b]

$$\int_{\pi/6}^{\pi/4} \frac{\cos 2x}{\cos^2 x \sin^2 x}\,\mathrm{d}x.$$

(v) Find[a]

$$\int (1+|x|)^2\,\mathrm{d}x.$$

[a] Hint: First, you need a function whose graph is the circle or part of the circle – use the formula $x^2+y^2=R^2$ for that.

[b] Answer: The value you obtained is correct if it rounds to 0.309.

[a] Hint: It is useful to consider the area function with $a=0$, $A(x)=\int_0^x (1+|t|)^2\,\mathrm{d}t$.

3.7 Improper Integrals

Definition 3.29 — Improper Integral. A definite integral $\int_a^b f(x)\,\mathrm{d}x$ is called ***improper*** if

(I) the interval of integration is infinite, i.e. $a=-\infty$, or $b=\infty$, or both; or if

(II) the interval of integration contains a ***singularity*** of f, that is, a point where f is not defined, e.g. a zero of the denominator.

If $b=\infty$ (type I) or b is a singularity (type II), then

$$\int_a^b f(x)\,\mathrm{d}x := \lim_{\substack{t\to b\\ t<b}} \int_a^t f(x)\,\mathrm{d}x,$$

and the improper integral is said to exist / not exist depending on whether the limit on the right-hand side exists.

Similarly if a is the boundary that causes the integral to be improper. If both boundaries are infinite (type I) or if the singularity lies inside the interval (type II), then the improper integral should be split up, cf. example 3.32 (i) below.

Example 3.30 [Type I]

(i) $\displaystyle\int_{-\infty}^0 \mathrm{e}^{2x}\,\mathrm{d}x$

(ii) $\displaystyle\int_1^\infty \frac{1}{x}\,\mathrm{d}x$

Remark 3.31

(i) The first example states that the area under the curve

$$f(x) = e^{2x} \qquad x \leq 0$$

is finite! Here, the word "area" refers to the total amount of area – as in "square footage" – rather than the set of points under that curve. The latter is an infinitely long "spike".

(ii) Improper integrals of type I are similar to series, and (ii) corresponds to the fact that the harmonic series $\sum 1/n$ diverges.

(iii) Recalling the definition of the area function $A(x)$, we see that the limit that is to be found for an improper integral of a continuous function on $[a, \infty)$ is that of the area function:

$$\int_a^\infty f(x)\,dx = \lim_{R\to\infty} A(R).$$

Example 3.32 [Type II]

(i) Integrate $f(x) = 1/\sqrt[3]{x^2}$ over the interval $[-1, +1]$.

(ii) $\displaystyle\int_0^{\pi/2} \frac{1}{\sin^2 x}\,dx$

Exercise 3.33 (i) As usual: Work through the material from the lectures and the improper integrals on the tutorial sheet.

(ii) For which powers $p \in \mathbb{R}$ do the integrals

$$\int_0^1 x^p \, \mathrm{d}x, \qquad \int_1^\infty x^p \, \mathrm{d}x$$

exist[a]?

(iii) Evaluate the improper integral[b]

$$I_n = \int_0^\infty x^n \mathrm{e}^{-x} \, \mathrm{d}x.$$

[a] Answer: Taking the union of the set of p values, for which the first improper exists, with the set of p's for the second, you should obtain the whole real line with only one point missing.

[b] Hint: Compute I_0 directly and use integration by parts to find a recursive formula for I_n. Can you derive an explicit formula!

3.8 Integrals of Functions of Several Variables

Example 3.34 As an introduction to this section, we compute a simple double-integral. That is done systematically starting from the innermost integral. In the following example, a $\mathrm{d}x$ integral is to be found first. For this step, the other variable, y, is treated like a constant, as it is not the variable with respect to which the current integration is carried out:

$$I = \int_0^1 \int_0^1 x + y \, \mathrm{d}x \, \mathrm{d}y$$

Remark 3.35

(i) Just as the integral $\int_a^b f(x)\,\mathrm{d}x$ is the area between the curve $y = f(x)$ and the interval $[a,b]$ of the x-axis, the double integral

$$\int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x$$

is the volume between the surface $z = f(x,y)$ and the rectangle

$$[a,b]\times[c,d] = \{(x,y)\in\mathbb{R}^2 \,|\, a\le x\le b,\, c\le y\le d\}$$

of the xy-plane.

(ii) To check whether this interpretation agrees with the result $I = 1$ we have found above, note that the graph of the integrand $f(x,y) = x+y$ is a plane and has heights

$$f(0,0) = 0, \quad f(0,1) = 1, \quad f(1,0) = 1, \quad f(1,1) = 2,$$

over the corner points of the domain of integration, $D = [0,1]\times[0,1]$. This means that the volume that is to be found is that of a cuboid of size $1\times 1\times 2$ that is diagonally cut in half. This solid has the volume

$$V = \frac{1\cdot 1\cdot 2}{2} = 1,$$

which we had also obtained with the integration above.

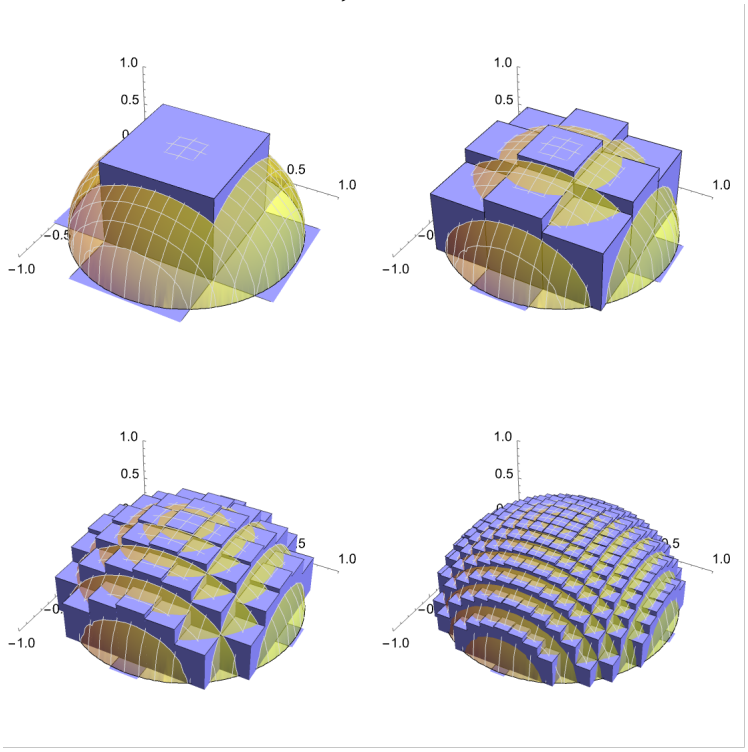
(iii) We write $\mathrm{d}A$ for $\mathrm{d}x\mathrm{d}y$,

$$\mathrm{d}A = \mathrm{d}x\mathrm{d}y,$$

meaning, roughly, that the change of area is equal to the change of x times the change of y. $\mathrm{d}A$ is

called the *area element*. The order of the integrations $\mathrm{d}x$ and $\mathrm{d}y$ can be swapped, but one needs to be careful about the boundaries, cf. later examples.

(iv) There also is a formulation of Riemann sums for functions of several variables. We will not address this further in MTH1002, but the idea is as follows:



Example 3.36 Integrate the function $f(x,y) = {}^{y}/_{x}$ over the domain $D = [3,6] \times [1,2]$.

All domains of integration so far have been rectangles. In this case, the x and y boundaries are constant, and the order of integration can be swapped easily – convince yourself of that by

re-doing one of the problems above integrating with respect to y and then w.r.t. x. Next we study non-rectangular domains of integration, for which the boundaries of the inner integral depend on the outer variable.

Example 3.37

$$I = \int_0^1 \int_0^{x^2} 1 \, \mathrm{d}y \, \mathrm{d}x = \int_0^1 \left[\int_0^{x^2} 1 \, \mathrm{d}y \right] \mathrm{d}x$$
$$= \int_0^1 \left(y|_0^{x^2} \right) \mathrm{d}x = \int_0^1 \left(x^2 - 0 \right) \mathrm{d}x = \int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3}.$$

Remark 3.38

(i) In the previous example, the domain of integration was

$$D = \{(x,y) \mid 0 \leq x \leq 1, \, 0 \leq y \leq x^2\}.$$

Integrating the function $f(x,y) = 1$ over a subset of $\mathbb{R}^2$ gives the area of that set. This is similar to 1D integration: integrating $f(x) = 1$ over an interval gives the length of that interval.

(ii) Our choice to integrate first w.r.t. y and then w.r.t. x in the previous example corresponds to the following steps: (1) for each $x \in [0,1]$, integrate over each vertical line in the sketch, then (2) collect those values in the x direction. We could also find $I = \iint_D 1 \, \mathrm{d}A$ by integrating the other way around. For this, one has to solve the equations that define the boundaries for the other variable:

$$I = \int_0^1 \int_{\sqrt{y}}^1 1 \, \mathrm{d}x \, \mathrm{d}y$$

– check that this gives the same result.

Example 3.39 Integrate $f(x,y) = y\sin x$ over the triangle D with corner points $(0,0)$, $(\pi,0)$, and $(\pi,1)$.

Sol.:

It is helpful to always start a computation of a 2D integral with a sketch of the domain of integration. Labelling its boundaries with the formulas that describe them helps to correctly write out $\iint_D f(x,y)\,\mathrm{d}A$ as $\int_{\cdot}^{\cdot}\int_{\cdot}^{\cdot} f(x,y)\,\mathrm{d}y\,\mathrm{d}x$. The integration can be carried out in either order, but the next example and the corresponding exercise below show that for some domains, one order of integration is easier than the other.

Example 3.40 (i) Let D be the region bounded by the line $y = x+1$ and by the parabola $y = x^2 - 1$. Find

$$I = \iint_D xy + 2\,\mathrm{d}A.$$

Sol.: The domain of integration is the region between the line and the parabola below. We find the intersection points as follows:

$$\begin{aligned} x+1 &\stackrel{!}{=} x^2 - 1 \\ \rightarrow \quad 0 &= x^2 - x - 2 \\ \Longrightarrow \quad x_1 &= -1, \quad x_2 = 2. \end{aligned}$$

Let us integrate with respect to x first and then w.r.t. y; that is, the outside integral is w.r.t. y. The sketch shows that y ranges from $y = -1$ to $y = 3$. Note that the

right x boundary is $x = +\sqrt{y+1}$ for any $y \in [-1,3]$. However, for the left x boundary, we have to distinguish two cases: $x = -\sqrt{y+1}$ for any $y \in [-1,0]$, and $y = x+1 \;\leftrightarrow\; x = y-1$ for any $y \in [0,3]$. We therefore split D along $y = 0$ to obtain

$$
\begin{aligned}
I &= \iint_{D_-} xy + 2 \,\mathrm{d}A + \iint_{D_+} xy + 2 \,\mathrm{d}A \\
&= \int_{-1}^{0} \int_{-\sqrt{y+1}}^{+\sqrt{y+1}} xy + 2 \,\mathrm{d}x \,\mathrm{d}y + \int_{0}^{3} \int_{y-1}^{+\sqrt{y+1}} xy + 2 \,\mathrm{d}x \,\mathrm{d}y \\
&= \int_{-1}^{0} \left(\frac{x^2 y}{2} + 2x \Big|_{-\sqrt{y+1}}^{+\sqrt{y+1}} \right) \mathrm{d}y + \int_{0}^{3} \left(\frac{x^2 y}{2} + 2x \Big|_{y-1}^{\sqrt{y+1}} \right) \mathrm{d}y \\
&= \int_{-1}^{0} \frac{(y+1)y}{2} + 2\sqrt{y+1} - \frac{(y+1)y}{2} + 2\sqrt{y+1} \,\mathrm{d}y \\
&\qquad + \int_{0}^{3} \frac{(y+1)y}{2} + 2\sqrt{y+1} - \frac{(y-1)^2 y}{2} - 2y + 2 \,\mathrm{d}y \\
&= \int_{-1}^{0} 4\sqrt{y+1} \,\mathrm{d}y + \int_{0}^{3} \frac{-y^3 + 3y^2 - 4y + 4}{2} + 2\sqrt{y+1} \,\mathrm{d}y \\
&= \frac{8}{3} \left((y+1)^{3/2} \Big|_{-1}^{0} \right) + \frac{1}{2} \left(-\frac{y^4}{4} + \frac{3y^3}{3} - \frac{4y^2}{2} + 4y \Big|_{0}^{3} \right) + \frac{4}{3} \left((y+1)^{3/2} \Big|_{0}^{3} \right) \\
&= \left(\frac{8}{3} - 0 \right) + \frac{1}{2} \left(-\frac{3^4}{4} + 3^3 - 2 \cdot 3^2 + 4 \cdot 3 - 0 \right) + \frac{4}{3} \left(4^{3/2} - 1 \right) \\
&= \frac{8}{3} + \frac{3}{8} + \frac{28}{3} = \frac{99}{8}.
\end{aligned}
$$

(ii) For $D = [0,2] \times [0,1]$, find

$$I = \iint_D \left| (x+y)^2 - 1 \right| \,\mathrm{d}A.$$

Sol.: Let $f(x,y) = (x+y)^2 - 1$. In order to integrate its *absolute value* over D, one first has to split D into the subset D_+ on which f is positive and the subset D_- where it is negative, and then use the definition of the absolute value,

$$|f| = \begin{cases} f, & \text{when } f \geq 0, \\ -f, & \text{when } f < 0. \end{cases}$$

The boundary between these subsets is the level set $f =$

0:

$$(x+y)^2-1=0 \quad\Longrightarrow\quad \begin{cases} y_1 &= 1-x, \\ y_2 &= -1-x, \end{cases}$$

but only y_1 intersects D. This gives

$$\begin{aligned} I &= \iint_{D_-} -\left[(x+y)^2-1\right]\,\mathrm{d}A + \iint_{D_+} +\left[(x+y)^2-1\right]\,\mathrm{d}A \\ &= \int_0^1\int_0^{1-y} 1-(x+y)^2\,\mathrm{d}x\,\mathrm{d}y + \int_0^1\int_{1-y}^2 (x+y)^2-1\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_0^1 \left(x-\frac{(x+y)^3}{3}\Big|_0^{1-y}\right)\mathrm{d}y + \int_0^1 \left(\frac{(x+y)^3}{3}-x\Big|_{1-y}^2\right)\mathrm{d}y \\ &= \int_0^1 1-y-\frac{1^3}{3}-0+\frac{y^3}{3}+\frac{(2+y)^3}{3}-2-\frac{1^3}{3}+1-y\,\mathrm{d}y \\ &= \int_0^1 -\frac{2}{3}-2y+\frac{y^3}{3}+\frac{(2+y)^3}{3}\,\mathrm{d}y = \cdots = \frac{23}{6}. \end{aligned}$$

Note that you do not need to multiply out $(2+y)^3/3$ to integrate it: similar to the integration of $(x+y)^2$ earlier in the computation, its antiderivative is $(2+y)^4/4\cdot 3$ – check using differentiation in case you have doubts!

Exercise 3.41 (i) Work through the material from the lectures and the integrals on the tutorial sheet that correspond to this section.

(ii) Complete the computation in 3.39. Also re-do example 3.40 (i), now integrating the other way around: first w.r.t. y and then w.r.t. x – this is a very important exercise, as it provides crucial insight on how to best choose the order of integration.

(iii) Using integration, find the volume of the pyramid with base $[-2,2]\times[-2,2]$ and height[a] $h=3$. Compare your result to the volume obtained with geometry formulas.

(iv) Let $f(x)$ be continuous on $[0,1]$ with

$$\int_0^1 f(x)\,\mathrm{d}x = \alpha.$$

Find[a]

$$I = \int_0^1 \int_x^1 f(x)\cdot f(y)\,\mathrm{d}y\,\mathrm{d}x.$$

[a] Hint: The pyramid is quite symmetric – this allows to consider a sub-domain S of the base and later multiply the volume that is sitting on top of it by a suitable factor. Choose S so that the walls of the pyramid do not have any edges over it – i.e. the restriction of the shell to S is a plane!

[a] Hint: Sketch the domain and the smallest rectangle that contains it, and argue that the integrand $f(x)\cdot f(y)$ integrates to $2I$ over the rectangle.

3.9 Change of Variables and Integration in Polar Coordinates

Remark 3.42

(i) The next theorem states a change-of-variables formula for two-dimensional integrals, and it is very similar to theorem 3.19. It is first stated in full generality, (i), and then for the special case when the transformation of variables is that from polar coordinates to Cartesian coordinates, (ii). The derivation of (ii) from (i) is given as an exercise, but both this derivation and (i) itself are not examinable. You do have to be able to integrate in polar coordinates though, i.e., you do have to know (ii).

(ii) Suppose we need to find the volume between the paraboloid $z = f(x,y) = x^2+y^2$ and the disk $D = \{x^2+y^2 \le a^2\}$ in the xy-plane. Note that both the function and the domain of integration are simpler in polar coordinates, $(x,y) = (r\cos\theta, r\sin\theta)$:

$$\begin{aligned} f(x,y) = x^2+y^2 &= (r\cos\theta)^2 + (r\sin\theta)^2 \\ &= r^2\left(\cos^2\theta + \sin^2\theta\right) = r^2, \end{aligned}$$

and D corresponds to a rectangular domain in po-

lar coordinates,

$$(x,y) \in D \quad \leftrightarrow \quad (r,\theta) \in [0,a] \times [0,2\pi].$$

You know from the previous section that 2D integrals over rectangular domains are the easier ones, and the next theorem allows us to "pull back" the integrand f and the domain of integration D into that setting.

Theorem 3.43 — Substitution in Higher Dimensions. (i) Let a differentiable and injective (or "one-to-one") transformation

$$\begin{array}{rccccc} \Phi & : & U & \to & \mathbb{R}^2 & (U \subseteq \mathbb{R}^2) \\ & & (u,v) & \mapsto & (x,y) & \end{array}$$

and a continuous function $f = f(x,y)$ be given. Then we have

$$\iint\limits_{\Phi(U)} f(x,y)\,\mathrm{d}x\mathrm{d}y = \iint\limits_{U} f(\Phi(u,v))\,\left|\det J_\phi(u,v)\right|\,\mathrm{d}u\mathrm{d}v,$$

where J_Φ is the *Jacobian* of the transformation $(u,v) \rightsquigarrow (x,y)$,

$$J_\Phi = \begin{bmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{bmatrix}.$$

(ii) Consider the transformation

$$\begin{array}{rccccc} \Phi & : & \mathbb{R}_0^+ \times [0,2\pi] & \to & \mathbb{R}^2 & \\ & & (r,\theta) & \mapsto & (x,y) & = (r\cos\theta, r\sin\theta) \end{array}$$

from polar coordinates to Cartesian coordinates, and let a continuous function $f = f(x,y)$ be given. Then we have for some set U in the domain of Φ that

$$\iint\limits_{\Phi(U)} f(x,y)\,\mathrm{d}x\mathrm{d}y = \iint\limits_{U} f(r\cos\theta, r\sin\theta)\,r\,\mathrm{d}r\mathrm{d}\theta.$$

That is, the area element is

$$\mathrm{d}A = r\,\mathrm{d}r\mathrm{d}\theta$$

in polar coordinates.

Remark 3.44

You may have noticed that the transformation $\Phi : \mathbb{R}_0^+ \times [0, 2\pi] \to \mathbb{R}^2$ in (ii) above is not injective. For example, all points $(r, \theta) = (0, \theta)$ are mapped to $(x, y) = (0, 0)$, and $(r, 0), (r, 2\pi)$ are mapped to the same points $(x, y) = (r, 0)$. This is not problematic, as the sets on which this non-injectiveness happens are of lower dimension and therefore do not contribute to the integral. Also, the choice of domain for the angle θ is flexible and can more generally be taken as $[\theta_0, \theta_0 + 2\pi]$, where $\theta_0 \in \mathbb{R}$, cf. example (ii) below.

Example 3.45 (i) Find

$$I = \iint_D x^2 + y^2 \,\mathrm{d}A,$$

where D is the disk of radius a, $D = \{x^2 + y^2 \leq a^2\}$.

Sol.:

(ii) Let

$$D = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0,\ |y| \leq x,\ 9 \leq x^2 + y^2 \leq 25\},$$

and find $\iint_D x \, \mathrm{d}A$.

Sol.:

(iii) Find the area A enclosed by one loop of the four-leaved rose $r = \cos(2\theta)$.

Example 3.46 Find the volume within the ball of radius R in $\mathbb{R}^3$ (centred at the origin).

Sol.: Denote that ball by B_R,

$$B_R = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq R^2\} \subseteq \mathbb{R}^3.$$

Besides using a basic geometry formula, there are two approaches to finding the volume of B_R: (1) compute the volume of the upper hemisphere by integrating the 2D function whose graph forms the surface of the upper hemisphere over the the disk D_R of radius R, and then multiply by 2; or (2) find the integral of the 3D function $f(x,y,z) = 1$ over B_R. In maths terms:

$$(1) \qquad V = 2 \iint_{D_R} \sqrt{R^2 - x^2 - y^2}\, \mathrm{d}A,$$

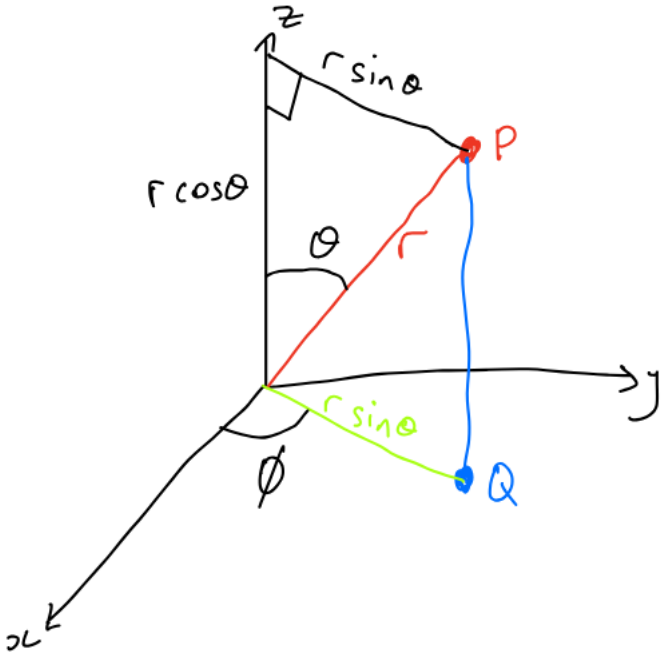
$$(2) \qquad V = \iiint_{B_R} 1\, \mathrm{d}V.$$

(1) can be carried out nicely in polar coordinates, but we will take the second approach now, as it is a good opportunity to apply the general form of theorem 3.43.

For this, we need *spherical polar coordinates*, i.e. a form of polar coordinates for $\mathbb{R}^3$. You do not need to know this transformation for the MTH1002 exam, but it is good to preview it for your second year in the programme:

$$\begin{array}{rcccl} \Phi & : & \mathbb{R}_0^+ \times [0,2\pi] \times [0,\pi] & \to & \mathbb{R}^3 \\ & & \begin{pmatrix} r \\ \phi \\ \theta \end{pmatrix} & \mapsto & \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r\cos\phi\sin\theta \\ r\sin\phi\sin\theta \\ r\cos\theta \end{pmatrix} \end{array}$$

For example, for fixed $r = r_0$, the points (x,y,z) will traverse the surface of the ball of radius r_0, with $\theta = 0$ corresponding to the north pole, $\theta = \pi$ to the south pole, and $\theta = {}^{\pi}/_{2}$ to the



equator.

The Jacobian for this transformation is

$$J_\Phi = \begin{bmatrix} \cos\phi\sin\theta & -r\sin\phi\sin\theta & r\cos\phi\cos\theta \\ \sin\phi\sin\theta & r\cos\phi\sin\theta & r\sin\phi\cos\theta \\ \cos\theta & 0 & -r\sin\theta \end{bmatrix},$$

whose determinant has an absolute value of $|\det J_\Phi| = r^2\sin\theta$. Hence the *volume element* (the 3D version of the area element) is

$$\mathrm{d}V = r^2\sin\theta\,\mathrm{d}r\mathrm{d}\phi\mathrm{d}\theta.$$

This allows to find the volume of the ball of radius R with 3D integration:

$$\begin{aligned} V &= \iiint_{B_R} 1\,\mathrm{d}V \\ &= \int_0^\pi \int_0^{2\pi} \int_0^R r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}\theta \\ &= \int_0^\pi \int_0^{2\pi} \sin\theta \left(\frac{r^3}{3}\Big|_0^R\right) \mathrm{d}\phi\,\mathrm{d}\theta \\ &= \frac{R^3}{3}\int_0^\pi \sin\theta \int_0^{2\pi} 1\,\mathrm{d}\phi\,\mathrm{d}\theta \\ &= \frac{2\pi R^3}{3}\int_0^\pi \sin\theta\,\mathrm{d}\theta = \frac{2\pi R^3}{3}\left(-\cos\theta|_0^\pi\right) = \frac{4\pi}{3}R^3. \end{aligned}$$

Exercise 3.47 (i) As usual, work through the material from the lectures and the corresponding problems on the tutorial sheet. You might also want to review polar coordinates using the material from MTH1000 – e.g., you can practise converting to and from polar coordinates, working with polar curves, and identifying polar domains using the interactive worksheet "Polar coordinates".

(ii) Evaluate

$$\iint_D \left| \sin\left(\sqrt{x^2+y^2}\right) \right| \, \mathrm{d}A,$$

where D is the upper half of the disk of radius[a] 2π,

$$D = \{y \geq 0,\, x^2 + y^2 \leq (2\pi)^2\}.$$

(iii) In theorem 3.43, derive the special case (ii) from the general formula (i).

(iv) Evaluate the integral

$$\iiint_V r \, \mathrm{d}V,$$

where V is the volume between the the two spheres of radius 1 and 2 in $\mathbb{R}^3$ (both centred at the origin)[b].

[a] Answer: The value you obtained is correct if it rounds to 39.478.

[b] Hint: You will need the general version of theorem 3.43 and "spherical polar coordinates" for this.
Answer: The value you obtained is correct if it rounds to 47.124.

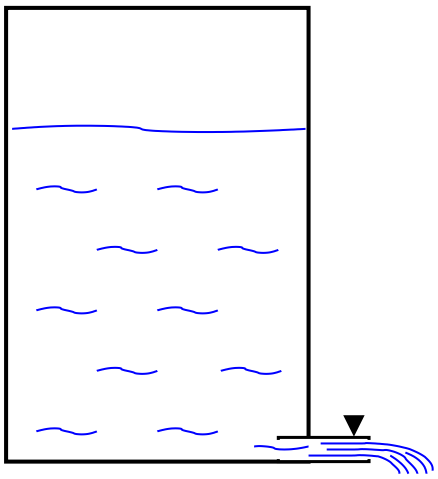
4 Differential Equations

You are perfectly familiar with solving equations involving an unknown variable x and its powers. For example, $x = 2$ solves $x^2 - x - 2 = 0$. We now study equations involving an unknown function and its derivatives. That is, we will be studying *differential equations* (DEs). For example, $f(x) = e^{2x}$ solves $f''(x) - f'(x) - 2f(x) = 0$. Differential equations are difficult to solve, but it is important to keep in mind that checking whether a given function is a solution is more straightforward: you can verify that $f(x) = e^{2x}$ solves $f''(x) - f'(x) - 2f(x) = 0$ solely with your knowledge of differentiation. Which of $f(x) = e^x$ and $f(x) = e^{-x}$ is another solution of that differential equation?

Differential equations for functions of one variable are called *ordinary differential equations* (ODEs), and for functions of several variables, they are called *partial differential equations* (PDEs). Almost all natural processes obey differential equations, and they are therefore very important for physics and in engineering. However, besides classical physics and engineering applications such as describing the motion of objects, many other phenomena can be modelled with differential equations as well: for example, the growth of populations and the spread of diseases.

Example 4.1 A water tank is being drained, and we are asked to find a formula for the height $h(t)$ of the water level as a function of time.

The behaviour of the height of the water level over time depends on how quickly water is leaving the tank, i.e. on the rate of change $h'(t)$. The problem now is the following: $h'(t)$ in turn depends on $h(t)$ – for example, if the level is very high, the water pressure at the bottom of the tank will be very high and water will be forced out at a very high velocity.



From physical principles, one can derive the differential equation

$$h'(t) = -\alpha \cdot \sqrt{h(t)}$$

for the height of the water level, where α is some positive constant that depends on the size of the tank and other parameters. This differential equation is known as Torricelli's law, and we will now learn how to solve it.

4.1 First-Order Ordinary Differential Equations

Definition 4.2 — ODEs, order, IVPs, solutions. (i) An ***ordinary differential equation (ODE)*** is an equation involving an unknown function $y(x)$ and its derivatives

$$y'(x),\ y''(x),\ y'''(x),\ y^{(4)}(x),\ \ldots,\ y^{(n)}(x)$$

and other known functions of x.

(ii) The order of the highest derivative appearing in an ordinary differential equation is called the *order* of the ODE. For example, a first-order ODE is of the form

$$y' = f(x, y),$$

where f is a known function. The argument x of y is often omitted, but it is important to be aware

that y is a function.

(iii) A first-order ODE together with an ***initial condition*** of the form

$$y(x_0) = y_0,$$

where the numbers x_0, y_0 are given, is called an ***initial-value problem (IVP)***.

(iv) A function $y = y(x)$ that satisfies a given ODE is called a *solution*. If y contains parameters (constants) such that varying them covers all possible solutions to the ODE, then y is called the *general solution*. A solution that also satisfies given initial conditions is called a *particular solution*. For now, i.e. until section 4.8, we can omit the specification 'ordinary'.

Example 4.3 (i) The DE

$$y''(x) + 4y(x) = 0$$

has $y(x) = \sin(2x)$ as a solution. Indeed, we can check

$$\begin{aligned} & y(x) = \sin(2x) \\ \rightarrow \quad & y'(x) = 2\cos(2x) \\ \rightarrow \quad & y''(x) = -4\sin(2x) \\ \rightarrow \quad & y''(x) + 4y(x) = -4\sin(2x) + 4\sin(2x) = 0 \quad \checkmark. \end{aligned}$$

We will see later that the general solution is

$$y(x) = c_1 \sin(2x) + c_2 \cos(2x),$$

which gives our guessed solution via the choice $(c_1, c_2) = (1, 0)$ of constants.

(ii) For the DE

$$y^{(4)} = 0,$$

one can guess solutions $y(x) = 1$, $y(x) = x^3$, $y(x) =$

$2x^2 - 3x + 5$, and hence the general solution

$$y(x) = Ax^3 + Bx^2 + Cx + D$$

– the polynomials of degree less than 4.

Remark 4.4

(i) As for integration: While solving DEs can be tricky, verifying solutions using differentiation is straightforward.

(ii) Differential equations of the form

$$y' = f(x),$$

i.e. when the function $f(x, y)$ in definition 4.2 (ii) does not depend on y, are the easiest to solve – by integration:

$$y(x) = \int f(x)\,\mathrm{d}x.$$

For example, the function $y(x) = \ln(x) + c$ solves the DE $y' = {}^1/_x$. We will refer to this type of DE as *directly integrable*.

(iii) The general solution of a nth-order DE usually contains n constants. You can see that in the examples above. For directly integrable DEs, one can argue as follows: To solve

$$y^{(n)} = f(x),$$

we integrate $f(x)$ n times, obtaining a constant of integration each time. For this module, you can always work with the assumption that the general solution of a nth-order DE contains n constants. Proving this for certain types of DEs – it is not true for all DEs – is beyond the scope of MTH1002.

(iv) Our first non-trivial technique for solving ODEs is for those of the form

$$y' = h(x) \cdot g(y).$$

4.2 Separable DEs

Example 4.5 (i) To solve the DE

$$y' = x \cdot y,$$

we first write it as

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x \cdot y$$

and then *separate variables*. That is, bring all expressions with an x on one side and all expressions with y on the other side. For that, we treat dx like a number and 'multiply' the DE by it. Integrating gives

$$\int \frac{1}{y}\,\mathrm{d}y = \int x\,\mathrm{d}x, \tag{4.1}$$

which leads to

$$\ln|y| = \frac{x^2}{2} + c_1 \qquad (c_1 \in \mathbb{R}).$$

Applying $\exp(.)$ on both sides gives

$$|y| = \mathrm{e}^{x^2/2 + c_1} = \mathrm{e}^{x^2/2}\,\mathrm{e}^{c_1} = c_2\,\mathrm{e}^{x^2/2} \qquad (c_2 \in \mathbb{R}^+),$$

where we let $c_2 = \mathrm{e}^{c_1}$. Now, the absolute value on the left is either a factor of $+1$ or -1, depending on whether y is positive or negative. Absorbing this potential negative sign into the constant, we have

$$y = c_3\,\mathrm{e}^{x^2/2} \qquad (c_3 \in \mathbb{R}\backslash\{0\}).$$

Even though $c_3 \neq 0$ is excluded, we note that a constant of 0 works as well: Then we have $y(x) = 0$ and

$$y'(x) = 0 = x \cdot y.$$

Hence we include the possibility $c = 0$ and find the general solution to be

$$y(x) = c\,\mathrm{e}^{x^2/2} \qquad (c \in \mathbb{R}).$$

Note that after (4.1), one should have stated that the function $y(x)$ can not have any zeros, since otherwise the integrand on the left-hand side would not be defined. Hence one solves the DE only on intervals on which y is either always positive or always negative, and considers the case of y having a zero separately. If $y(x_0) = 0$, then $y'(x_0) = 0$ by the original DE, and this suggests that the constant function $y(x) = 0$ is a solution. Now, for the remaining cases, y is either always positive or always negative, which allows to absorb the potential negative from $|y|$ into the constant c_2. However, we will not always discuss the solution of DEs and the ranges of any constants involved that carefully. For example, in the next computation we might change the meaning of constants from one line to the next without explicitly renaming them.

(ii) Solve the IVP

$$\begin{cases} 1 + xyy' = y^2 + yy' \\ \quad y(0) = -1/2. \end{cases}$$

Sol.:

$$\begin{aligned} &\to \quad (xy - y)\frac{\mathrm{d}y}{\mathrm{d}x} = y^2 - 1 \quad \overset{\text{sep. of var.}}{\to} \quad \int \frac{y}{y^2-1}\,\mathrm{d}y = \int \frac{1}{x-1}\,\mathrm{d}x \\ &\overset{u=y^2-1}{\to} \quad \frac{1}{2}\int \frac{1}{u}\,\mathrm{d}u = \ln|x-1| + c \quad \overset{\text{integrate}}{\to} \quad \frac{1}{2}\ln\left|y^2-1\right| = \ln|x-1| + c \\ &\overset{\cdot 2}{\to} \quad \ln\left|y^2-1\right| = \ln(x-1)^2 + c \quad \overset{\exp(.)}{\to} \quad \left|y^2-1\right| = \mathrm{e}^{\ln(x-1)^2+c} \\ &\overset{\text{as in (i)}}{\to} \quad y^2 - 1 = c\,(x-1)^2 \quad \to \quad y^2 + c\,(x-1)^2 = 1. \end{aligned}$$

This is the general solution of the DE, but it is in implicit form, i.e. not solved for y (it is not an explicit formula for $y(x)$). Next, we use the given initial data $(x_0, y_0) = (0, -1/2)$:

$$\left(-\frac{1}{2}\right)^2 + c\,(0-1)^2 = \frac{1}{4} + c \overset{!}{=} 1,$$

which gives $c = 3/4$ and the explicit solution

$$y = \pm\sqrt{1 - 3/4\,(x - 1^2)}.$$

For the particular solution to the given IVP, it remains to decide between $+$ and $-$. We choose

$$y(x) = -\sqrt{1 - {}^3/_4\,(x - 1^2)}$$

– why not the positive solution*?

Remark 4.6

(i) The implicit general solution $y^2 + c\,(x-1)^2 = 1$ in (ii) describes an ellipse for $c > 0$ (centred at $(1,0)$; a circle for $c = 1$), a hyperbola for $c < 0$, and a pair of horizontal lines for $c = 0$.

4.3 Linear DEs

Definition 4.7 — First-Order Linear, Homogeneity. A first-order ODE of the form

$$y' + p(x)\,y = r(x)$$

is called ***linear***.

It is further called ***homogeneous*** if $r(x) = 0$, and ***inhomogeneous*** otherwise.

Multiplying this DE by its *integrating factor*

$$h(x) := \mathrm{e}^{P(x)},$$

where P is any antiderivative of p, we obtain

$$h\,r = h\,y' + h\,p\,y = h\,y' + h'\,y = (h\,y)',$$

which can be solved by integrating both sides:

$$h(x)\cdot y(x) = \int h(x)\cdot r(x)\,\mathrm{d}x.$$

Example 4.8 (i) Solve

$$y' + \frac{5}{x}\,y = \cos\left(x^6\right).$$

*Hint: Check the initial condition for both possible choices.

Sol.: The DE is linear, and the integrating factor

$$h(x) = \mathrm{e}^{\int 5/x \, \mathrm{d}x} = \mathrm{e}^{5 \ln x} = x^5.$$

Note that the constant of integration was omitted in the integration of $p(x)$ – that is because the method using the integrating factor works for any antiderivative, and hence one can choose the easiest one: the antiderivative with $c = 0$. Next, we find

$$x^5 y = \int x^5 \cos\left(x^6\right) \mathrm{d}x \overset{u=x^6}{=} \cdots = \frac{1}{6}\left(\sin\left(x^6\right) + c\right).$$

Finally, dividing by the integrating factor gives the solution:

$$y(x) = \frac{\sin\left(x^6\right) + c}{6x^5}.$$

(ii) Solve the IVP

$$\begin{cases} (1+x^2)y' - 2xy = 1 + x^2 \\ \qquad\qquad y(1) = 6. \end{cases}$$

Sol.: First bring the DE in the correct form:

$$y' - \frac{2x}{1+x^2} y = 1.$$

Now one can read off the antiderivative $P(x) = -\ln(1+x^2)$ of $p(x)$. This gives the integrating factor

$$h(x) = \mathrm{e}^{-\ln(1+x^2)} = \frac{1}{1+x^2},$$

and therefore the general solution

$$y(x) = (1+x^2) \int \frac{1}{1+x^2} \cdot 1 \, \mathrm{d}x = (1+x^2)(\arctan x + c).$$

The initial condition gives

$$6 = (1+1^2)\left(\frac{\pi}{4} + c\right) \quad \to \quad y(x) = (1+x^2)\left(\arctan x + 3 - \frac{\pi}{4}\right).$$

Exercise 4.9 (i) Work through the first-order DEs on the tutorial sheet for chapter 4.

(ii) Letting $\alpha = 1$ and $h(0) = 100$, solve the differential equation for the water tank from the beginning of this chapter. Plot your solution – does it

make sense[a]? How long does it take for the tank to drain[b]?

(iii) Solve[c]

$$xy' = y + x\sec\left(\frac{y}{x}\right), \quad y(1) = \frac{\pi}{3}.$$

(iv) Solve[d]

$$y' = y\tan x + \sin x, \quad y(0) = 0.$$

[a] Hint: Initially, it does not make sense, but setting the solution equal to 0 from a certain t_0 on, one obtains a function for the water level $h(t)$ that does agrees with how one would imagine a tank to drain.

[b] Answer: 20 seconds.

[c] Answer: $y(x) \approx x \cdot \arcsin(\ln x + 0.866)$, but you should obtain the constant exactly.

[d] Answer: $y(x) = \frac{\sin^2 x}{\cos x}$.

4.4 Linear Ordinary Differential Equations of Higher Order

Definition 4.10 — Second-Order Linear ODEs. The standard form for a ***second-order linear ODE is***

$$y'' + p(x)y' + q(x)y = r(x).$$

A LODE is called homogeneous if $r(x) = 0$, inhomogeneous otherwise.

Remark 4.11

(i) If we set

$$L[y] = y'' + p(x)y' + q(x)y,$$

the DE reads $L[y] = r(x)$. Here, L is a *differential operator* – it takes a function as input and outputs

a function. For example, the differential operator

$$\widetilde{L} = 7\frac{\mathrm{d}}{\mathrm{d}x}$$

would map the input x^2 to $14x$.

(ii) For LODEs: If y_1, y_2 are functions and α, β constants, then

$$L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2].$$

(iii) If y_1 and y_2 are solutions to a homogeneous s-o LODE and they are *linearly independent* (i.e. $y_2 \neq c \cdot y_1$), then

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

is the general solution.

(iv) However, the previous statement, (iii), is not true in the inhomogeneous case: If

$$L[y_1] = r(x) \quad \text{and} \quad L[y_2] = r(x),$$

then

$$L[y_1 + y_2] = L[y_1] + L[y_2] = 2r(x),$$

that is, $y_1 + y_2$ does not satisfy $L[y] = r(x)$ (unless $r(x) = 0$, which is the homogeneous case (iii)).

(v) The following technique allows us to find the general solution if one solution is known or can be guessed.

4.5 Reduction of Order

Example 4.12 Given the solution $y_1(x) = x$ of

$$x^2y'' + xy' - y = 0,$$

find y_2 and hence the general solution.

Sol.: First, one should verify that y_1 really is a solution:

$$x^2y_1'' + xy_1' - y_1 = x^2 \cdot 0 + x \cdot 1 - x = x - x = 0 \quad \checkmark.$$

Now make the ansatz $y(x) = v(x) \cdot y_1(x)$, or, abbreviated,

$$y = vy_1.$$

The idea behind this ansatz is: the other solutions should be structurally similar to y_1, as they satisfy the same differential equation. From this viewpoint, it seems possible that using the knowledge of y_1 and analysing its ratio with other solutions might reduce the complexity of the problem. We have

$$y = vy_1, \quad y' = v'y_1 + vy_1', \quad y'' = v''y_1 + 2v'y_1' + vy_1'',$$

and therefore

$$\begin{aligned} L[y] = L[vy_1] &= x^2 \cdot \left[v''y_1 + 2v'y_1' + vy_1''\right] + x \cdot \left[v'y_1 + vy_1'\right] - \left[vy_1\right] \\ &= x^2v''y_1 + 2x^2v'y_1' + xv'y_1 + v \cdot \underbrace{\left[x^2y_1'' + xy_1' - y_1\right]}_{=L[y_1]=0} \\ &= x^2v''y_1 + 2x^2v'y_1' + xv'y_1 + v \cdot 0 \qquad \text{(now use } y_1 = x,\ y_1' = 1,\ y_1'' = 0\text{)} \\ &= x^3v'' + 3x^2v' \overset{w=v'}{=} x^3w' + 3x^2w \overset{!}{=} 0. \end{aligned}$$

That is, if $w = w(x)$ satisfies the first-order LODE

$$x^3w' + 3x^2w = 0,$$

then $y = vy_1$ will satisfy the original second-order LODE. We find

$$\begin{aligned} w(x) &= \frac{\alpha}{x^3}, \\ v(x) &= -\frac{\alpha}{2}x^{-2} + \beta, \\ y(x) &= \beta x - \frac{\alpha}{2}\frac{1}{x}, \end{aligned}$$

where $\alpha, \beta \in \mathbb{R}$, and, after renaming the constants,

$$y(x) = c_1 \cdot x + c_2 \cdot \frac{1}{x}.$$

This is the general solution of the DE, and it is in the form mentioned in remark 4.11 (iii), with $y_2 = {}^1/_x$.

Remark 4.13

We recommend writing out $L[y] = L[vy_1] = \dots$ as above, rather than using the formula for y_1, which is already known at this point. Then, *after* the term $v \cdot L[y_1]$ has dropped out of the computation, one can substitute in the formula for y_1 and its derivatives. Using the formula for y_1 from the beginning, i.e. $L[y] = L[v \cdot x] = \dots$ in this case, would often make the computation more complicated, e.g. if y_1 is a rational function with bulky derivatives.

4.6 Constant-Coefficient Homogeneous

Consider the second-order LODE

$$L[y] = y'' + py' + qy = 0, \tag{4.2}$$

where p, q are *constants*, not functions of x. Making the ansatz

$$y(x) = \mathrm{e}^{\lambda x},$$

we find

$$0 \stackrel{!}{=} L[\mathrm{e}^{\lambda x}] = \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\mathrm{e}^{\lambda x}\right) + p\,\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\lambda x}\right) + q\,\mathrm{e}^{\lambda x} = \mathrm{e}^{\lambda x}\left(\lambda^2 + p\,\lambda + q\right).$$

That is, if λ satisfies the *characteristic equation* (or *auxiliary equation*)

$$\lambda^2 + p\,\lambda + q = 0, \tag{4.3}$$

then $y = \mathrm{e}^{\lambda x}$ is a solution of (4.2). The solutions of (4.3) are

$$\lambda_{1/2} = \frac{-p \pm \sqrt{p^2 - 4q}}{2},$$

and we now have to consider three cases:

(I) If (4.3) has two distinct real solutions λ_1 and λ_2 – this happens when $p^2 - 4q > 0$ – then we have solutions

$$y_1(x) = \mathrm{e}^{\lambda_1 x}, \qquad y_2(x) = \mathrm{e}^{\lambda_2 x},$$

of (4.2). From earlier theory, we conclude that the general solution is

$$y(x) = c_1\,\mathrm{e}^{\lambda_1 x} + c_2\,\mathrm{e}^{\lambda_2 x}.$$

(II) If (4.3) has one real solution λ (i.e. a double root, $\lambda_1 = \lambda_2$) – this happens when $p^2 - 4q = 0$ – then we have solutions

$$y_1(x) = \mathrm{e}^{\lambda x}, \qquad y_2(x) = x\,\mathrm{e}^{\lambda x}.$$

That y_1 is a solution follows again from λ solving the characteristic equation, and y_2 is obtained via reduction of order. Note that $\lambda = -p/2$ in this case. The general solution is

$$y(x) = c_1\,\mathrm{e}^{\lambda x} + c_2\,x\,\mathrm{e}^{\lambda x}.$$

(III) If (4.3) has no real solutions – this happens when $p^2 - 4q < 0$ – then it has complex solutions

$$\lambda_{1/2} = a \pm ib \qquad (a, b \in \mathbb{R}),$$

and the solutions to the differential equation are

$$y_1(x) = \mathrm{e}^{ax}\cos(bx), \qquad y_2(x) = \mathrm{e}^{ax}\sin(bx).$$

The derivation of this is outlined in the exercises below. The general solution is

$$y(x) = \mathrm{e}^{ax}\left[c_1\cos(bx) + c_2\sin(bx)\right].$$

Definition 4.14 — IVPs and BVPs. For a second-order ODE, $L[y] = r(x)$, the system of equations

$$\begin{cases} L[y] = r(x) \\ y(a) = A \\ y'(a) = B \end{cases}$$

is called an ***initial-value problem*** (IVP). A system of

the form

$$\begin{cases} L[y] = r(x) \\ y(a) = A \\ y(b) = B \end{cases}$$

is called an *boundary-value problem* (BVP).

Example 4.15 (i)

$$\begin{cases} y'' - 3y' + {}^9/_4\, y = 0 \\ y(0) = 0 \\ y\left({}^2/_3\right) = 1 \end{cases}$$

Sol.: We first consider the DE only, which is second-order, linear, and homogeneous with constant coefficients. Its characteristic equation is

$$0 = \lambda^2 - 3\lambda + \frac{9}{4} = (\lambda - {}^3/_2)(\lambda - {}^3/_2),$$

which has the double root $\lambda = {}^3/_2$, i.e. we have case (II) from the discussion above. This value of λ implies that the general solution is

$$y(x) = c_1\, \mathrm{e}^{{}^3/_2\, x} + c_2\, x\, \mathrm{e}^{{}^3/_2\, x},$$

and it remains to determine the constants so that y satisfies the given boundary conditions:

$$\begin{aligned} 0 &\overset{!}{=} y(0) = c_1 \cdot 1 + c_2 \cdot 0 \cdot 1 = c_1 \\ 1 &\overset{!}{=} y\left({}^2/_3\right) = c_1 \cdot \mathrm{e} + c_2 \cdot {}^2/_3 \cdot \mathrm{e} = {}^{2\mathrm{e}}/_3\, c_2, \end{aligned}$$

which leads to the particular solution

$$y(x) = \frac{3x}{2\mathrm{e}}\, \mathrm{e}^{{}^2/_3\, x}.$$

(ii)

$$\begin{cases} y'' - y' + {}^{17}/_4\, y = 0 \\ y(0) = 0 \\ y'(0) = 10 \end{cases}$$

Sol.: The characteristic equation is

$$\lambda^2 - \lambda + \frac{17}{4} = 0,$$

which has solutions

$$\lambda_{1/2} = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot {}^{17}/_4}}{2} = \frac{1}{2} \pm \frac{\sqrt{-16}}{2} = \frac{1}{2} \pm i \cdot 2,$$

and therefore leads to the general solution

$$y(x) = \mathrm{e}^{x/2} \left(c_1 \cos(2x) + c_2 \sin(2x) \right).$$

In order to find the solution of the IVP, we need to set the values of y and y' at $x = 0$ equal to the given values. Therefore, $y'(x)$ needs to be found:

$$y'(x) = \frac{1}{2} \mathrm{e}^{x/2} \left(c_1 \cos(2x) + c_2 \sin(2x) \right) + \mathrm{e}^{x/2} \left(-2c_1 \sin(2x) + 2c_2 \cos(2x) \right).$$

Now we can solve for the constants:

$$\begin{cases} 0 = y(0) = \mathrm{e}^0 (c_1 \cos 0 + c_2 \sin 0) = c_1 \\ 10 = y'(0) = {}^1/_2 \, \mathrm{e}^0 (c_1 \cos 0 + c_2 \sin 0) + \mathrm{e}^0 (-2c_1 \sin 0 + 2c_2 \cos 0) = {}^1/_2 \, c_1 + 2c_2, \end{cases}$$

which gives $c_1 = 0$, $c_2 = 5$ and hence

$$y(x) = 5 \, \mathrm{e}^{x/2} \sin(2x).$$

(iii)

$$\begin{cases} y'' + y' - 6y = 0 \\ y(0) = 1 \\ y'(0) = 12 \end{cases}$$

Sol.:

$$y(x) = 3 \, \mathrm{e}^{2x} - 2 \, \mathrm{e}^{-3x}$$

Remark 4.16

(i) At the beginning of this chapter, in remark 4.4, it was stated that the general solution of a nth-order DE contains n constants. From this, one can deduce that it takes n conditions to specify a particular solution – this agrees with the examples we have seen so far: one condition in section 4.1 and two conditions (either IVP or BVP) for the second-order DEs in the previous example.

(ii) The difference between an IVP and an BVP can be illustrated as follows. Many natural processes obey second-order differential equations, and the motion of objects is a prominent example. Suppose a ball is kicked vertically in the air, and you are shown a snapshot of it. That is, you are given the height at a certain time, $h(t_1)$. From this, you cannot say what will happen next – for example, you do not know whether the ball was on its way up or down when the picture was taken. However, if you are given a second piece of information, you can – with your knowledge from this module – determine the exact trajectory $h(t)$ of the ball. This second piece of information could be the velocity $h'(t_1)$ of the ball ($\rightarrow$ IVP; how fast and in which direction was it moving when the picture was taken) or the position at some other specified time ($\rightarrow$ BVP; e.g., another snapshot, taken exactly one second later).

(iii) The method of using the characteristic equation remains applicable for LODEs of higher orders – however, then the characteristic equation will be of higher order as well, and it might therefore not be readily solvable. Sometimes, one is able to guess one solution and then reduce its order, cf. the next example.

Example 4.17

$$y''' - \frac{1}{2}y'' - \frac{41}{2}y' + 35\,y = 0$$

Sol.: The characteristic equation is

$$\lambda^3 - \frac{1}{2}\lambda^2 - \frac{41}{2}\lambda + 35 = 0,$$

and after some experimentation, one finds the solution $\lambda = 2$, since $8 - 2 - 41 + 35 = 0$. This means that $(\lambda - 2)$ is a factor

of the cubic expression on the left – dividing by it (long division of polynomials), we obtain

$$0 = \lambda^3 - \frac{1}{2}\lambda^2 - \frac{41}{2}\lambda + 35 = (\lambda - 2)\left(\lambda^2 + \frac{3}{2}\lambda - \frac{35}{2}\right) = (\lambda - 2)(\lambda - {}^7\!/_2)(\lambda + 5),$$

and therefore the general solution

$$y(x) = c_1\,\mathrm{e}^{2x} + c_2\,\mathrm{e}^{{}^7\!/_2 x} + c_3\,\mathrm{e}^{-5x}.$$

4.7 Constant-Coefficient Inhomogeneous

The steps for solving an *inhomogeneous* linear second-order DE with constant coefficients,

$$L[y] = y'' + py' + qy = r(x), \qquad p, q \in \mathbb{R},$$

are:

(1) Solve the corresponding homogeneous DE, $L[y] = 0$. Its general solution is called the *complementary function*:

$$y_{CF} = c_1 y_1 + c_2 y_2.$$

(2) Find one solution of the inhomogeneous DE, $L[y] = r(x)$. This solution is denoted y_{PI} and called the *particular integral*. Finding y_{PI} will be discussed below.

(3) The general solution of the inhomogeneous DE is

$$y = y_{CF} + y_{PI} = c_1 y_1 + c_2 y_2 + y_{PI}.$$

(4) If the DE was given as an IVP or a BVP, i.e. with two additional conditions, then determine the constants in the general solution (3) so that the resulting function satisfies those conditions. This function is the solution to the IVP/BVP.

Let us check whether the function from step (3) above really is the general solution of $L[y] = r(x)$: The function

$$y = y_{CF} + y_{PI} = c_1 y_1 + c_2 y_2 + y_{PI}$$

does contain two constants – as we expect for the general solution of a second-order DE – and

$$\begin{aligned} L[y] = L[c_1 y_1 + c_2 y_2 + y_{PI}] &= c_1 L[y_1] + c_2 L[y_2] + L[y_{PI}] \\ &= c_1 \cdot 0 + c_2 \cdot 0 + r(x) = r(x) \quad \checkmark. \end{aligned}$$

Example 4.18 Solve the IVP

$$\begin{cases} y'' - y' - 12y = e^{2x} \\ y(0) = -2\,. \\ y'(0) = 2 \end{cases}$$

Sol.:

(1) First we consider the homogeneous version of the DE,

$$y'' - y' - 12y = 0.$$

Its characteristic equation is

$$0 = \lambda^2 - \lambda - 12 = (\lambda - 4)(\lambda + 3),$$

and it leads to the two solutions

$$y_1(x) = e^{4x}, \qquad y_2(x) = e^{-3x},$$

of the homogeneous DE. Hence the complimentary function is

$$y_{CF}(x) = c_1\, e^{4x} + c_2\, e^{-3x}.$$

(2) Next, we need one solution of the inhomogeneous DE, the particular integral, y_{PI}. The idea is to make an ansatz containing parameters for it and then substitute it into $L[y] = e^{2x}$ to determine those parameters. Choosing the correct form for y_{PI} is key:

Attempt 1: Take $y_{PI}(x) = \sqrt{Ax^2 + Bx + C}$. However, thinking about the action of the operator L, we see that

$$\begin{aligned} L[y_{PI}] &= L\left[\sqrt{Ax^2 + Bx + C}\right] \\ &= \frac{d^2}{dx^2}\left[\sqrt{Ax^2 + Bx + C}\right] - \frac{d}{dx}\left[\sqrt{Ax^2 + Bx + C}\right] \\ &\qquad - 12\left[\sqrt{Ax^2 + Bx + C}\right] = \ldots \end{aligned}$$

will not lead to an expression that can be made equal to e^{2x} by making suitable choices for A, B, C.

Attempt 2: For $y_{PI}(x) = A\cos x + B\sin x$, application of L will give a combination of trigonometric functions $\cos x$ and $\sin x$, not e^{2x}.

Attempt 3: The previous attempts suggest that one should make an ansatz of the same form as the inhomogeneity $r(x)$. Hence try $y_{PI}(x) = A\mathrm{e}^{2x}$:

$$L[y_{PI}] = L\left[A\mathrm{e}^{2x}\right] = A\left[4\mathrm{e}^{2x} - 2\mathrm{e}^{2x} - 12\mathrm{e}^{2x}\right] = A(-10)\mathrm{e}^{2x} \stackrel{!}{=} \mathrm{e}^{2x},$$

and therefore $A = {}^{-1}/_{10}$,

$$y_{PI}(x) = \frac{-1}{10}\mathrm{e}^{2x}.$$

General advice on how to make the ansatz for the particular integral will be given below.

(3) The general solution of the original inhomogeneous DE is

$$y = y_{CF} + y_{PI} = c_1\,\mathrm{e}^{4x} + c_2\,\mathrm{e}^{-3x} - \frac{1}{10}\,\mathrm{e}^{2x}.$$

(4) The initial conditions give rise to the system

$$\begin{cases} -2 = y(0) = c_1 + c_2 - {}^{1}/_{10} \\ 2 = y'(0) = 4c_1 - 3c_2 - {}^{2}/_{10}, \end{cases}$$

which has solutions $c_1 = -{}^{1}/_{2}$, $c_2 = -{}^{7}/_{5}$. This gives the solution to the IVP as

$$y = -\frac{1}{2}\,\mathrm{e}^{4x} - \frac{7}{5}\,\mathrm{e}^{-3x} - \frac{1}{10}\,\mathrm{e}^{2x}.$$

Remark 4.19

(i) The following table and remarks should help you make the ansatz for the particular integral:

Right-hand side $r(x)$	Ansatz for y_{PI}
polynomial of degree m: $\alpha_m x^m + \ldots \alpha_1 x + \alpha_0$	polynomial of degree m: $A_m x^m + \ldots A_1 x + A_0$
exponential: $\alpha e^{\lambda x}$	exponential: $A e^{\lambda x}$
comb. of trig. functions: $\alpha \cos(\lambda x) + \beta \sin(\lambda x)$	comb. of trig. functions: $A \cos(\lambda x) + B \sin(\lambda x)$
... (e.g.: comb. of hyp. trig. fcns	... comb. of hyp. trig. fcns)
a sum of any of the above: $r(x) = r_1(x) + \cdots + r_k(x)$	find $y_{PI,j}$ for each r_j and then: $y_{PI} = y_{PI,1} + \cdots + y_{PI,k}$

Here it is important to always have full generality on the ansatz side: E.g., if $r(x) = 2\sin(3x)$, i.e. there is no $\cos(3x)$ term in the inhomogeneity, one still needs $y_{PI} = A\cos(3x) + B\sin(3x)$ rather than an ansatz with a sine term only. If, for example, $r(x) = \sin(5x) + \cos(2x)$, the rule in the middle row of the table needs to be applied twice – once for $\lambda = 5$ and once for $\lambda = 2$ – and the outcomes are then to be combined as in the last row. If you have written out the ansatz y_{PI} based on $r(x)$, but this y_{PI} happens to already be a solution to the homogeneous version of the DE, then multiply it by x.

(ii) The linearity of the ODE is crucial for the above approach of finding all solutions of the homogeneous DE and then adding one solution of the inhomogeneous DE. Let us compare this to another linear situation: An inhomogeneous linear equation for the variables x, y, z: Solve

$$x + 2y - 5z = 2. \tag{4.4}$$

Sol.:

(1) For the homogeneous version of the equation,

$$x + 2y - 5z = 0,$$

we can find the two linearly independent (cf. 1.25 for the general definition of this term) solutions

$$\vec{v}_1 = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}, \qquad \vec{v}_2 = \begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix}$$

and hence the general solution

$$\vec{v}_{CF} = c_1 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix}.$$

(2) We now just need a single vector that satisfies the inhomogeneous equation (4.4). For example,

$$\vec{v}_{PI} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix},$$

as

$$\begin{bmatrix} 1 & 2 & -5 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = 2 \quad \checkmark$$

(the entries of the row vector are the coefficients of (4.4)).

(3) The general solution of the inhomogeneous equation is

$$\vec{v} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix}.$$

This is a parametrisation of the plane (4.4).

Example 4.20 Find the general solution of

$$L[y] = y'' - 3y' + {}^9\!/_4\, y = \mathrm{e}^{3x/2}.$$

Sol.:

(1) We have already found the solution to the homogeneous version of that DE in example 4.15 (i):

$$y_{CF}(x) = c_1\,\mathrm{e}^{3/2\,x} + c_2\,x\,\mathrm{e}^{3/2\,x}.$$

(2) Based on the form of the right-hand side, $r(x) = \mathrm{e}^{3x/2}$, one would make the ansatz $y_{PI} = A\mathrm{e}^{3x/2}$. However, this will be mapped to 0 by L, as it is a solution of $L[y] = 0$ – after all, the function $A\mathrm{e}^{3x/2}$ is already contained in the complementary function; for $(c_1, c_2) = (A, 0)$. According to the last sentence of remark 4.19 (i), we multiply by x to obtain $y_{PI} = Ax\mathrm{e}^{3x/2}$. Again, this already solves the homogeneous DE, and we therefore multiply by x one more time:

$$y_{PI}(x) = Ax^2\mathrm{e}^{3x/2}.$$

Substituting this into $L[y] = \mathrm{e}^{3x/2}$ gives

$$\begin{aligned} L\left[Ax^2\mathrm{e}^{3x/2}\right] &= A\left[2\mathrm{e}^{3x/2} + 6x\mathrm{e}^{3x/2} + \frac{9}{4}x^2\mathrm{e}^{3x/2}\right] \\ &\quad - 3A\left[2x\mathrm{e}^{3x/2} + \frac{3}{2}x^2\mathrm{e}^{3x/2}\right] + \frac{9}{4}A\left[x^2\mathrm{e}^{3x/2}\right] \\ &= A\left[2 + 6x + \frac{9}{4}x^2 - 6x - \frac{9}{2}x^2 + \frac{9}{4}x^2\right]\mathrm{e}^{3x/2} = 2A\mathrm{e}^{3x/2} \overset{!}{=} \mathrm{e}^{3x/2}, \end{aligned}$$

and hence

$$y_{PI}(x) = \frac{x^2}{2}\mathrm{e}^{3x/2}.$$

(3) Steps (1) and (2) lead to the general solution

$$y(x) = \mathrm{e}^{3x/2}\left(c_1 + c_2 x + \frac{x^2}{2}\right).$$

Exercise 4.21 (i) Solve the higher-order ODEs on the tutorial sheet.

(ii) Consider the s-o LODE

$$y'' + 4y = 0$$

from example 4.3 and the solution $y_1(x) = \sin(2x)$ we had already guessed and verified. Using reduction of order, find the general solution. Then also solve the DE using the theory from section 4.6.

(iii) Solve the BVP[a]

$$\begin{cases} y'' - 2y' + 2y = 10 \\ y(0) = 7 \\ y(\pi/2) = 4. \end{cases}$$

(iv) Find the general solution of[b]

$$y'' + 2y' + y = \cosh(3x) + \mathrm{e}^{-x}.$$

(v) Show that the function y_2 in (II) of remark 4.6 and both functions $y_{1/2}$ in (III) really are solutions of (4.2) in the respective cases. The former can be done with reduction of order, and the latter by carrying out (I) for complex λ, e.g. $y_1 = \mathrm{e}^{(a+ib)x}$ – the usual argument involving the characteristic equation remains applicable, and the laws of exponentials are true in the complex case as well – and then using *Euler's identity*,

$$\mathrm{e}^{ix} = \cos(x) + i\sin(x)$$

and renaming the constants. A different approach would be, of course, to simply verify that the functions that are given satisfy the constant-coefficient homogeneous second-order LODE (4.2). Euler's

identity is very significant and perhaps surprising: for $x = \pi$ we obtain

$$e^{i\pi} = -1,$$

combining three important mathematical constants on the left – that are quite non-trivial (irrational or imaginary) and seemed unrelated until now – to a rather simple number on the right.

[a] Answer: The general solution is $y_{\text{gen}}(x) = A\,e^x \cos x + B\,e^x \sin x + 5$, and you can verify the constants you have found for the particular solution by checking whether it satisfies the boundary conditions.

[b] Answer: $y(x) = A\,e^{-x} + B\,xe^{-x} + {}^1\!/_2\,x^2e^{-x} + {}^5\!/_{32}\cosh(3x) - {}^3\!/_{32}\sinh(3x)$, where the hyperbolic functions could also be written out as a sum of exponential expressions.

4.8 Partial Differential Equations

Partial differential equations (PDEs) are differential equations for multivariate functions, i.e. equations involving the partial derivatives of an unknown function. Many natural processes obey partial differential equations, for example: The temperature in a room as a function of time can be modelled with

$$u = u(t, x, y, z)$$

and satisfies the *heat equation*

$$u_t - \alpha\left(u_{xx} + u_{yy} + u_{zz}\right) = 0.$$

The heat equation is a very important PDE and techniques on how to solve it will be discussed in MTH2003.

Example 4.22 (i) Solve the PDE

$$\frac{\partial u}{\partial x} = 0,$$

where $u = u(x, y)$.

Sol.: Let us just integrate that equation with respect to x:

$$\int \frac{\partial u}{\partial x}\,dx = u = \int 0\,dx = c(y).$$

Here, the constant of integration can be a function of y, since any expression that does not depend on x has 0 as its x-derivative. Renaming the "constant", we find the solution

$$u(x,y) = w(y),$$

where w is an arbitrary differentiable single-variable function. Since this is a new type of computation, and since it might be surprising to have so much freedom of choice in the general solution – for first-order ODEs, that would be a constant; here it is a function that can be chosen freely – let us check:

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left(w(y)\right) = 0 \quad \checkmark.$$

(ii)

$$\frac{\partial^2 u}{\partial x \partial y} = 0$$

Sol.:

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = 0 \qquad \overset{\int \cdot \,\mathrm{d}x}{\to} \qquad \frac{\partial u}{\partial y} = \int 0 \,\mathrm{d}x = w(y)$$

$$\overset{\int \cdot \,\mathrm{d}y}{\to} \qquad u = \int w(y)\,\mathrm{d}y \overset{W'=w}{=} W(y) + v(x)$$

$$\overset{\text{rename}}{\to} \qquad u(x,y) = f(x) + g(y),$$

where f and g are arbitrary differentiable single-variable functions. Check this result using partial differentiation!

As hinted on above, the set of solution is *much* bigger for PDEs than for ODEs of the same order. You will see that when solving the exercises below and the PDEs on the tutorial sheet.

The previous examples are analogous to directly integrable ODEs – they can be solved directly using integration. The next example can not be solved as readily, but we will bring it into directly integrable form using theory from Chapter 2.

Example 4.23 Find $u = u(x,y)$ that satisfies

$$L[u] = 6\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial y^2} = 0.$$

Sol.: Note that we can not simply integrate the PDE, as $L[u]$ is a sum of different partial derivatives of u. The idea is to define new variables,

$$\xi = x + ay, \qquad \eta = x + by, \qquad (\xi \to \text{“xi”},\ \eta \to \text{“eta”})$$

and then choose the parameters a and b so that the PDE can be integrated directly in the new variables (we had already carried out such a transformation in chapter 2: example 2.21 (iv)). The partials transform as follows:

$$\begin{aligned}
\frac{\partial u}{\partial x} &= u_\xi + u_\eta, \qquad \frac{\partial u}{\partial y} = a\,u_\xi + b\,u_\eta, \\
\frac{\partial^2 u}{\partial x^2} &= u_{\xi\xi} + 2\,u_{\xi\eta} + u_{\eta\eta}, \\
\frac{\partial^2 u}{\partial y^2} &= a^2\,u_{\xi\xi} + 2ab\,u_{\xi\eta} + b^2\,u_{\eta\eta}, \\
\frac{\partial^2 u}{\partial x \partial y} &= a\,u_{\xi\xi} + (a+b)\,u_{\xi\eta} + b\,u_{\eta\eta},
\end{aligned}$$

and hence $L[u]$ transforms to

$$\begin{aligned}
L[u] &= 6\,u_{\xi\xi} + 12\,u_{\xi\eta} + 6\,u_{\eta\eta} + a\,u_{\xi\xi} + (a+b)\,u_{\xi\eta} + b\,u_{\eta\eta} - a^2\,u_{\xi\xi} - 2ab\,u_{\xi\eta} - b^2\,u_{\eta\eta} \\
&= \left[6 + a - a^2\right] u_{\xi\xi} + \left[12 + a + b - 2ab\right] u_{\xi\eta} + \left[6 + b - b^2\right] u_{\eta\eta} \\
&=: c_1\,u_{\xi\xi} + c_2\,u_{\xi\eta} + c_3\,u_{\eta\eta} \overset{!}{=} c \cdot u_{\xi\eta}.
\end{aligned}$$

The last line states that we assigned the names c_1, c_2, c_3 to the expressions in square brackets in the previous line. We would then like this combination of second-order partials with respect to ξ and η to be of the form some constants times $u_{\xi\eta}$, as this is a PDE that we can solve, cf. the previous example. For $L[u] = 0$ to be equivalent to $u_{\xi\eta} = 0$, we need $c_1 = 0$, $c_3 = 0$, and $c_2 \neq 0$, because then

$$L[u] = 0 \quad \Longleftrightarrow \quad c_2\,u_{\xi\eta} = 0 \quad \Longleftrightarrow \quad u_{\xi\eta} = 0.$$

Setting $c_1 = 0$ and $c_3 = 0$ leads to

$$a \in \{-2, 3\} \quad \text{and} \quad b \in \{-2, 3\},$$

which a-priori leaves four choices for the combination of parameters (a, b). However, if $a = b$, then

$$c_2 = 12 + 2a - 2a^2 = 2[6 + a - a^2] = 2 \cdot 0 = 0,$$

which shows that $a = b$ is not a suitable choice. (If we chose $a = b$, we would have $\xi = \eta$, and we would therefore be looking for functions that can effectively be written as functions of one variable – this would not give us full generality for solutions of $L[u] = 0$.) We are left with two equivalent choices for (a, b). Take

$$\xi = x + 3y, \qquad \eta = x - 2y.$$

The new DE is

$$u_{\xi\eta} = 0,$$

which we have already solved in the previous example (well, the variables were called x and y then). Its solution is $f(\xi) + g(\eta)$, and we therefore obtain the solution

$$u(x, y) = f(x + 3y) + g(x - 2y)$$

in the original variables. Here, f and g are arbitrary single-variable functions (twice differentiable).

Exercise 4.24 (i) Work through the PDEs on the tutorial sheet. Check the solution from the previous example by finding its second-order partial derivatives and check whether they add to 0 when combined in way prescribed by the PDE. This will require the 1D chain rule, e.g.:

$$\begin{aligned}\frac{\partial}{\partial y}\left(f(x + 3y)\right) &= f'(x + 3y)\,\frac{\partial}{\partial y}\,(x + 3y)\\ &= 3f'(x + 3y).\end{aligned}$$

(ii) Find $u = u(x, y)$ that satisfies[a]

$$2\frac{\partial^2 u}{\partial x^2} + 5\frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0.$$

(iii) Find the function $u = u(x,y)$ with[a]

$$\begin{cases} \dfrac{\partial^2 u}{\partial y^2} = 0 \\ u(x,0) = \cosh(x) \\ u(x,-1) = 0. \end{cases}$$

(iv) Solve the first-order PDE[b]

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0.$$

[a] Answer: No need to provide the answer here – just check it yourself in the way described in the previous exercise. This is a very good thing to do, as it shows you what we are actually doing here.

[a] Answer: $u(x,y) = (y+1)\cdot\cosh x$, but, again, this would have been easy to check – always try to check your answers yourself!

[b] Hint: Use the same change of variables as for s-o PDEs, and then make a suitable choice for (a,b). It is less straightforward what to do now, but with some flexibility, you will find a solution (containing a single-variable function) that can be easily verified.

4.9 *Outlook: Coupled Oscillator*

This section describes an application that connects differential equations to matrices. However, it is optional – it is not part of the syllabus for MTH1002 and it will not be on the exam.

Suppose we have two separate oscillators (e.g., pendulums, weights on springs) described by functions $x(t)$ and $y(t)$. Each of those two functions obeys a second-order ODE, for example an ODE of the form $x''(t) = c \cdot x(t)$. Now suppose we couple the two (e.g., connect them with a spring) – then the position of y will affect the motion of x and vice versa. This can be described by a system of ODEs. An example would be

$$\begin{cases} x'' &= -2x + y \\ y'' &= x - 2y, \end{cases} \tag{4.5}$$

where $'$ stands for the derivative with respect to t.

We approach solving this system by writing it in matrix form:

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

and

$$w'' = Aw, \tag{4.6}$$

where $w(t) = \begin{bmatrix} x(t) & y(t) \end{bmatrix}^\top$ and A is the matrix from the previous equation. Next, we compute the eigenvalues and eigenvectors of A:

$$\lambda_1 = -3, \qquad v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

$$\lambda_2 = -1, \qquad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The ansatz

$$w(t) = f_1(t)v_1 = f_1(t) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} f_1(t) \\ -f_1(t) \end{bmatrix},$$

where $f_1(t)$ is a function to be determined, is now very useful for solving (4.6) because (a) the derivatives in (4.6) affect only

$f_1(t)$, as v_1 is constant; (b) the matrix product affects only v_1, as $f_1(t)$ is a scalar factor that can be swapped with the matrix multiplication; and (c) this matrix multiplication is simplified by v_1 being an eigenvector. We obtain

$$A\left(w(t)\right) = A(f_1(t)v_1) = f_1(t)Av_1 = f_1(t)\lambda_1 v_1 = \lambda_1 f_1(t)\cdot v_1$$

and

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}w(t) = \frac{\mathrm{d}^2}{\mathrm{d}t^2}(f_1(t)v_1) = \frac{\mathrm{d}^2}{\mathrm{d}t^2}(f_1(t))\cdot v_1. \qquad (4.7)$$

Equating the the two leads to

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}(f_1(t))v_1 = \lambda_1 f_1(t)v_1 \Longrightarrow \begin{cases} f_1''(t)\cdot 1 & = -3f_1(t)\cdot 1 \\ f_1''(t)\cdot(-1) & = -3f_1(t)\cdot(-1) \end{cases} \Longrightarrow f_1'' = -3f_1.$$

This gives

$$f_1(t) = A\cos\left(\sqrt{3}t\right) + B\sin\left(\sqrt{3}t\right),$$

and therefore a first solution

$$w_1(t) = \left[A\cos\left(\sqrt{3}t\right) + B\sin\left(\sqrt{3}t\right)\right]\cdot v_1.$$

Similarly, one finds

$$f_2(t) = C\cos(t) + D\sin(t)$$

for $f_2(t)$ in the ansatz $w_2(t) = f_2(t)v_2$.

We have therefore found the general solution:

$$\begin{aligned} w(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} &= A\cos\left(\sqrt{3}t\right)v_1 + B\sin\left(\sqrt{3}t\right)v_1 + C\cos(t)\,v_2 + D\sin(t)\,v_2 \\ &= A\begin{bmatrix} \cos(\sqrt{3}t) \\ -\cos(\sqrt{3}t) \end{bmatrix} + B\begin{bmatrix} \sin(\sqrt{3}t) \\ -\sin(\sqrt{3}t) \end{bmatrix} + C\begin{bmatrix} \cos(t) \\ \cos(t) \end{bmatrix} + D\begin{bmatrix} \sin(t) \\ \sin(t) \end{bmatrix}. \end{aligned} \qquad (4.8)$$

The constants A, B, C, D may now be chosen so that $w(t)$ satisfies any initial conditions the system (4.5) might have come with. If in doubt about any of the vector steps above – e.g. taking the derivative of a vector – one could always re-write them

as pairs of ordinary equations. For example, one could write out (4.7) as

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} w = \frac{\mathrm{d}^2}{\mathrm{d}t^2} \begin{bmatrix} f_1 \\ -f_1 \end{bmatrix} = \begin{bmatrix} f_1'' \\ -f_1'' \end{bmatrix} = f_1'' \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = f_1'' \cdot v_1.$$

Exercise 4.25 Suppose the motion of the first oscillator in (4.5) is given as

$$x(t) = 2\cos\left(\sqrt{3}t\right) - \sin\left(\sqrt{3}t\right) + \cos(t) - 3\sin(t).$$

Differentiate $x(t)$ twice, and then find $y(t)$ using the first equation of (4.5). (This approach to find y does not use any of the theory above.) Could you have read off $y(t)$ immediately from the general solution (4.8) and the given $x(t)$?